# The undivided energy and its differentiation

$E = mc^2$ in the split-signature arena: the unitary energy, the symmetric conversion $c^2$, the differentiated mass, the place of the Yang–Mills $b$-field, and the duplication from $\mathrm{Cl}(3,3)$ to $\mathrm{Cl}(6,6)$

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**Abstract**

We test, inside the split-signature arena $\mathrm{Cl}(3,3)$ of [1], a reading of the rest-energy relation proposed by the author of that programme: in $E = mc^2$, $c^2$ is the symmetric part, $m$ is what breaks the symmetry, the energy is one undivided six-dimensional entity whose internal differentiation produces the mass, and the mass is generated by the Yang–Mills $b$-field. The undivided energy exists: the only rotation- and duality-invariant quadratic energy of the six frame components is the positive form $E_6 = \frac{1}{2}(|\mathbf{B}|^2 + |\mathbf{E}|^2)$, whose pointwise symmetry group is $U(3)$, and in the closure of the force law it is a Casimir of the momentum field. It differentiates without any interaction, $E_6^2 = E_0^2 + |\mathbf{S}|^2$, with $E_0 = \frac{1}{2}|q|$ the complete invariant of Lorentz transformations and duality and the minimum of $E_6$ over boosts; this is the field-point form of $E^2 = (pc)^2 + (mc^2)^2$. The constant $c$ is a conversion forced by Schur's lemma, not a symmetry; the clause "$c^2$ is the symmetric part" holds exactly for the $\eta$-self-adjoint summand $\kappa G$ of the shift, $(\kappa G)^2 = -c^2/\ell^2$, which is the only summand that binds. A nonzero rest energy breaks duality to $\mathbb{Z}_2$ and, for matter, the chiral algebra to $\mathfrak{u}(1)_J$; it breaks neither Lorentz invariance nor $\mathrm{Spin}(3,3)$, since the Dirac mass form $e_1e_2e_3$ is invariant under the whole six-dimensional group. The $b$-field is the antisymmetric, connection summand of the same shift: classically it never generates the binding rate, a part commuting with $G$ only splits the rest frequency into charge-labelled lines, and at three constituents the compatible algebra is $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$. What remains for it is the one scale the algebra leaves free, which dimensional transmutation could supply; that identification is a conjecture. We also give the arena's answers on particles and antiparticles (the two senses of one rotation, exchanged by charge conjugation), the vacuum and its zero-point energy, and the Goldstone bosons.

A second part takes up two further remarks of the author: that the $b$-particle is a deal breaker when two manifolds or more are in question, and that a unitary energy should result in a duplication, from $\mathrm{Cl}(3,3)$ to $\mathrm{Cl}(6,6)$. The duplication is forced on the algebra, as the graded tensor product whose spinor module is a Fock space. It is chosen on the energy, within the three-sphere of a lossless beam splitter, which the metric cuts to one circle, the copy rotation. The duplicate carries a copy charge, a non-rigid vacuum and a balance surface on which massive constituents turn at a constant rate. But no covariant duplication makes a massive pair out of one null field, and the rest energy of one copy is gauge dependent unless the copies are distinguished. Between manifolds the $b$-field exists: under a common Lorentz action it is the copy algebra $\mathfrak{so}(n,\mathbb{C})$, with compact part $\mathfrak{so}(n)$. For unequal rates it cannot be removed. It repels and confines, and from three manifolds on it carries a time-odd invariant, the analogue of Jarlskog's. It never creates or changes a rate, because the rates are gauge invariants, and the copy isospin is integral. The deal breaker is one of transmission, not of creation. The differentiation of the rates is what makes the $b$-field physical, the reverse of the order in the hypothesis. Every finite-dimensional statement was checked twice, by independent scripts.


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# 1 The question

## 1.1 The hypothesis in its author's words

The starting point of this note is a reading of the rest-energy relation proposed by the author of the programme:

> *$E = mc^2$: consider $c^2$ as the symmetrical part of the equation and $m$ as what breaks this symmetry; $m$ is generated from a $b$ particle, $b$ standing for the b-particle of Yang–Mills.*

Three clarifications fix the intended reading. First, the energy is to be taken as *one undivided entity*, and the mass is to emerge through an *internal differentiation* of that unity, before any external interaction is applied. Second, $c^2$ corresponds to the variable $X$ of the former momentum–force chapter, whose successor is the interior dynamics of Part IV of the paper [1] (Sections 15–17), where the shift operators are $\alpha_\pm = \partial_\tau \pm \kappa G$ with $\kappa = c/\ell$. Third, the energy is six-dimensional. The $b$-field is the non-abelian gauge potential of Yang and Mills [63], whose quanta carry isotopic spin; Yang and Mills left open whether those quanta are massive.

An earlier exploratory session settled several points that we take as given and do not repeat. The scalar $c^2$ is not by itself a symmetry, and if $E$ and $c^2$ are invariant under a group then so is $m = E/c^2$, so a mass cannot *be* the breaking of a symmetry that $E$ and $c^2$ share. A nonzero invariant mass does not break Lorentz invariance. The free six-dimensional scalar has an energy unbounded below ([1], Proposition 27.1), and positive four-dimensional mass cannot be obtained from momentum along the surplus times, since $m_4^2 = \mu_6^2 - (p_u^2 + p_v^2)/c^2$. Non-orthogonality of $\mathbf{E}$ and $\mathbf{B}$ generates no mass ([1], Section 14). The same session was rightly criticised for jumping to Yang–Mills interactions, condensates and gap equations before the unity of the energy and its differentiation had been built. This note does the work in the order the hypothesis asks for: the undivided energy first, its differentiation second, the dynamics of the differentiation third, and the $b$-field last, where it belongs.

### 1.2 The second question: duplication, and the $b$-field between manifolds

After the first four sections had established that, on one manifold, the $b$-field can split or destabilise the binding but never generate it, the author added two remarks that define the second half of this note: *the b-particle is a deal breaker when two manifolds or more are in question*, and *one should investigate what physical and mathematical process makes a unitary energy result in a duplication, from* $\mathrm{Cl}(3,3)$ *to* $\mathrm{Cl}(6,6)$. We read the first as the claim that a $b$-field acting *between* copies of the arena can do what it cannot do on one copy, and the second as the question of which constructions and processes take the undivided energy of one copy to two, and what the duplicate can do that a single copy cannot. Sections 7 to 9 answer them and Section 10 collects the answers.

### 1.3 The dictionary under test

All constructions are made inside the arena of [1]: $V = \mathbb{R}^{3,3}$ with $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$, the realisation of $\mathrm{Cl}(3,3)$ on $\mathbb{R}^8$, the binding root $G$ ($Ge_i = e_{i+3}$, $Ge_{i+3} = -e_i$), which squares to $-\mathbf{1}$, is self-adjoint for $\eta$ and acts as the Hodge star in the field-strength reading $X = \sum_i (B_i e_i + E_i e_{i+3})$, $z = \mathbf{B} + i\mathbf{E}$, $q(z) = z \cdot z$ ([1], Theorem 12.3). The working dictionary is

| hypothesis | candidate object in the arena |
|---|---|
| $E$, undivided, six-dimensional | the positive, duality-invariant energy $E_6 = \frac{1}{2}\lvert z\rvert^2 = \frac{1}{2}(\lvert\mathbf{B}\rvert^2 + \lvert\mathbf{E}\rvert^2)$ of a frame vector; in the closure of the force law, the Casimir $\lvert p\rvert^2$ |
| $c^2$, the symmetric part | the conversion $C = c\,\mathbf{1}$ between the temporal and the spatial triple forced by Schur's lemma ([1], Theorem 3.2), and the $\eta$-self-adjoint, factorising part $\kappa G$ of the shift, $(\kappa G)^2 = -c^2/\ell^2$ |
| $m$, the differentiation | the Lorentz-class invariant $E_0 = \frac{1}{2}\lvert q\rvert$, the rest-frame energy density of the field; for Dirac matter the mass form $T = e_1 e_2 e_3$ |
| $b$, the Yang–Mills field | the $\eta$-antisymmetric (transport) part of the shift, $\mathrm{End}(V) = \mathfrak{so}(3,3) \oplus \{X = X^\dagger\}$ ([1], Theorem 17.3); at three constituents the binding-compatible algebra $\mathfrak{so}(4)_\Delta \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ ([1], Theorem 17.21) |

Every entry is a hypothesis to be tested, and several of them turn out to need correction.

### 1.4 How claims are classified

We follow the deflationary principle of [1] (Corollary 4.4): a statement about the arena carries information only if it could have come out otherwise. Every claim below belongs to one of six classes, named in the text. A claim is *fixed* when it is an identity or theorem of the algebra that could have failed; *constructed* when the algebra admits the object and nothing selects it; *obstructed* when it is impossible under stated hypotheses; *open* when it is neither established nor refuted. To these four classes of the paper we add two that the subject requires: *inherited*, for facts of ordinary relativistic physics that hold in any notation and are reproduced only because something else is stated in terms of them, and *interpretive*, for readings and dictionary entries, which are not theorems at all. The relation $E = mc^2$ itself is inherited. What can be new is only how the arena realises its three ingredients, and which of the realisations are forced.

Every finite-dimensional statement was checked by exact or high-precision computation. Each section has a verification script in `investigations/rest_energy/verify/`, written independently of a second, adversarial check script by a different author; the scripts print one `PASS` line per claim and exit with status zero only if all pass (Section 11).

## 2 The undivided energy and its differentiation

The hypothesis reads $E = mc^2$ [13] with $c^2$ as the symmetric part of the equation and $m$ as what breaks that symmetry; the agreed clarification asks that energy first be one undivided, six-dimensional entity, and that mass appear through an internal differentiation of that unity before any interaction is applied. We follow that order: the undivided energy is built from the arena alone and differentiated with no other field and no coupling, $c$ is restored, and the Yang–Mills field $b$ [63] enters only in Section 2.9. The points settled in Section 1 ($c^2$ alone is no symmetry; a mass breaks no Lorentz invariance; Proposition 27.1, Example 27.3 and Section 14.4 of [1]) are used without repetition. Most of what follows is inherited electrodynamics or restates results of [1], and is labelled so; the verdicts are in Section 2.10.

**Conventions.** We use the field-strength reading of Theorem 12.3 of [1]: $X = \sum_i (B_i e_i + E_i e_{i+3})$, $z = \mathbf{B} + i\mathbf{E} \in \mathbb{C}^3$, the binding root $G$ ($Ge_i = e_{i+3}$, $Ge_{i+3} = -e_i$) acts as $i$, $q(z) = z \cdot z = |\mathbf{B}|^2 - |\mathbf{E}|^2 + 2i\,\mathbf{E}\cdot\mathbf{B}$, $\eta(X,X) = \operatorname{Re} q$ and $\eta(X, GX) = -2\,\mathbf{E}\cdot\mathbf{B}$. The frame variable is minus the textbook field, $\mathbf{B}_{\mathrm{t}} = -\mathbf{B}$ (Section 1.4 of [1]). With $h(z,w) = \sum_j \bar{z}_j w_j$ we put

$$E_6 := \tfrac{1}{2}|z|^2 = \tfrac{1}{2}(|\mathbf{B}|^2 + |\mathbf{E}|^2), \qquad E_0 := \tfrac{1}{2}|q(z)|, \qquad \mathbf{S} := \mathbf{E}\times\mathbf{B} = \tfrac{i}{2}\,\bar{z}\times z = -\mathbf{S}_{\mathrm{t}}, \quad \mathbf{S}_{\mathrm{t}} := \mathbf{E}\times\mathbf{B}_{\mathrm{t}}. \tag{1}$$

$\mathbf{S}$ is the frame Poynting vector of Section 12 of [1], $\mathbf{S}_{\mathrm{t}}$ the textbook one. Applied to the frame field $X$ these are the frame field's energies; for the momentum field $\pi = \eta^{-1}p$ of the closure we write $E_6(\pi)$, $E_0(\pi)$, $\mathbf{S}(\pi)$. Units are Heaviside–Lorentz with $c = 1$ until Section 2.5. For cyclic $(i,j,k)$ the diagonal rotations and boosts of $\mathfrak{l}_G$ are $L_i = \frac{1}{2}(e_{j+3}e_{k+3} - e_j e_k)$ and $K_i = \frac{1}{2}(e_k e_{j+3} - e_j e_{k+3})$, acting on vectors by the commutator: $L_i$ by $\delta\mathbf{B} = e_i \times \mathbf{B}$, $\delta\mathbf{E} = e_i \times \mathbf{E}$, and $K_i$ by $\delta\mathbf{E} = e_i \times \mathbf{B}$, $\delta\mathbf{B} = -e_i \times \mathbf{E}$; $\mathfrak{p}_G$ is the part of $\mathfrak{so}(3,3)$ anticommuting with $G$, so $\mathfrak{so}(3,3) = \mathfrak{l}_G \oplus \mathfrak{p}_G$. We write the Lorentz-class inertia of Theorem 17.11 of [1] as $N = (\nu_1\eta^{-1} + \nu_2 G\eta^{-1})/m$, the paper's $(a,b)$ renamed, so that $b$ denotes only the Yang–Mills field. These laws, Theorem 12.3, Example 12.4 and Lemma 3.4 of [1] ($\mathrm{Ad}_{e_1e_2e_3}|_V = \eta$, $G = -\mathcal{K}\circ\mathrm{Ad}_{e_1e_2e_3}$) were checked in the realisation of Section 2 of [1] (d1-0); identifiers d1-$n$ refer to the verification script.

### 2.1 One undivided energy, with pointwise symmetry $U(3)$

**Theorem 2.1** (Uniqueness of the undivided energy). *1. The symmetric forms on $V$ invariant under the diagonal rotations are spanned by $|\mathbf{B}|^2$, $|\mathbf{E}|^2$, $\mathbf{B}\cdot\mathbf{E}$, and split as $\mathbb{R}\,\mathbf{1} \oplus \operatorname{span}\{\eta, \eta G\}$, the first summand commuting and the second anticommuting with $G$.*

2. *A quadratic energy invariant under rotations and under duality $e^{\vartheta G}$ is a multiple of $E_6$, which is positive definite.*

3. *The forms invariant under all of $\mathfrak{l}_G$ are* $\operatorname{span}\{\eta, \eta G\}$*, that is $|\mathbf{B}|^2 - |\mathbf{E}|^2$ and $\mathbf{E}\cdot\mathbf{B}$; none is definite.*

*Proof.* The diagonal rotations act on $V = V^+ \oplus V^-$ as $\mathbb{R}^2 \otimes \mathbb{R}^3$ with $SO(3)$ on the second factor, which is absolutely irreducible, so by Schur's lemma [51] the invariant forms are $N = \left(\begin{smallmatrix} \alpha & \gamma \\ \gamma & \beta \end{smallmatrix}\right) \otimes \mathbf{1}$. With $G = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) \otimes \mathbf{1}$, $\frac{1}{2}(N - GNG) = \frac{1}{2}(\alpha + \beta)\mathbf{1}$ and $\frac{1}{2}(N + GNG) = \frac{1}{2}(\alpha - \beta)\eta - \gamma\,\eta G$. Since $G^\top G = \mathbf{1}$ and $G^\top = -G$, $N$ is $e^{\vartheta G}$-invariant iff $G^\top N + NG = 0$ iff $[N, G] = 0$, which gives (2). For (3) the boost law gives $\delta(\alpha|\mathbf{B}|^2 + \beta|\mathbf{E}|^2 + 2\gamma\,\mathbf{B}\cdot\mathbf{E}) = -2(\alpha + \beta)\,\mathbf{v}\cdot(\mathbf{E}\times\mathbf{B})$, the $\gamma$ term being unchanged, so $\beta = -\alpha$. □

This restates Theorem 17.11(iii) and Proposition 17.4 of [1] on $V$, and is inherited: for any binding root $G' = bGb^{-1}$, $b \in SO(3,3)$, the forms invariant under the compact part of $\mathfrak{l}_{G'}$ and under $G'$ are the multiples of the positive $b^{-\top}b^{-1}$, so the statement could not have failed; it is the uniqueness of the energy density among rotation- and duality-invariant forms. The arena adds only that this form is the Euclidean one of its basis: $G$ is $\eta$-self-adjoint and anticommutes with $\mathrm{Ad}_{e_1e_2e_3}|_V = \eta$, so $G^\top G = \eta G \eta G = \mathbf{1}$ (Lemma 3.4 of [1]; d1-1a, with an exact conjugated root). $E_6$ has duality weight 0, while $e^{\vartheta G}$ turns $(\operatorname{Re} q, \operatorname{Im} q)$ through $2\vartheta$: $E_6$ is undivided, distinguishing neither the magnetic triple from the electric one nor, as we now show, rest from motion.

**Proposition 2.2** (The pointwise symmetry of the undivided energy is $U(3)$)**.** *(1) The maps of $V$ preserving $E_6$ and commuting with $G$ form $\mathfrak{u}(3)$, of dimension 9: $A = \left(\begin{smallmatrix} P & Q \\ -Q & P \end{smallmatrix}\right)$ acts on $z$ by the anti-Hermitian matrix $P - iQ$. (2) The Lie algebra of $O(3,\mathbb{C}) \times U(1)_G$, the $G$-linear maps preserving $q$ up to a phase, is $\mathfrak{l}_G \oplus \mathbb{R}G$, of dimension 7. (3) Their intersection is $\mathfrak{so}(3) \oplus \mathbb{R}G$, of dimension 4, the maximal compact subalgebra of $\mathfrak{l}_G \oplus \mathbb{R}G$; and $\mathfrak{u}(3) \cap \mathfrak{l}_G = \mathfrak{so}(3)$. (4) The stabiliser of the function $E_0$ in $U(3)$ is $U(1)\cdot O(3) = \{e^{i\varphi}O\}$; for instance the unitary map $(1,0,0) \mapsto \frac{1}{\sqrt{2}}(1,i,0)$ carries a field at rest ($E_0 = E_6 = \frac{1}{2}$) to a null field ($E_0 = 0$) at the same $E_6$.*

*Proof.* (1) Commuting with $G$ forces the block form, which is multiplication by $P - iQ$ on $z$; $A^\top = -A$ means $P^\top = -P$, $Q^\top = Q$, so $P - iQ$ is anti-Hermitian; $3 + 6 = 9$. (2) $q(e^{tM}z) = e^{i\varphi t}q(z)$ gives $M^\top + M = i\varphi\,\mathbf{1}$, so $M = \frac{i\varphi}{2} + M_0$ with $M_0 \in \mathfrak{so}(3,\mathbb{C}) = \mathfrak{l}_G$ (Theorem 12.1 of [1]), and $i\,\mathbf{1}$ is $G$. (3) A complex antisymmetric anti-Hermitian matrix is real. (4) If $U \in U(3)$ preserves $|q|$, the form $z^\top U^\top U z$ vanishes on the cone $q = 0$; $q$ is irreducible, so $U^\top U = e^{i\varphi}\mathbf{1}$, and $O = e^{-i\varphi/2}U$ is unitary and complex orthogonal, hence real. □

Parts (1)–(3) are inherited Hermitian linear algebra, which makes "unitary" literally true of the pointwise form. Part (4) is inherited as well, the stabiliser of $|z\cdot z|$ among unitary maps of the Riemann–Silberstein vector; read as a statement about the rest energy, it is the exact sense in which $E_0$ "breaks" something: $U(3)$ is reduced to $U(1)\cdot O(3)$, and the five-dimensional orbit $U(3)/U(1)\cdot O(3)$ parametrises the real structures $z \mapsto \bar{z}$ up to phase, of which the arena has chosen one (Proposition 2.11; d1-1b, d1-8b). But $U(3)$ is the symmetry of the pointwise form $E_6$ only. It is not a symmetry of Maxwell's equations on the frame, $\partial_t z = -i\nabla\times z$ (Proposition 12.10 of [1]): the unitary of (4) carries the wave $(1,i,0)e^{ik(x_3-t)}$ of Example 12.11 to $\frac{1+i}{\sqrt{2}}(1,1,0)e^{ik(x_3-t)}$, which does not solve them. Nor is it one of the split-form interior, since $G = i\,\mathbf{1} \in U(3)$ reverses $\eta$ and $U(3) \cap O(\eta) = O(3)$; of the closures it is a symmetry only for $N = \mathbf{1}/m$, the $U(3)$-invariant forms being the multiples of $\mathbf{1}$.

## 2.2 The rest energy is the complete invariant of Lorentz and duality

**Theorem 2.3** (Orbits)**.** *(1) Each level set $\{|q| = r^2\}$, $r > 0$, is one orbit of $O(3,\mathbb{C}) \times U(1)_G$, of real dimension 5, with two-dimensional stabilisers; every non-null $z$ is carried to $(r,0,0)$,*

$r = \sqrt{2E_0}$. *(2) The nonzero null fields form one orbit of* $SO(3,\mathbb{C})$ *alone, of dimension* 4*, and at a null point* $Gz$ *is tangent to the Lorentz orbit. (3) Hence* $E_0$ *is a complete invariant of Lorentz and duality on non-null fields, and* $E_6$ *is the complete invariant of* $U(3)$*, whose orbits are the spheres.*

*Proof.* (1) Write $q(z) = r^2 e^{i\varphi}$ and $w = e^{-i\varphi/2} z/r$ (a duality rotation), so $q(w) = 1$. Since $q$ is nondegenerate on $w^\perp = \{f : f \cdot w = 0\}$, some $f \in w^\perp$ has $f \cdot f = 1$; with $f_3 = w \times f$, $(a \times b) \cdot (a \times b) = (a \cdot a)(b \cdot b) - (a \cdot b)^2$ gives $f_3 \cdot f_3 = 1$, so $A = (w\, f\, f_3)^\top \in SO(3,\mathbb{C})$ and $Ae^{-i\varphi/2} z = re_1$. An element $M_0 + \frac{i\varphi}{2}$ fixes $e_1$ iff $\varphi = 0$ and $M_0$ is a complex rotation about $e_1$: the stabiliser is $\mathfrak{so}(2,\mathbb{C})$ and the orbit has dimension $7 - 2 = 5$. (2) A rotation carries a null $z$ to $\lambda(1, i, 0)$, $\lambda > 0$; $K_3$ acts by $\delta z = i\, e_3 \times z$ and $i\, e_3 \times (1, i, 0) = (1, i, 0)$, so the Doppler boost $e^{\chi K_3}$, $\chi = -\log \lambda$, finishes. The Lorentz tangent space at a null $z$ is its $q$-orthogonal complement, which contains $iz$; at $(1, i, 0)$, $iz = -e_3 \times z$ is the rotation $L_3$, the helicity of Example 13.11 of [1]. (3) follows. □

Inherited: the classification of a field strength at a point [54, 32] (d1-2a, d1-2b). Without duality the complete invariant is $q$, whose phase (twice the complexion of Proposition 12.12 of [1]) duality removes.

### 2.3 The differentiation identity

**Proposition 2.4** (Differentiation)**.** *For every* $z \in \mathbb{C}^3$

$$|z|^4 = |z \cdot z|^2 + |\bar{z} \times z|^2, \qquad \textit{equivalently} \qquad E_6^2 = E_0^2 + |\mathbf{S}|^2. \tag{2}$$

*The rank-one Hermitian* $H = zz^\dagger$ *carries all three terms:* $\operatorname{tr} H = 2E_6$, $(\operatorname{Im} H)_{jk} = \epsilon_{jkl} S_l$, $\operatorname{tr}(H\bar{H}) = |q|^2 = 4E_0^2$, $H^2 = (\operatorname{tr} H)H$. *Hence* $E_0 \le E_6$*, with equality iff* $\mathbf{S} = 0$*, iff* $\bar{z} \parallel z$*, iff* $\mathbf{B} \parallel \mathbf{E}$ *or one of them vanishes.*

*Proof.* $\bar{z} \times z = -2i\, \mathbf{E} \times \mathbf{B}$, and $(a \times b) \cdot (c \times d) = (a \cdot c)(b \cdot d) - (a \cdot d)(b \cdot c)$ at $(\bar{z}, z, z, \bar{z})$ gives $|\bar{z} \times z|^2 = |z|^4 - |q|^2$. Next $H_{jk} = B_j B_k + E_j E_k + i\epsilon_{jkl}(\mathbf{E} \times \mathbf{B})_l$ and $\operatorname{tr}(H\bar{H}) = |\sum_j z_j^2|^2$. The equality case is Cauchy–Schwarz for $|h(\bar{z}, z)| \le |z|^2$. □

Inherited, a complex Lagrange identity; the relation $u^2 - |\mathbf{S}|^2 = \frac{1}{4}|q|^2$ is stated in Section 14.3 of [1] (d1-3a). In the user's terms it is exact: the undivided $E_6$ is the hypotenuse of a right triangle whose legs are a Lorentz-invariant rest part $E_0$ and a motion part $|\mathbf{S}|$, and the split needs no interaction and no second field; in Minkowski terms $E_0$ is the Lorentz length of $(E_6, \mathbf{S})$, although $(E_6, \mathbf{S}_\mathrm{t})$ is not a four-vector (Proposition 2.5).

**Proposition 2.5** (Rainich identity)**.** *For the stress–energy tensor of* $(\mathbf{E}, \mathbf{B}_\mathrm{t})$*,* $T^{00} = \frac{1}{2}(|\mathbf{E}|^2 + |\mathbf{B}_\mathrm{t}|^2) = E_6$*,* $T^{0i} = (\mathbf{S}_\mathrm{t})_i$*,* $T^{ij} = T^{00}\delta_{ij} - E_i E_j - B_{\mathrm{t}i} B_{\mathrm{t}j}$*, one has* $T^\mu{}_\mu = 0$ *and*

$$T^\mu{}_\lambda T^\lambda{}_\nu = E_0^2\, \delta^\mu{}_\nu, \qquad T_{\mu\nu} T^{\mu\nu} = 4E_0^2. \tag{3}$$

$T^{\mu\nu}$ *is duality invariant, and for every unit timelike observer* $t$ *the density* $P(t)^\mu = T^\mu{}_\nu t^\nu$ *is future causal with* $\eta(P(t), P(t)) = E_0^2$*.*

*Proof.* Duality invariance is a polynomial identity; both sides of (3) are covariant, so by Theorem 2.3 it suffices to take $z = (r, 0, 0)$, where $T^\mu{}_\nu = \frac{r^2}{2}\operatorname{diag}(1, 1, -1, -1)$, and null fields follow by continuity. Causality of $P(t)$ is the dominant energy condition [22]. □

Inherited [49, 54] (d1-3b). The mass of a field point is the Lorentz scalar of the tensor whose 00 component is the undivided energy, but $(u, \mathbf{S}_\mathrm{t})$ is not a four-vector: each observer sees a different density $P(t)$, all of length $E_0$.

## 2.4 The rest energy as a minimum, and two boost factors

**Theorem 2.6** (Variational form of the rest energy)**.** *1. For non-null $z$, $E_0$ is the minimum of $E_6$ over the Lorentz orbit, and over the Lorentz–duality orbit, attained exactly where $\mathbf{S} = 0$; for null $z \neq 0$ the infimum 0 is not attained. Moreover $E_6 = E_0/\sqrt{1 - v_E^2}$ with $v_E := |\mathbf{S}|/E_6 < 1$.*

*2. If $\mathbf{B} \parallel \mathbf{E} \parallel \mathbf{n}$ and the boost $e^{\chi K}$ of rapidity $\chi$ has axis orthogonal to $\mathbf{n}$, then $E_6 = E_0 \cosh 2\chi$, $|\mathbf{S}| = E_0 \sinh 2\chi$ and $v_E = \tanh 2\chi = 2\beta/(1+\beta^2)$, $\beta = \tanh \chi$; a boost along $\mathbf{n}$ changes neither $E_6$ nor $\mathbf{S} = 0$.*

*3. The observer moving with $\mathbf{v}_{\text{rest}} = \mathbf{S}_{\text{t}}/(E_6 + E_0)$ sees $\mathbf{S} = 0$ and $E_6 = E_0$; $|\mathbf{v}_{\text{rest}}| = \beta$.*

*Proof.* (1) By (2) $E_6 \geq E_0$ with equality iff $\mathbf{S} = 0$, and Theorem 2.3 supplies points with $\mathbf{S} = 0$ in both orbits; for null $z$ the Doppler boost drives $E_6$ to 0. Divide (2) by $E_6^2$. (2) Take $\mathbf{n} = e_1$, $X = \sigma_1 e_1 + \sigma_2 e_4$. By Example 12.4 of [1], $[K_3, e_1] = e_5$, $[K_3, e_5] = e_1$, $[K_3, e_4] = -e_2$, $[K_3, e_2] = -e_4$, so $\operatorname{ad}_{K_3}^3 = \operatorname{ad}_{K_3}$, $e^{\chi \operatorname{ad}_{K_3}} = \mathbf{1} + \sinh\chi \operatorname{ad}_{K_3} + (\cosh\chi - 1)\operatorname{ad}_{K_3}^2$, $\mathbf{B} = (\sigma_1 \cosh\chi, -\sigma_2 \sinh\chi, 0)$, $\mathbf{E} = (\sigma_2 \cosh\chi, \sigma_1 \sinh\chi, 0)$, and $\mathbf{S} = -E_0 \sinh 2\chi\, e_3$; likewise for $K_2$, while $K_1$ fixes $e_1$ and $e_4$. (3) $|\mathbf{S}|/(E_6 + E_0) = \sinh 2\chi/(\cosh 2\chi + 1) = \tanh\chi$. For the direction: $e^{\chi K_3}$ is the passage to a frame moving with $-\tanh\chi\, e_3$ (Section 12 of [1]), in which the rest frame moves along $+e_3$, and so does $\mathbf{S}_{\text{t}} = -\mathbf{S}$. □

Inherited [32]; verified with the paper's own $K_i$, symbolically and numerically; the boost built from $\mathbf{E} \times \mathbf{B}$ instead of $\mathbf{S}_{\text{t}}$ leaves a Poynting vector of at least $0.2\,E_0$ (d1-4a–d1-4c). This is the rest-frame velocity $\mathbf{S}/(u + \frac{1}{2}|q|)$, with $\frac{1}{2}|q| = u\sqrt{1 - v_E^2}$, of Section 14.4 of [1], where $\mathbf{S}$ is the textbook $\mathbf{S}_{\text{t}}$, as Section 1.4 of [1] stipulates for formulas from the literature; with the frame vector $\mathbf{S} = \mathbf{E} \times \mathbf{B}$, which Section 12 also calls the Poynting vector, the direction would be reversed. A field point carries two boost factors, $E_6 = E_0 \cosh 2\chi$ against $E = mc^2 \cosh\chi$ for a particle, because the energy density is a component of a second-order tensor and is quadratic in the boost (the standard behaviour of $T^{00}$): the rest frame moves with $\beta$, the energy with $\beta \oplus \beta$, and an observer riding with the energy velocity sees the same $E_6$ and the reversed Poynting vector. $E_0$ is a rest-energy density of a field point, not the rest energy of a particle carried by the flux.

## 2.5 Restoring $c$: where $c^2$ enters

Adding the dimensionless $G$ to a derivative along a clock coordinate needs a length $\ell$ (Section 3 of [1]):

$$\kappa = c/\ell, \qquad \alpha_\pm = \partial_\tau \pm \kappa G. \tag{4}$$

**Proposition 2.7** ($c^2$ is an isotropic weight)**.** *Let the temporal triple carry $\mathbf{B}$ and the spatial triple $\mathbf{E}$, $C : V^+ \to V^-$ the conversion and $G_C = \left(\begin{smallmatrix} 0 & -C^{-1} \\ C & 0 \end{smallmatrix}\right)$, $G_C^2 = -\mathbf{1}$. (1) $G_C$ commutes with the diagonal rotations iff $C = c\,\mathbf{1}$. (2) For $C = c\,\mathbf{1}$ the forms invariant under the diagonal rotations and under $e^{\vartheta G_c}$ are the multiples of $c^2|\mathbf{B}|^2 + |\mathbf{E}|^2$. (3) $(\kappa G)^2 = -c^2/\ell^2$, and $\ell$ is not fixed.*

*Proof.* (1) The lower-left block of $[G_C, R \oplus R]$ is $CR - RC$, and Schur's lemma gives $C = c\,\mathbf{1}$: Theorem 3.2 of [1]. (2) For $N = \left(\begin{smallmatrix} \alpha & \gamma \\ \gamma & \beta \end{smallmatrix}\right) \otimes \mathbf{1}$, $G_c^\top N + N G_c = \left(\begin{smallmatrix} 2c\gamma & c\beta - \alpha/c \\ c\beta - \alpha/c & -2\gamma/c \end{smallmatrix}\right) \otimes \mathbf{1}$, zero iff $\gamma = 0$ and $\alpha = c^2\beta$. (3) $G^2 = -\mathbf{1}$. □

Part (1) is Theorem 3.2 of [1], fixed there; part (2) is inherited; verified exactly, including the failure for $C = \operatorname{diag}(c_1, c_2, c_3)$ with distinct entries (d1-5b). Even the weight is a unit convention: $\Phi = \operatorname{diag}(c, c, c, 1, 1, 1)$ conjugates $G_c$ to $G$ and turns $c^2|\mathbf{B}|^2 + |\mathbf{E}|^2$ into $|\Phi X|^2$, so with $\mathbf{B}$ measured as $c\mathbf{B}$ (Heaviside–Lorentz units) $c$ disappears from $E_6$ and appears only in $\rho_0 = u_0/c^2$ below.

**Proposition 2.8** (SI form)**.** *With* $\varepsilon_0\mu_0c^2 = 1$, $z_c := c\mathbf{B} + i\mathbf{E}$, $u = \frac{\varepsilon_0}{2}|z_c|^2$, $\mathbf{g} = \mathbf{S}_\mathrm{t}/c^2$ *with* $\mathbf{S}_\mathrm{t} = \mathbf{E}\times\mathbf{B}_\mathrm{t}/\mu_0$, $q_c := z_c\cdot z_c = c^2|\mathbf{B}_\mathrm{t}|^2 - |\mathbf{E}|^2 - 2ic\,\mathbf{B}_\mathrm{t}\cdot\mathbf{E}$ *and* $u_0 := \frac{\varepsilon_0}{2}|q_c|$, *Proposition 2.4 applied to* $z_c$ *gives*

$$u^2 = c^2|\mathbf{g}|^2 + u_0^2, \qquad \rho_0 := u_0/c^2, \qquad [u] = [c\,\mathbf{g}] = [u_0] = \mathrm{J\,m^{-3}}, \quad [\rho_0] = \mathrm{kg\,m^{-3}}. \tag{5}$$

Inherited (d1-5a); calling $\rho_0$ a rest-mass density is a reading, since $(u, c\,\mathbf{g})$ is not a four-vector. This is the exact content of "$c^2$ is the symmetric part": $c$ is the unique conversion between the triples that commutes with their common rotations, the residual symmetry of both $E_6$ and $E_0$; in SI units it enters $E_6$ as the isotropic weight $c^2$ of $|\mathbf{B}|^2$, which a choice of units removes. It is a unit conversion, not a symmetry, and nothing breaks it: $u_0$ uses the same $c$, and $\rho_0 = u_0/c^2$ only converts units.

## 2.6 Finite configurations: the mass of a whole

**Proposition 2.9** (Mass of a whole)**.**

1. *For a free field with compactly supported Cauchy data,* $(U,\mathbf{P}) = \int(u,\mathbf{S}_\mathrm{t})\,d^3x$ *is a four-vector and* $M^2 := U^2 - |\mathbf{P}|^2 \geq U\int(u - |\mathbf{S}|)\,d^3x \geq 0$, *so* $M > 0$ *whenever* $q \neq 0$ *on a set of positive measure.*

2. *In every frame* $M \geq \int E_0\,d^3x$, *with equality iff the densities* $(u,\mathbf{S}_\mathrm{t})(x)$ *are all proportional.*

3. *A plane wave has* $M = 0$. *The waves* $E_x = a\cos(kz \mp \omega t)$, $cB_{\mathrm{t}y} = \pm a\cos(kz \mp \omega t)$ *superpose to* $E_x = 2a\cos kz\cos\omega t$, $cB_{\mathrm{t}y} = 2a\sin kz\sin\omega t$; *per cell (unit area, length* $2\pi/k$*)* $U = 2\pi\varepsilon_0a^2/k$ *and* $\mathbf{P} = 0$, *so* $Mc^2 = U$, *although each constituent has* $q_c = 0$ *and* $U_\pm = |\mathbf{P}_\pm|c$. *The total has* $q_c = -2a^2(\cos 2kz + \cos 2\omega t)$, $\int_\mathrm{cell} T^{zz} = U$, $\int_\mathrm{cell} T^{xx} = -U\cos 2\omega t$, *and the integrated rest energy* $\int_\mathrm{cell} u_0$ *oscillates between* $\frac{2}{\pi}U$ *and* $U$.

4. *With* $c = 1$ *and* $f(s) = (1-s^2)^2$ *on* $|s| \leq 1$, $0$ *elsewhere,* $E_x = f(z-t) + f(z+t)$, $B_{\mathrm{t}y} = f(z-t) - f(z+t)$ *has, per unit transverse area,* $U = \frac{512}{315}$ *and* $\mathbf{P} = 0$ *at all times, so* $M = U$, *with* $(U, P_z)$ *a two-vector under boosts along* $z$ *only; for* $|t| \geq 1$ *the pulses are disjoint,* $q \equiv 0$ *and* $\int E_0 = 0$.

*Proof.* (1) Conservation and finite propagation speed make $\int_\Sigma T^{\mu\nu}d\Sigma_\nu$ independent of the spacelike hyperplane $\Sigma$ [34]; $u - |\mathbf{S}| = E_0^2/(u + |\mathbf{S}|) \geq 0$, and $M^2 = (U - |\mathbf{P}|)(U + |\mathbf{P}|)$. (2) Each $(u,\mathbf{S}_\mathrm{t})(x)$ is future causal of length $E_0(x)$ (Proposition 2.5), and future causal vectors obey $|P_1 + P_2| \geq |P_1| + |P_2|$, with equality iff proportional. (3) Elementary cell integrals; for the last statement, with $s = \sin^2\omega t$ one has $q_c = 4a^2(s - \cos^2 kz)$, so $u_0 = 2\varepsilon_0a^2|s - \cos^2 kz|$, whose mean is least at the median $s = \frac{1}{2}$ of the arcsine law, giving $\frac{2}{\pi}U$. (4) $u = f(z-t)^2 + f(z+t)^2$ and $S_z = f(z-t)^2 - f(z+t)^2$, the cross terms cancelling, and $q = -4f(z-t)f(z+t)$. □

Items (1) and (3) are inherited, (1) being the statement of Section 14.4 of [1]. The cell in (3) is not isolated: $\int T^{zz} = U$ is the pressure on whatever confines it, so its $(U,\mathbf{P})$ is a four-vector only with the walls included, as for radiation in a box [34]. Items (2) and (4) refute the reading, proposed in the working dictionary of this investigation, that the mass of a whole "emerges from the interference that makes $q \neq 0$": interference suffices for $M > 0$ but is not necessary. A whole gets its mass from the non-collinearity of its constituents' energy–momenta, as two photons do; the interference reading is obstructed. $M$ is also blind to orthogonality, by the example of Section 14.4 of [1]: $(\mathbf{E},\mathbf{B}_\mathrm{t}) = (0,\mathbf{F})$ and $(\mathbf{F}\sin\vartheta, \mathbf{F}\cos\vartheta)$ have the same $T^{\mu\nu}$ (d1-6a–d1-6c).

## 2.7 The momentum reading of "$E$ is six-dimensional"

**Proposition 2.10** (Splitting the temporal energy lowers the mass)**.** *Let* $P = \sum_i(\mathsf{E}_i/c)e_i + \sum_a p_a e_{a+3}$ *with* $\eta(P,P) = \mu^2c^2$, $\widehat{W}$ *the symmetric clock of Proposition 3.5 of [1],* $\mathsf{E}_W := (\mathsf{E}_1 +$

$\mathsf{E}_2+\mathsf{E}_3)/\sqrt{3}$ *and* $\mathsf{E}_\perp$ *the part of* $(\mathsf{E}_1,\mathsf{E}_2,\mathsf{E}_3)$ *orthogonal to* $(1,1,1)$*, the relative-time energies. The mass read by the clock,* $m_4^2c^4:=\mathsf{E}_W^2-|\mathbf{p}|^2c^2$*, is*

$$m_4^2c^4=\mu^2c^4-|\mathsf{E}_\perp|^2\le\mu^2c^4;$$

*for the clock* $e_1$*,* $m_4^2c^4=\mu^2c^4-\mathsf{E}_2^2-\mathsf{E}_3^2$*, and* $\mathsf{E}=(3,4,0)$*,* $\mathbf{p}=(5,0,0)$*,* $c=1$ *give* $\mu=0$*,* $m_4^2=-16$*.*

*Proof.* $|(\mathsf{E}_1,\mathsf{E}_2,\mathsf{E}_3)|^2=\mathsf{E}_W^2+|\mathsf{E}_\perp|^2$ and $\mu^2c^4=c^2\eta(P,P)=|(\mathsf{E}_1,\mathsf{E}_2,\mathsf{E}_3)|^2-|\mathbf{p}|^2c^2$. □

Obstructed; this restates Proposition 27.1 and Example 27.3 of [1] in clock-covariant form (d1-7). In the momentum reading the differentiation of the temporal energy subtracts from the mass and can make it tachyonic, and under the surplus constraints of Theorem 30.1 of [1], $\mathsf{E}_\perp=0$, nothing is left to differentiate; in the field-strength reading the motion part adds, $E_6\ge E_0$. Only the latter carries the structure the user describes.

### 2.8 Three candidates for $m$ in the arena's own structures

**Proposition 2.11** (The mass form acts on the frame as parity)**.** *Let* $T=e_1e_2e_3$ *and* $P:=\mathrm{Ad}_T|_V$*. (1)* $P=\eta$ *fixes* $\mathbf{B}$ *and reverses* $\mathbf{E}$*: it is parity on the frame; it commutes with the* $L_i$ *and anticommutes with* $G$ *and the* $K_i$*, so it normalises* $\mathfrak{l}_G$ *without being* $\mathfrak{l}_G$*-invariant. (2)* $PX$ *corresponds to* $\bar z$ *and* $q(z)=h(Pz,z)$*, so* $E_6=\frac12h(z,z)$ *and* $E_0=\frac12|h(Pz,z)|=\frac12\max_\vartheta\eta(e^{\vartheta G}X,e^{\vartheta G}X)$*. (3) As a Grassmann mass kernel on spinors* $T$ *is invariant under all of* $\mathfrak{so}(3,3)$*,* $\mathfrak{l}_G$ *included, in the sense of Definition 18.2 of [1], and breaks nothing; only its sign changes, under conjugation by a spatial generator* $e_4,e_5,e_6$*.*

*Proof.* (1) Lemma 3.4 of [1]; $\eta G\eta=-G$ blockwise, the $L_i$ preserve each triple and the $K_i$ exchange them. (2) $z(\eta X)=\mathbf{B}-i\mathbf{E}$, $h(\bar z,z)=\sum_j z_j^2$, and $\eta(e^{\vartheta G}X,e^{\vartheta G}X)=\mathrm{Re}(e^{2i\vartheta}q)$. (3) By Lemma 2.1 of [1], $e_ae_b$ commutes with $e_1e_2e_3$ when $a,b$ have the same sign and anticommutes otherwise, and $T^\top=-T$; this is Corollary 18.4 of [1]. □

The identities are inherited or restate [1] (d1-8a); their consequence for the dictionary is negative, but as a type mismatch: "$m\leftrightarrow T$" fails on frame vectors, where the adjoint action of $T$ is parity, while as a kernel on spinors $T$ is $\mathfrak{l}_G$-invariant, (3). The Dirac mass lives on the interior reading, which Proposition 12.6 of [1] excludes on the same six directions, and even there it breaks nothing. What survives is a reading: the rest energy is the overlap of a field with its mirror image, and $P$, the arena's splitting into triples, is the real structure whose stabiliser in $U(3)$ is $U(1)\cdot O(3)$ (Proposition 2.2(4)).

**Proposition 2.12** (The interior energy is not a rest energy)**.** *For the split-form interior with the binding root (Theorem 16.2 of [1]) every solution is* $z(\tau)=c_-e^{-i\kappa\tau}+c_+e^{i\kappa\tau}$*, and* $\ddot X=-\kappa^2X$ *does not involve* $m$*. The Noether energy* $\mathcal{E}=2m\kappa^2\,\mathrm{Re}(c_-\cdot c_+)$ *(Proposition 16.5(iii) of [1]) is indefinite and vanishes identically on the duality rotations of both senses,* $z=e^{-i\kappa\tau}c_-$ *(the orbits without momentum,* $c_+=0$*, of Proposition 16.5(i)) and* $z=e^{i\kappa\tau}c_+$ *(which carry the momentum* $2i\kappa c_+e^{i\kappa\tau}\neq0$*); on both, the frame field's* $E_6$ *and* $E_0$ *are constant. In general*

$$|z|^2=|c_-|^2+|c_+|^2+2\,\mathrm{Re}\big(h(c_-,c_+)e^{2i\kappa\tau}\big),\qquad q(z)=e^{-2i\kappa\tau}c_-\cdot c_-+2c_-\cdot c_++e^{2i\kappa\tau}c_+\cdot c_+,$$

*so the frame field's* $E_6$ *is constant iff* $h(c_-,c_+)=0$*, and its* $E_0$ *oscillates:* $c_-=e_1$*,* $c_+=e_2$ *give* $\mathcal{E}=0$*,* $E_6=1$ *and* $E_0=|\cos2\kappa\tau|$*.*

*Proof.* With $\eta(GX,GX)=-\eta(X,X)=-\,\mathrm{Re}\,q$, $\mathcal{E}=\frac{m}{2}\,\mathrm{Re}(\dot z\cdot\dot z+\kappa^2z\cdot z)=2m\kappa^2\,\mathrm{Re}(c_-\cdot c_+)$; the momentum is $\dot z+i\kappa z=2i\kappa c_+e^{i\kappa\tau}$; expand $\bar z\cdot z$ and $z\cdot z$. □

Obstructed as an identification, verified symbolically (d1-9): in the interior $m$ is an overall factor and an input independent of $\kappa$, the pure duality motion of either sense carries zero energy, and, with $X$ a length and $\kappa = c/\ell$, $\mathcal{E} = mc^2 \cdot 2\,\mathrm{Re}(c_- \cdot c_+)/\ell^2$ has the form $mc^2$ times a dimensionless amplitude, but the amplitude is indefinite, vanishes on the pure duality orbits, and is selected by nothing, since $\ell$ is free. The entry "$c^2 \leftrightarrow \kappa G$" is right that $\kappa G$ lies in the factorising summand of Theorem 17.3 of [1], but nothing in the interior turns $\kappa^2$ into a rest energy.

**Theorem 2.13** (In the closure the rest energy of the momentum field breaks the duality translation)**.** *In the Lie–Poisson closure of Proposition 17.8 of [1], $H = h^2/(2I_0) + \frac{1}{2}p^\top Np$, with momentum field $\pi = \eta^{-1}p$:*

1. *for every $N$, $E_6(\pi) = \frac{1}{2}|p|^2$, $E_0(\pi) = \frac{1}{2}|w(p)|$ and $\mathbf{S}(\pi)$ are Casimirs of the momentum field (constant already in the clamped interior, where $p = e^{-\kappa\tau G}p_0$);*

2. *for Lorentz-class $N = (\nu_1\eta^{-1} + \nu_2 G\eta^{-1})/m$ the reduced Hamiltonian is*

$$H = \frac{h^2}{2I_0} + \frac{|\nu_1 + i\nu_2|\,E_0(\pi_0)}{m}\cos(2\theta + \varphi_0), \qquad R_0 e^{i\varphi_0} := (\nu_1 + i\nu_2)\,w(p_0),$$

   *so the rotor's translation $\theta \mapsto \theta + \mathrm{const}$, the duality rotation of the momentum field, is broken with amplitude $|\nu_1+i\nu_2|E_0(\pi_0)/m$; for $(\nu_1,\nu_2) \neq (0,0)$ the rate $\kappa = h/I_0$ is constant and nonzero iff $E_0(\pi_0) = 0$ and $h(0) \neq 0$ (Corollary 17.14 of [1]); for $E_0(\pi_0) > 0$ and $(\nu_1,\nu_2) \neq (0,0)$ it is constant only at the equilibria $h(0) = 0$, $2\theta(0) + \varphi_0 \in \pi\mathbb{Z}$, where it vanishes; for $(\nu_1,\nu_2) = (0,0)$ it is constant for all data;*

3. *for the duality-class $N = \mathbf{1}/m$ the potential $\frac{1}{2}p^\top Np$ (the "frame energy" of [1]) is $E_6(\pi)/m$, the undivided energy of the momentum field, and the rate is conserved for all data;*

4. *the frame field is not conserved: with a free rotor in the rotating regime, Lorentz-class $N$, $(\nu_1,\nu_2) \neq (0,0)$ and $E_0(\pi_0) > 0$, the frame field's $E_6$ and $E_0$ grow without bound.*

*Proof.* (1) Along the flow $p = e^{-\theta G}p_0$, so $\pi = e^{\theta G}\pi_0$ and $z(\pi) = e^{i\theta}z(\pi_0)$, a duality rotation, which preserves $E_6$, $E_0$, $\mathbf{S}$; and $|p| = |\pi|$ as $\eta$ is orthogonal. (2) is Theorem 17.13 of [1], $R_0 = |\nu_1 + i\nu_2|\,|w(p_0)| = 2|\nu_1 + i\nu_2|E_0(\pi_0)$, and $\dot{h} = (R_0/m)\sin(2\theta + \varphi_0)$: if $R_0 > 0$ and $h$ is constant, $\sin(2\theta + \varphi_0) \equiv 0$, so $\theta$ is constant and $h = I_0\dot{\theta} = 0$; conversely these points are equilibria; the rest is Corollary 17.14. (3) $N = \mathbf{1}/m$ commutes with $G$, so $\dot{h} = -p^\top GNp = 0$. (4) By Theorem 17.16 of [1] the frame $x$ (with $X = e^{-\theta G}x$, so $E_6(X) = E_6(x)$, $E_0(X) = E_0(x)$) is displaced by $\Delta x = \lambda e^{-\varphi_0 G}Np_0$, $\lambda > 0$, in each period of the rate, and $x = x_0 + n\Delta x$ after $n$ periods, since $\dot{\theta}$ and $\dot{x} = e^{\theta G}Ne^{-\theta G}p_0$ depend on $\theta$ with period $\pi$. Since $G$ acts as $i$, $z(Np_0) = (\nu_1 + i\nu_2)z(\pi_0)/m$ and $q(\Delta x) = \lambda^2 e^{-2i\varphi_0}(\nu_1 + i\nu_2)^2 q(\pi_0)/m^2 \neq 0$, so $|q(x)|$ and $|x|^2$ grow like $n^2$. □

Parts (1)–(3) restate Proposition 17.8, Theorem 17.13 and Corollary 17.14 of [1], fixed there, with $\frac{1}{2}|w|$ renamed $E_0(\pi)$; part (4) is a direct consequence of Theorem 17.16. Nothing in the theorem is new. Verified by integrating the full system (conservation to $10^{-9}$), at the exact equilibria $\pi_0 = e_1$, $h(0) = 0$, $(\nu_1,\nu_2) = (\pm 1, 0)$, where the rate is constant with $E_0(\pi_0) = \frac{1}{2}$, and with the frame included, where the frame field's $E_0$ grows as $n^2$ (d1-10a, d1-10b). Read through a dictionary that is interpretive, not a theorem (the clamped rate $\kappa = c/\ell$, with $\ell$ free, for the user's $c$, and $E_0(\pi)$, which has the dimension of a momentum squared, for a rest energy), the momentum field's rate keeps its symmetry (constancy, invariance under the rotor's duality translation) when the momentum field is null, its rest part $E_0(\pi)$ is the amplitude that breaks it, and $E_6(\pi)$ is a Casimir and breaks nothing. But the breaking is explicit, from an inertia that Theorem 17.11 of [1] forbids from also conserving the rate; what is broken is the constancy of the rotor rate, a duality property, not Lorentz invariance and nothing carried by $c$; and with the undivided energy as Hamiltonian ($N = \mathbf{1}/m$) the rate is constant at the price of the frame drift of Theorem 17.12.

### 2.9 What transport by $b$ does to the two energies

An admissible Yang–Mills generator is compact: Theorem 2.3 and Corollary 2.4 of [1] restrict gauge generators to $\mathfrak{so}(3)_T \oplus \mathfrak{so}(3)_S \subset \mathfrak{so}(3,3)$, the rotations of the two triples. Write such a $b$ as $(A+D, A-D)$, $A+D$ acting on the temporal triple and $A-D$ on the spatial one; the diagonal part $(A, A)$ lies in $\mathfrak{l}_G$ and the anti-diagonal part $(D, -D)$ in $\mathfrak{p}_G$. Boost-valued connections are legitimate metric connections (Proposition 16.7 of [1]) but not admissible Yang–Mills potentials.

**Proposition 2.14** (What transport preserves)**.** *(1) $Y \in \mathfrak{so}(3,3)$ preserves $\mathbf{E}\cdot\mathbf{B}$ for all $X$, equivalently $q$, equivalently $E_0$, iff $[Y, G] = 0$, that is $Y \in \mathfrak{l}_G$. (2) A compact $b = (A+D, A-D)$ preserves $E_6$ and $\operatorname{Re} q = |\mathbf{B}|^2 - |\mathbf{E}|^2$; it preserves $E_0$ for all $X$ iff $D = 0$; the anti-diagonal rotation through $\frac{\pi}{4}$ about $e_3$ carries the null field $(1, i, 0)$ to the field $\frac{1+i}{\sqrt{2}}(1,1,0)$ at rest, $E_0$ going from 0 to 1 at $E_6 = 1$. (3) At three constituents the part of $\mathfrak{ps}_W$ commuting with $G_3$, $\mathfrak{so}(4)_\Delta \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ of Theorem 17.21 of [1], consists of paired rotations, so it preserves the $\mathfrak{so}(9,\mathbb{C})$ invariants $\frac{1}{2}|Z|^2 = \frac{1}{2}\sum_k |z^{(k)}|^2$ and $q_3(Z) = \sum_k z^{(k)}\cdot z^{(k)}$, hence $\frac{1}{2}|q_3|$. (4) It preserves neither the constituents' rest energies nor that of their summed density: the rotation through $\pi/4$ generated by $\frac{1}{2}\mathrm{rot}(7,13)$ maps $(z^{(2)}, z^{(3)}) = ((1,i,0),(1,0,0))$ to $((0,i,0),(\sqrt{2},0,0))$, keeping $\frac{1}{2}|Z|^2 = \frac{3}{2}$ and $q_3 = 1$, while $\sum_k E_0^{(k)}$ goes from $\frac{1}{2}$ to $\frac{3}{2}$ and $(\sum_k E_6^{(k)}; \sum_k \mathbf{S}_{\mathrm{t}}^{(k)})$ from $(\frac{3}{2}; 0, 0, 1)$ to $(\frac{3}{2}; 0, 0, 0)$, its Lorentz length from $\frac{\sqrt{5}}{2}$ to $\frac{3}{2}$.*

*Proof.* (1) For $Y \in \mathfrak{so}(3,3)$, $Y^\top \eta G + \eta G Y = \eta[G, Y]$. If $Y$ preserves $\operatorname{Re} q$ and $|q|$, then $\operatorname{Im} q \cdot \delta(\operatorname{Im} q) = 0$, so $\delta(\operatorname{Im} q) = 0$ since polynomials form an integral domain. (2) $b$ rotates each triple orthogonally, so it preserves $|\mathbf{B}|^2$ and $|\mathbf{E}|^2$; $(A, A)$ commutes with $G$ and $G(D, -D)G^{-1} = (-D, D)$, so by (1) $E_0$ is preserved for all $X$ iff $D = 0$; the example is $\mathbf{B} = (1,0,0) \mapsto \frac{1}{\sqrt{2}}(1,1,0)$, $\mathbf{E} = (0,1,0) \mapsto \frac{1}{\sqrt{2}}(1,1,0)$. (3) By Theorem 17.19 of [1] the elements of $\mathfrak{so}(4)_\Delta$ are paired rotations, with $S = 0$ and $R$ real antisymmetric, hence they preserve $|Z|^2$ and $q_3$. (4) Direct, with $[e_A e_B, e_C] = 2\eta_{BC} e_A - 2\eta_{AC} e_B$, checked against the realisation. □

Verified exactly (d1-11a, d1-11b). Parts (1) and (2) are inherited linear algebra, and they cut both ways. Pointwise, the anti-diagonal part of an admissible $b$ does turn a null field into one at rest at fixed undivided energy, while the diagonal part, a rotation in $\mathfrak{l}_G$, changes no invariant. But the anti-diagonal part is the part that fails to commute with $G$: along a worldline it cannot be gauged away, unlike the diagonal part (Larmor's theorem [33]), and in the interior it produces Jordan blocks and secular growth unless screened by a diagonal part at equal rates, outside one exceptional ratio (Theorem 5.10); nothing here selects an amount of rest energy. Parts (3)–(4) are an obstruction for the $\mathfrak{so}(9,\mathbb{C})$ invariants only: no transport valued in the paired rotations changes $\frac{1}{2}|Z|^2$ or $q_3$. They are not one for the rest energy of the whole in the diagonal Lorentz reading, which the example changes, and $\mathrm{rot}(7,13)$ does not commute with the diagonal rotations; which is physical depends on the Lorentz action at three constituents, which is open (Theorem 17.20 of [1]); and $\mathfrak{so}(4)_\Delta$ is the part of $\mathfrak{ps}_W$ commuting with $G_3$, not the only part with bounded orbits (Section 5). Here $b$ appears only as a constant transport element or a holonomy, with no field strength or $b$-field energy, so "where the $b$-field belongs" is, at this level, interpretive.

### 2.10 Assessment

Most of what was asked is inherited (Table 1), ordinary electrodynamics in the arena's variables, which by Corollary 4.4 of [1] predicts nothing. Nothing in this section is fixed beyond what [1] already fixes: one conversion constant (Theorem 3.2), the relation $G = -\mathcal{K} \circ \mathrm{Ad}_{e_1 e_2 e_3}$ that makes the binding root orthogonal for the arena's Euclidean form (Lemma 3.4), and the duality pendulum with $R_0 = 2|\nu_1 + i\nu_2|E_0(\pi_0)$ (Theorem 17.13). The uniqueness of $E_6$ and the stabiliser $U(1)\cdot O(3)$ are inherited. Five readings are obstructed. The $c^2$ clause gets less support than its wording suggests: in $E_6$ the weight $c^2$ is removed by the unit choice $\mathbf{B} \to c\mathbf{B}$; in the interior $c^2$

| claim | statement | class |
|---|---|---|
| d1-1a | $E_6$ is the unique rotation- and duality-invariant energy | inherited (Thm. 17.11(iii) of [1]) |
| d1-1b | pointwise symmetry $U(3)$; $\mathfrak{u}(3) \cap (\mathfrak{l}_G \oplus \mathbb{R}G) = \mathfrak{so}(3) \oplus \mathbb{R}G$ | inherited |
| d1-2a,b | $E_0$ complete invariant of Lorentz $\times$ duality, $E_6$ of $U(3)$ | inherited |
| d1-3a,b | $E_6^2 = E_0^2 + \lvert\mathbf{S}\rvert^2$; Rainich identity | inherited |
| d1-4a–c | $E_0 = \min E_6$; $E_6 = E_0 \cosh 2\chi$; rest-frame velocity with $\mathbf{S}_\mathrm{t}$ | inherited |
| d1-5a | $u^2 = c^2\lvert\mathbf{g}\rvert^2 + u_0^2$ | inherited |
| d1-5b | $C = c\,\mathbf{1}$; the weight $c^2$ of $\lvert\mathbf{B}\rvert^2$ | fixed in [1] (Thm. 3.2); weight conventional |
| d1-6a,c | standing wave $Mc^2 = U$; $M$ blind to orthogonality | inherited |
| d1-6b | the mass of a whole is not interference; $M \geq \int E_0$ | obstructed |
| d1-7 | the momentum reading lowers the mass | obstructed (restates Prop. 27.1) |
| d1-8a | $T$ acts on the frame as parity; "$m \leftrightarrow T$" fails there | obstructed (type mismatch) |
| d1-8b | $\mathrm{Stab}_{U(3)}(E_0) = U(1) \cdot O(3)$ | inherited; reduction reading interpretive |
| d1-9 | the interior Noether energy is not a rest energy | obstructed |
| d1-10a | closure: $E_0(\pi)$ is the duality-breaking amplitude | restates [1] Thm. 17.13; reading interpretive |
| d1-10b | the frame field's $E_6$, $E_0$ grow under the free-rotor drift | restates [1] Thm. 17.16 (consequence) |
| d1-11a | compact $b$ changes $E_0$ only through its anti-diagonal part | inherited |
| d1-11b | $\mathfrak{so}(4)_\Delta$ preserves $\lvert Z\rvert^2$, $q_3$; changes $\sum_k E_0^{(k)}$ and the summed density's length | obstructed (for $q_3$, $\lvert Z\rvert^2$ only) |
| d1-8c | "$E$ undivided, $m$ differentiated, $c^2$ symmetric" | interpretive |

Table 1: The claims of Section 2 and their classes; identifiers refer to the verification script.

enters as $\kappa^2 = c^2/\ell^2$ and the Noether energy is $mc^2$ times an indefinite, unselected amplitude; and in the closure what $E_0(\pi)$ breaks is the constancy of the rotor rate, not anything carried by $c$.

Corrected so that it is true, the user's sentence becomes the following interpretive statement (d1-8c). The undivided energy is $E_6 = \frac{1}{2}|z|^2$, the one rotation- and duality-invariant quadratic energy of a six-component frame vector, with pointwise symmetry $U(3)$. It differentiates internally, with no interaction, as $E_6^2 = E_0^2 + |\mathbf{S}|^2$, into a Lorentz-invariant rest energy $E_0 = \frac{1}{2}|z \cdot z|$ and a motion part $|\mathbf{S}|$. What differentiates is the arena's splitting into a temporal and a spatial triple, the real structure $\mathrm{Ad}_{e_1e_2e_3}$, which reduces $U(3)$ to $U(1) \cdot O(3)$ (a symmetry of the pointwise form, not of any dynamics); nothing Lorentzian is broken. $c$ is a conversion, not a symmetry, its weight in $E_6$ is a unit convention, and $m = E_0/c^2$ is a rest-mass density of a field point. The reading fails in three respects: a field point is not a particle (two boost factors; the mass of a composite is not interference); in the closure only the momentum field's energies are conserved, while the frame field's oscillate or grow; and $b$ generates no selected amount of $m$: pointwise its anti-diagonal part moves a field between null and rest at fixed $E_6$, and at three constituents $\mathfrak{so}(4)_\Delta$ moves rest energy between constituents at fixed $\frac{1}{2}|Z|^2$ and $q_3$ while changing the Lorentz length of the summed density. Whether some dynamics selects the rest frame or the value of $E_0$ is open; nothing here does.

# 3 What a mass breaks: duality, chirality, and the two senses of rotation

The hypothesis reads $m$ as what breaks a symmetry of which $c^2$ is the symmetric part. The prior question is which symmetry a mass actually breaks in the arena, and which it leaves alone. The arena has two things called mass. In the field-strength reading (Theorem 12.3 of [1]) a vector $X = \sum_i (B_i e_i + E_i e_{i+3})$, $z(X) = \mathbf{B} + i\mathbf{E}$, has energy density $u(X) = \frac{1}{2}|z|^2$ (the $E_6$ of Section 2), complex invariant $q(X) = z \cdot z$ and rest-frame energy density $E_0(X) = \frac{1}{2}|q|$ (Section 14 of [1]). In the interior these pointwise functions are evaluated on two fields, the frame field $X(\tau)$ and the momentum field $\pi = \eta^{-1}p$; we always say which. In the Dirac reading the mass is the kernel $T = e_1e_2e_3 = \gamma^0 J$ of the Grassmann bilinear (Proposition 4.5 and Corollary 18.4 of [1]). In brief: a mass breaks no Lorentz generator and, in the Dirac reading, no generator of $\mathfrak{so}(3,3)$; it breaks the duality rotation generated by the binding root $G$, in the orbit-space sense of Proposition 3.1 and Theorem 3.4, and the chiral and Pauli–Gürsey symmetries of the massless kinetic term, down to $\mathfrak{u}(1)_J$; and the two senses of the Dirac rest rotation are exchanged by charge conjugation, those of the frame's binding rotation by parity. Units are $\hbar = c = 1$ unless a symbol is shown. Every finite-dimensional statement proved here was checked, exactly or on dyadic-exact float64 data, by `verify/d2_symmetry_breaking.py`, and independently by `verify/x2_check.py`; the checks are listed in Table 4.

## 3.1 Duality and the rest energy of a field

Write $D_\vartheta = e^{\vartheta G} = \cos\vartheta\,\mathbf{1} + \sin\vartheta\,G$ (since $G^2 = -\mathbf{1}$). As $G$ acts on $z$ as multiplication by $i$ (proof of Theorem 12.1 of [1]), $D_\vartheta$ is the duality rotation.

**Proposition 3.1** (Duality on the invariants)**.** *For every $X$ and $\vartheta$:*

*(i)* $z(D_\vartheta X) = e^{i\vartheta} z(X)$, $u(D_\vartheta X) = u(X)$, $q(D_\vartheta X) = e^{2i\vartheta} q(X)$;

*(ii)* $D_\vartheta^\top D_\vartheta = \mathbf{1}$ *and* $D_\vartheta^\top \eta\, D_\vartheta = \cos 2\vartheta\,\eta + \sin 2\vartheta\,\eta G$*: $D_\vartheta$ preserves the positive form for every $\vartheta$, and $\eta$ only for $\vartheta \in \pi\mathbb{Z}$;*

*(iii) if $q(X) = 0$ every $\vartheta$ fixes $q$; if $q(X) \neq 0$ exactly $\vartheta \in \pi\mathbb{Z}$ do;*

*(iv) if $q \neq 0$, there are Lorentz coordinates in which $z = (f_m + if_e)\hat{e}_1$ with $f_m, f_e$ real; then $q = (f_m^2 + f_e^2)e^{2i\varphi}$ with $\varphi = \arg(f_m + if_e)$, so $\tan\varphi = f_e/f_m$: $\arg q$ is twice the Rainich complexion, and $D_{-\varphi}$ makes the field purely magnetic, $z = \sqrt{f_m^2 + f_e^2}\,\hat{e}_1$.*

*Proof.* (i) is $G \leftrightarrow i$ and the quadratic character of $q$. (ii) $G$ is antisymmetric, so $D_\vartheta$ is orthogonal; $G$ is $\eta$-self-adjoint, so $G^\top = \eta G \eta$, and expanding $(\cos\vartheta + \sin\vartheta\,\eta G\eta)\eta(\cos\vartheta + \sin\vartheta\,G)$ with $G^2 = -\mathbf{1}$ gives the formula; $\eta$ and $\eta G$ are linearly independent. (iii) is $e^{2i\vartheta} - 1 = 2i\sin\vartheta\,e^{i\vartheta}$. (iv) Proposition 12.12 of [1] supplies coordinates in which the field is an electric plane of strength $f_e$ orthogonal to a magnetic one of strength $f_m$; then $e^{-i\varphi}(f_m + if_e) = |f_m + if_e|$. The principal branch of $\arctan(f_e/f_m)$ would give $-|f_m + if_e|$ for $f_m < 0$. □

The proposition is inherited: (i) and (iii) are the classical duality rotation of the Maxwell invariants [49, 5, 10], restated from Section 14.3 of [1]; (iv) restates Proposition 12.12 of [1] with the branch of $\varphi$ fixed; and (ii), recorded in Section 14.3 and after Theorem 17.23 of [1], is the textbook rotation of $\frac{1}{2}F_{\mu\nu}F^{\mu\nu}$ into $\frac{1}{2}F_{\mu\nu}{\star}F^{\mu\nu}$ under the isometry $X \mapsto F_X$ of Theorem 12.3 of [1], which carries $G$ to the Hodge star. What the arena adds (Theorems 12.3, 16.2 and 17.3 of [1]) is that the generator of this rotation is the binding root, which lies in the $\eta$-self-adjoint summand of $\mathrm{End}(V) = \mathfrak{so}(3,3) \oplus \{A = A^\dagger\}$. For $\vartheta \notin \pi\mathbb{Z}$, $D_\vartheta \notin O(3,3)$ lifts to no element of $\mathrm{Pin}(3,3)$ (as noted after Theorem 17.23 of [1]). Duality is a symmetry of the positive form $u$; the split-form interior Lagrangian is rotated by it into its $\eta G$ partner.

**Proposition 3.2** (The rest energy is the doubly invariant modulus)**.** *Let* $\mathbf{S} = \mathbf{E} \times \mathbf{B}$ *be the frame Poynting vector; as* $\mathbf{B}_\mathrm{t} = -\mathbf{B}$ *(Section 1.4 of [1]),* $\mathbf{S} = -\mathbf{E} \times \mathbf{B}_\mathrm{t} = -\mathbf{S}_\mathrm{t}$ *is minus the textbook one. Then* $u^2 - |\mathbf{S}|^2 = \frac{1}{4}|q|^2$*, so* $u \geq E_0$*, with equality exactly when* $\mathbf{S} = 0$*. Of* $u$*,* $\operatorname{Re} q$*,* $\operatorname{Im} q$ *and* $E_0$*, the first is invariant under* $U(1)_G$ *but not under* $\mathfrak{l}_G$*, the next two under* $\mathfrak{l}_G$ *but not under* $U(1)_G$*, and* $E_0 = \frac{1}{2}|q|$ *under both.*

*Proof.* $u^2 - |\mathbf{E}\times\mathbf{B}|^2 = \frac{1}{4}(|\mathbf{B}|^2 + |\mathbf{E}|^2)^2 - |\mathbf{E}|^2|\mathbf{B}|^2 + (\mathbf{E}\cdot\mathbf{B})^2 = \frac{1}{4}(|\mathbf{B}|^2 - |\mathbf{E}|^2)^2 + (\mathbf{E}\cdot\mathbf{B})^2$. The six generators of $\mathfrak{l}_G$ commute with $G$ and lie in $\mathfrak{so}(3,3)$, so $\eta(X, AX) = \eta(X, GAX) = 0$; a boost $K_i$ is symmetric for the positive form and $X^\top K_i X \neq 0$ for generic $X$. The duality statements are Proposition 3.1. □

This restates Section 14.3 of [1] and is inherited; with units restored it reads $u^2 - c^2|\mathbf{g}|^2 = E_0^2$, $\mathbf{g} = \mathbf{S}_\mathrm{t}/c^2$, the field form of $E^2 - p^2c^2 = m^2c^4$. Its reading for the hypothesis is interpretive. Lorentz orbits of nonzero vectors are labelled by $q \in \mathbb{C}$, since $\mathfrak{l}_G \cong \mathfrak{so}(3,\mathbb{C})$ is transitive on each level set of $q$ minus the origin (proof of Theorem 14.5 of [1]); duality rotates the label through $2\vartheta$, fixes only the origin, and the stabiliser of a nonzero label is $\{D_0, D_\pi\} = \{\pm\mathbf{1}\}$. A nonzero rest energy thus breaks $U(1)_G$ to $\mathbb{Z}_2$ on the space of Lorentz orbits, and $\arg q$ is the angle along the broken direction. But the breaking is a property of a configuration, not of a vacuum ($u$ is minimised by the zero field and no potential selects $|q| \neq 0$, Section 14.3 of [1]), and the broken symmetry is not one of the interior Lagrangian. What the interior dynamics does with duality is sharper. Every solution of $\ddot{X} = -\kappa^2 X$ (Theorem 16.2 of [1]) is $z(\tau) = c_- e^{-i\kappa\tau} + c_+ e^{i\kappa\tau}$, $c_\pm \in \mathbb{C}^3$; the momentum is $p = m\eta(\dot{X} + \kappa GX)$, so the momentum field is $\pi = \eta^{-1}p = m\alpha_+ X$ with $\alpha_\pm = \partial_\tau \pm \kappa G$ (eq. (7) of [1]).

**Proposition 3.3** (Frame field and momentum field in the interior)**.** *Along every solution:*

*(i)* $\mathcal{E}_\mathbb{C} := \frac{m}{2}(q(\dot{z}) + \kappa^2 q(z)) = 2m\kappa^2\, c_- \cdot c_+$ *is conserved, and the Noether energy* $\frac{m}{2}\eta(\dot{X},\dot{X}) - \frac{m\kappa^2}{2}\eta(GX, GX)$ *is* $\operatorname{Re}\mathcal{E}_\mathbb{C}$*; it vanishes when either sense is absent. A constant duality rotation maps solutions to solutions and multiplies* $\mathcal{E}_\mathbb{C}$ *by* $e^{2i\vartheta}$*.*

*(ii) The momentum field is* $z(\pi) = 2im\kappa\, c_+ e^{i\kappa\tau}$*, i.e.* $\pi(\tau) = e^{\kappa\tau G}\pi_0$*: it moves by pure duality, and* $u(\pi) = \frac{1}{2}|p|^2 = 2m^2\kappa^2|c_+|^2$*,* $E_0(\pi) = 2m^2\kappa^2|c_+ \cdot c_+|$ *and* $\mathbf{S}(\pi)$ *are constant.*

*(iii) The frame field has* $u(X) = \frac{1}{2}(|c_-|^2 + |c_+|^2) + \operatorname{Re}(\bar{c}_- \cdot c_+\, e^{2i\kappa\tau})$*, constant exactly when* $\bar{c}_- \cdot c_+ = 0$*, and* $q(X) = c_- \cdot c_-\, e^{-2i\kappa\tau} + 2c_- \cdot c_+ + c_+ \cdot c_+\, e^{2i\kappa\tau}$*, so* $E_0(X)$ *oscillates in general.*

*(iv) Both pure senses,* $z = e^{-i\kappa\tau}c_-$ *and* $z = e^{i\kappa\tau}c_+$*, are duality rotations of the frame field, with Noether energy* 0 *and constant* $u(X)$ *and* $E_0(X)$*. On the first, the zero-momentum orbits,* $p = 0$ *while* $u(X) = \frac{1}{2}|c_-|^2 > 0$*; on the second* $|p|^2 = 8m^2\kappa^2 u(X)$*.*

*Proof.* $\dot{z} + i\kappa z = 2i\kappa c_+ e^{i\kappa\tau}$ is $z(\alpha_+ X)$, since $G$ acts as $i$; with $|\pi| = |p|$ ($\eta$ is orthogonal) this gives (ii), and $\overline{z(\pi)} \times z(\pi)$, $\overline{z(\pi)} \cdot z(\pi)$ and $|z(\pi) \cdot z(\pi)|$ are invariant under $z(\pi) \mapsto e^{i\kappa\tau}z(\pi)$. In $q(\dot{z}) + \kappa^2 q(z)$ the terms in $e^{\mp 2i\kappa\tau}$ cancel, leaving $4\kappa^2 c_- \cdot c_+$; $\eta(GX, GX) = -\eta(X, X)$ identifies the Noether energy with $\frac{m}{2}\operatorname{Re}(q(\dot{z}) + \kappa^2 q(z))$; $q$ has duality charge two. (iii) is the expansion of $|z|^2$ and $z \cdot z$, and (iv) is (i)–(iii) at $c_+ = 0$ or $c_- = 0$. Each identity was also checked exactly as a polynomial identity in the real and imaginary parts of $c_\pm$, modulo $\cos^2\kappa\tau + \sin^2\kappa\tau = 1$. □

Items (i) and (ii) restate Proposition 16.5(iii) and (ii) of [1] (fixed there), with the momentum field made explicit; the zero-momentum half of (iv) restates Proposition 16.5(i). Item (iii) and the other half of (iv) are added and fixed. The proposition corrects the working dictionary. The quantities that the interior conserves belong to the momentum field: $u(\pi)$, $E_0(\pi)$ and $\mathbf{S}(\pi)$ are constant already in the clamped interior, because $\pi$ moves by a duality rotation, and they are the Casimirs of the closure (Theorem 4.1). The same functions of the frame field are not conserved: $u(X)$ moves unless $\bar{c}_- \cdot c_+ = 0$, and $E_0(X)$ oscillates. Nor is $u(X)$ a function of $|p|^2$:

on the zero-momentum orbits $|p|^2 = 0$ while $u(X) > 0$, on the duality rotations of the other sense $|p|^2 = 8m^2\kappa^2 u(X)$. Duality acts along both pure senses, so duality orbits are not zero-momentum orbits; the zero-momentum sense is singled out by the momentum, not by duality. And the interior's conserved energy is not undivided: it is the real part of a constant of duality charge two, carried by neither sense alone.

### 3.2 Little groups: field points and momenta

On $z$, $\boldsymbol{\omega}\cdot L + \mathbf{v}\cdot K \in \mathfrak{l}_G$ acts as $z \mapsto (\boldsymbol{\omega} + i\mathbf{v})\times z$, that is $\delta\mathbf{B} = \boldsymbol{\omega}\times\mathbf{B} - \mathbf{v}\times\mathbf{E}$, $\delta\mathbf{E} = \boldsymbol{\omega}\times\mathbf{E} + \mathbf{v}\times\mathbf{B}$ (Theorem 12.3 of [1]): $\mathfrak{l}_G = \mathfrak{so}(3,\mathbb{C})$ acting by complex cross products, and $G$ acts as $i$.

**Theorem 3.4** (Stabilisers of field points)**.** *Let $z \neq 0$.*

*(i)* $\mathrm{Stab}_{\mathfrak{l}_G}(z) = \{(\zeta z)\times\cdot : \zeta \in \mathbb{C}\}$ *is abelian of real dimension two. If $q \neq 0$ and $\hat{z} = z/\sqrt{q}$, then $R = \hat{z}\times\cdot$ has $R^3 = -R$ (a rotation) and $(GR)^3 = GR$ (a boost) about the common complex axis $\hat{z}$: $\mathfrak{so}(2)\oplus\mathfrak{so}(1,1)$. If $q = 0$, every element $N$ of the stabiliser has $N^3 = 0 \neq N^2$: null rotations, $\mathbb{R}^2$.*

*(ii) In $\mathfrak{l}_G \oplus \mathbb{R}G$ a non-null $z$ has the same stabiliser, with no $G$ component. A null $z = \mathbf{n}_1 + i\mathbf{n}_2$ has a three-dimensional stabiliser $\cong \mathfrak{e}(2)$: the null rotations and $L_{\mathbf{n}} + G$, with $\mathbf{n} = \mathbf{n}_1\times\mathbf{n}_2/|\mathbf{n}_1|^2 = \mathbf{B}\times\mathbf{E}/|\mathbf{B}\times\mathbf{E}|$ the propagation axis, acting on the null rotations with eigenvalues $\pm i$. For $z = (1,i,0)$, $(L_3 + G)z = 0$ and $e^{\vartheta G}z = e^{-\vartheta L_3}z$.*

*(iii) In $\mathfrak{l}_{\mathrm{int}}$ on the interior four-plane $\langle e_1, e_4, e_5, e_6\rangle$,* $\mathrm{Stab}(e_1) = \mathfrak{so}(3)$ *and* $\mathrm{Stab}(e_1 + e_6) = \mathfrak{e}(2)$*, both three-dimensional.*

*Proof.* (i) $w \times z = 0$ iff $w \in \mathbb{C}z$. From $(z\times)^2 w = z(z\cdot w) - qw$, $(z\times)^3 = -q\,(z\times)$; with $\hat{z}\cdot\hat{z} = 1$ this is $R^3 = -R$, and $G \leftrightarrow i$ gives $(GR)^3 = GR$; for $q = 0$, $(z\times)^3 = 0 \neq (z\times)^2$. (ii) The condition is $w \times z + i\vartheta z = 0$; dotting with $z$ gives $i\vartheta q = 0$. For null $z$, $\mathbf{n}_1\cdot\mathbf{n}_2 = 0$ and $|\mathbf{n}_1| = |\mathbf{n}_2|$ give $\mathbf{n}\times\mathbf{n}_1 = \mathbf{n}_2$, $\mathbf{n}\times\mathbf{n}_2 = -\mathbf{n}_1$, so $\mathbf{n}\times z = -iz$, $(L_{\mathbf{n}} + G)z = 0$, and $[L_{\mathbf{n}}, z\times\cdot] = (\mathbf{n}\times z)\times\cdot = -i(z\times\cdot)$. (iii) is the classical computation. All items were also verified exactly at $z = (1+i, 1, i)$, $q = 2i$, and at the null vectors $(1,i,0)$ and $(1+2i, 2+i, 2-2i)$. □

The theorem is inherited: (iii) is Wigner's classification [58, 57], (i)–(ii) the classical stabilisers of a bivector and the identity of helicity and duality on null fields [44, 5], recorded on the frame in Theorem 13.8 (the axis of a null field) and Example 13.11 (the $G$-charge is the helicity) of [1]. It separates two notions the hypothesis runs together: a particle's little group (of its momentum) is three-dimensional with or without mass, a field point's stabiliser two-dimensional with or without $q$. A mass changes the type of a Lorentz stabiliser, never its dimension, and breaks no Lorentz generator, since $q$ and $P^2$ are invariants. What a nonzero $q$ removes is the duality generator: a null field point is duality-invariant up to a rotation about its axis, so Wigner's $\mathfrak{e}(2)$ reappears once duality is admitted, with helicity as compact generator; for a non-null point duality is broken to $\{\pm\mathbf{1}\}$.

### 3.3 The Dirac mass: invariant in six dimensions, breaking the kinetic algebra

With $\gamma^0 = e_1$, $\gamma^a = e_{a+3}$ the kinetic kernels are $K^\mu = -\gamma^0\gamma^\mu$ and the mass kernel is $T = \gamma^0 J$ (Proposition 4.5 of [1]); a Grassmann bilinear $\psi^\top M\psi$ is invariant under $\delta\psi = Y\psi$ iff $Y^\top M + MY = 0$.

**Proposition 3.5** (The mass form is a Spin(3,3) invariant; the kinetic form is not)**.** *On the spinor module $\mathbb{R}^8$:*

*(i) Each of the fifteen bivectors $Y = e_a e_b$ satisfies $Y^\top T + TY = 0$, and $(e^{sY})^\top T\, e^{sY} = T$ for all real $s$. The antisymmetric $M \in M_8(\mathbb{R})$ satisfying the fifteen conditions form exactly $\mathbb{R}\,T$.*

*(ii) An element $Y \in \mathfrak{so}(3,3)$ satisfies $Y^\top K^\mu + K^\mu Y = \Lambda^\mu{}_\nu K^\nu$ for some real $4\times 4$ matrix $\Lambda$ iff $Y \in \mathfrak{l}_{\text{int}} \oplus \mathbb{R}J$, of dimension 7.*

*Proof.* (i) By Lemma 2.1 of [1] the trivector $T$ commutes with a bivector sharing two or no indices with $\{1,2,3\}$ and anticommutes with a mixed one; by Theorem 2.3 the former are the antisymmetric and the latter the symmetric bivectors, so $Y^\top T + TY$ is $[T,Y]$ or $\{T,Y\}$ and vanishes. For an antisymmetric $Y$, $e^{sY} = \cos s + \sin s\, Y$ commutes with $T$; for a symmetric one $e^{sY} = \cosh s + \sinh s\, Y$ and $(e^{sY})^\top T = Te^{-sY}$. Uniqueness is Corollary 18.4 of [1], re-derived by solving the 960 linear conditions on the 28 antisymmetric entries. (ii) The 256 linear equations in the fifteen bivector coefficients and the sixteen entries of $\Lambda$ have a seven-dimensional solution space, projecting injectively to the bivectors; $\mathfrak{l}_{\text{int}}$ and $J$ lie in it by covariance and by $[J, \gamma^0\gamma^\mu] = 0$. □

Item (i) restates Corollary 18.4 of [1] (fixed there); item (ii) is added and fixed. Within $\mathrm{Spin}(3,3)$ the Dirac mass breaks nothing, and what reduces the six-dimensional symmetry to the interior one is the kinetic term, where the clock is converted into the spatial triple and $c$ enters. In the six-dimensional sense the entry $c^2$ *symmetric, $m$ breaking* is inverted; the breaking it describes lives in the internal symmetries of the kinetic term.

**Proposition 3.6** (What the Dirac mass breaks)**.** *(i) The $Y \in \mathrm{End}_{\mathbb{R}}(\mathbb{R}^8)$ with $Y^\top K^\mu + K^\mu Y = 0$ for $\mu = 0,\dots,3$ form the algebra $\mathfrak{g}_{\text{massless}} = \mathrm{span}\{\Omega, J, e_2\Omega, e_3\Omega\} = \mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2)$, with trace form $-8\,\mathbf{1}$; adding $Y^\top T + TY = 0$ leaves $\mathbb{R}J$.*

*(ii) On $\mathfrak{g}_{\text{massless}}$ the orbit map $Y \mapsto Y^\top T + TY$ has rank 3 and image $\mathrm{span}\{T\Omega, Te_2\Omega, Te_3\Omega\}$, inside the six-dimensional space of $\mathfrak{l}_{\text{int}}$-invariant kernels; $T\Omega$ commutes with $J$ (pseudoscalar mass), $Te_2\Omega$ and $Te_3\Omega$ anticommute with it (Majorana kernels).*

*(iii) For $M = T(m_s + m_p\Omega)$ the field equation $K^\mu\partial_\mu\Psi = M\Psi$ of $\frac{1}{2}\Psi^\top(K^\mu\partial_\mu - M)\Psi$, with $\partial_a\Psi = Jp_a\Psi$, is $\dot\Psi = -JH\Psi$ where*

$$H = \gamma^0\gamma^a p_a + \gamma^0(m_s + m_p\Omega), \qquad H^\top = H, \qquad [H,J] = 0, \qquad H^2 = (|\mathbf{p}|^2 + m_s^2 + m_p^2)\mathbf{1}. \tag{6}$$

*(iv) $e^{\xi\Omega}$ preserves each $K^\mu$ and sends $M \mapsto Me^{2\xi\Omega}$; the stabiliser of $M \neq 0$ in $U(1)_\Omega$ is $\{\pm\mathbf{1}\}$.*

*Proof.* (i) The 256 conditions on the 64 entries of $Y$ have exactly these solutions. At $\mu = 0$ the condition is $Y^\top = -Y$, so mass invariance is $[T,Y] = 0$, which only $J$ satisfies. (ii) For antisymmetric $Y$ the map is $[T,Y]$; by the blade rule $T$ anticommutes with $\Omega$, $e_2\Omega$, $e_3\Omega$ and commutes with $J$, giving $2T\Omega, 2Te_2\Omega, 2Te_3\Omega$; $e_2, e_3$ anticommute with $J$, $T$ and $\Omega$ commute with it. (iii) Varying gives $K^\mu\partial_\mu\Psi = M\Psi$, since $K^\mu$ is symmetric and $M$ antisymmetric. With $K^0 = -\mathbf{1}$ and $J$ commuting with $\gamma^0\gamma^a$, $\gamma^0$, $\Omega$: $\dot\Psi = (K^a Jp_a - M)\Psi = -JH\Psi$. The five matrices $\gamma^0\gamma^a$, $\gamma^0$, $\gamma^0\Omega = e_4e_5e_6$ are symmetric, square to $\mathbf{1}$ and pairwise anticommute, which gives $H^2$. (iv) $\Omega \in \mathfrak{g}_{\text{massless}}$ and $\{T,\Omega\} = 0$, so $e^{-\xi\Omega}Te^{\xi\Omega} = Te^{2\xi\Omega}$. □

Item (i) restates Proposition 5.5 of [1], and the count in (ii) restates Theorem 18.5(a) of [1] (both fixed there); the explicit broken directions of (ii) are added and fixed. Their physical content, that a Dirac mass breaks the axial $U(1)$ and the Pauli–Gürsey $SU(2)$ of a massless Dirac field to fermion number [42, 21], is inherited by Theorem 4.3 of [1]. Items (iii) and (iv) are inherited, the realification of $h = \boldsymbol{\alpha}\cdot\mathbf{p} + \beta(m_s + i\gamma_5 m_p)$ (Proposition 4.6 of [1]) and of the chiral rotation; (iii) was derived from the Grassmann kernels, and the commuting first-order form $\frac{1}{2}\Psi^\top J\dot\Psi - \frac{1}{2}\Psi^\top H\Psi$ of Section 4 of [1] with the same $H$ has the same flow. In both readings the mass is the modulus of an order parameter of charge two under a $U(1)$ generated by a complex structure commuting with the Lorentz algebra ($G$ for $\mathfrak{l}_G$, $\Omega$ for $\mathfrak{l}_{\text{int}}$), and a nonzero modulus leaves $\{\pm\mathbf{1}\}$; but the Dirac mass is a coupling, or a value selected by a potential (Theorem 18.5 of [1]), while the rest energy of a frame vector is a property of a configuration that no potential selects.

### 3.4 Particle and antiparticle: the two senses of one rotation

At $\mathbf{p} = 0$ the real first-order form $\frac{1}{2}\Psi^\top J\dot\Psi - \frac{1}{2}\Psi^\top H\Psi$ of Section 4 of [1] has $H(0) = m\gamma^0 = me_1$.

**Proposition 3.7** (The rest flow)**.** *$JH(0) = mT$, so $\Psi(\lambda) = e^{-m\lambda T}\Psi_0 = (\cos m\lambda - \sin m\lambda\, T)\Psi_0$. On the eigenspaces $e_1 = \pm 1$, each four-dimensional and $J$-invariant, $T = \pm J$: $\Psi(\lambda) = e^{\mp m\lambda J}\Psi_0$, with energy $\frac{1}{2}\Psi^\top H\Psi = \pm\frac{1}{2}m|\Psi|^2$. Moreover $\gamma^0 = -TJ$.*

*Proof.* $Je_1 = e_1e_2e_3$; $T = e_1J$ with $[e_1, J] = 0$ gives $T\Pi_\pm = \pm J\Pi_\pm$ for $\Pi_\pm = \frac{1}{2}(\mathbf{1} \pm e_1)$; $T^2 = -\mathbf{1}$ gives the exponential; $-TJ = -e_1J^2 = e_1$. □

This is inherited, the realification of $\psi = e^{\mp im\lambda}\psi_0$ on $\beta = \pm 1$, at angular frequency $mc^2/\hbar$ in restored units. The arena adds a name: the rest motion is the one-parameter group of the temporal volume element $T$, the unique $\mathrm{Spin}(3,3)$-invariant mass form, and the clock $\gamma^0 = -TJ$ is the quotient of that rotation by the charge rotation. Read interpretively, the single rotation $e^{-m\lambda T}$ splits into two senses of the charge rotation according to the eigenvalue of the clock. Below $\mathsf{C}, \mathsf{P}, \mathsf{T}$ name the operations; on the frame they are realised by $C_f = -\mathbf{1}$, $P_f = \eta$ ($\mathbf{E} \mapsto -\mathbf{E}$) and $T_f\colon X(\tau) \mapsto -\eta X(-\tau)$, a convention that bundles the reversal of $\tau$ with the field map $(\mathbf{B}, \mathbf{E}) \mapsto (-\mathbf{B}, \mathbf{E})$ of $C_fP_f$.

**Proposition 3.8** (Which operation reverses the sense)**.** *(i) Every unit vector $C$ of the surplus plane $U$ has $C^2 = \mathbf{1}$, $[C, T] = 0$, $\{C, J\} = \{C, e_1\} = 0$ and $C^\top H(0)C = -H(0)$; it maps the solution $e^{-m\lambda J}\Psi_0$ on $e_1 = +1$ to the solution $e^{+m\lambda J}C\Psi_0$ on $e_1 = -1$ of the same flow, and $C^\top JC = -J$, $C^\top H(\mathbf{p})C = -H(-\mathbf{p})$. The parities $\gamma^0$, $\gamma^0J$ and the time reversal $T_{\mathrm{rev}} = e_2e_4e_5e_6$ (Theorem 5.3 of [1]) keep the sign of the rest energy.*

*(ii) On the frame, $P_fGP_f^{-1} = -G$, $\mathcal{L}_\kappa(P_fX) = \mathcal{L}_{-\kappa}(X)$ and $\mathcal{L}_\kappa - \mathcal{L}_{-\kappa} = -2m\kappa\,\frac{d}{d\tau}(\mathbf{E}\cdot\mathbf{B})$, while $C_f$ and $T_f$ leave $\mathcal{L}_\kappa = \frac{m}{2}\eta(\alpha_+X, \alpha_+X)$ invariant. On $z = c_-e^{-i\kappa\tau} + c_+e^{i\kappa\tau}$: $P_f\colon (c_-, c_+) \mapsto (\bar c_+, \bar c_-)$, $C_f\colon (c_-, c_+) \mapsto (-c_-, -c_+)$, $T_f\colon (c_-, c_+) \mapsto (-\bar c_-, -\bar c_+)$.*

*(iii) The sense of the Dirac rest rotation is reversed by $\mathsf{C}$, $\mathsf{CP}$, $\mathsf{CT}$, $\mathsf{CPT}$; that of the frame's binding rotation by $\mathsf{P}$, $\mathsf{CP}$, $\mathsf{PT}$, $\mathsf{CPT}$.*

*Proof.* (i) $C = \cos\sigma\, e_2 + \sin\sigma\, e_3$; by the blade rule it anticommutes with $e_1$, $J$ and the $\gamma^a$ and commutes with $T$ and $\Omega$, so $C^\top e_1C = -e_1$, $C$ commutes with $e^{-m\lambda T}$, and $CH(\mathbf{p})C^{-1} = -H(-\mathbf{p})$ (Theorem 5.1 of [1]). $\gamma^0$, $\gamma^0J$ commute with $e_1$; $T_{\mathrm{rev}}$ commutes with $e_1$ and anticommutes with $J$ and $T$, so $T_{\mathrm{rev}}\Psi(-\lambda)$ solves the same flow on the same eigenspace. (ii) $\eta G\eta = G^\top = -G$ and $\eta$ is an $\eta$-isometry, so $\mathcal{L}_\kappa(P_fX) = \frac{m}{2}\eta(\dot X - \kappa GX, \dot X - \kappa GX)$; the total derivative is Theorem 16.6 of [1]; for $T_f$ the reversal of $\tau$ compensates the sign. On $z$: $P_f\colon z \mapsto \bar z$, $C_f\colon z \mapsto -z$, $T_f\colon z(\tau) \mapsto -\overline{z(-\tau)}$. (iii) Signs multiply under composition (also checked on the matrices). □

Item (i) is inherited, realised by the arena's $C$ (Theorems 5.1 and 5.3 of [1]); (ii) and (iii) are fixed, (ii) extending Theorem 16.6 of [1]. For Dirac matter the two senses are particle and antiparticle: solutions of one Lagrangian, of angular frequency $m$, each with energy of definite sign, exchanged by $C$, a symmetry of the Grassmann Lagrangian (it preserves the mass form and every $K^\mu$, Theorem 5.1 of [1]; on the commuting first-order form it reverses the sign of the action). For the frame the two senses are the kernels of the two factors of $\alpha_-\alpha_+ = \partial_\tau^2 + \kappa^2$: one carries no momentum, the other all of it; each alone carries zero Noether energy (Proposition 3.3); they are exchanged by parity, a symmetry only up to a total derivative; and their frequency $\kappa = c/\ell$ is a conversion rate, not a mass. In the interior $m$ and $\kappa$ are independent inputs ($m$ multiplies the Lagrangian, $\kappa$ sets the period), so reading the frame's $\hbar\kappa$ as $mc^2$ would be an added identification, not a result. That the frame's binding is not a particle–antiparticle pair in the Dirac sense is an interpretive conclusion from these fixed facts. In both readings the reversal is $\mathsf{CPT}$-odd.

### 3.5 Two vacua

**Proposition 3.9** (The Fock vacuum of one momentum)**.** *Let $\theta_1, \dots, \theta_8$ be antisymmetric $16\times16$ integer matrices with $\theta_i^2 = -\mathbf{1}$, pairwise anticommuting, and $\rho(A) = -\frac{1}{4}\sum A_{ij}\theta_i\theta_j$, and let $\varepsilon = (|\mathbf{p}|^2 + m^2)^{1/2}$. Then $\chi_{2\rho(J)}(t) = t^6(t^2+4)^4(t^2+16)$, $\chi_{2\rho(JH)}(t) = t^6(t^2+4\varepsilon^2)^4(t^2+16\varepsilon^2)$; the joint spectrum of $Q = -i\rho(J)$ and $H = -i\rho(JH(\mathbf{p}))$ is the table of Theorem 5.7 of [1], with a unique charge-zero vacuum at $-2\varepsilon$ and multiplicities $1, 4, 6, 4, 1$ for $H + 2\varepsilon$. Charge conjugation $C = e_2$ lifts to $\hat{C} = \theta(v_1)\cdots\theta(v_4)/4$, with $v_k$ orthogonal integer vectors, $|v_k|^2 = 2$, spanning the $-1$ eigenspace of $C$; $\hat{C}$ is orthogonal, $\hat{C}^2 = \mathbf{1}$, $\hat{C}Q\hat{C}^{-1} = -Q$, $\hat{C}H(\mathbf{p})\hat{C}^{-1} = H(-\mathbf{p})$.*

*Proof.* The $\theta_i$ are tensor words in the $2\times2$ matrices of eq. (1) of [1], found by search; their 256 monomials have rank 256, so their commutant is $\mathbb{R}\mathbf{1}$ (Theorem 5.6 of [1]); $\rho$ is a Lie homomorphism with $[\rho(A), \theta_k] = \sum_j A_{jk}\theta_j$. The characteristic polynomials were computed exactly at four momenta, the joint eigenspaces at two. $\hat{C}\rho(A)\hat{C}^{-1} = \rho(CAC^\top)$, with $CJC^\top = -J$ and $CJH(\mathbf{p})C^\top = JH(-\mathbf{p})$ (Theorem 5.1 of [1]). □

This restates Theorem 5.7 and Proposition 5.8 of [1] and is inherited: the Fock space of four fermionic modes of one Dirac momentum. Its energy, $-2m$ per momentum at rest ($-2mc^2$ in restored units), is minus half the rest energies of the four modes; only in that sense does the vacuum carry the rest energy with the opposite sign, and only this vacuum does.

**Proposition 3.10** (The frame vacuum)**.** *The interior energy is $\mathcal{E} = \frac{m}{2}(\eta(\dot{X},\dot{X}) + \kappa^2\eta(X,X)) = \sum_a \eta_{aa}\frac{m}{2}(\dot{x}_a^2 + \kappa^2 x_a^2)$; a spatial mode $x$ has $p_x = -m\dot{x}$ and Hamiltonian $-(p_x^2/2m + m\kappa^2x^2/2)$. With canonical, positive-norm quantisation a temporal mode contributes $\hbar\kappa(j+\frac{1}{2})$ and a spatial one $-\hbar\kappa(j+\frac{1}{2})$, $j = 0, 1, \dots$, so on a coordinate subspace with $k$ temporal and $l$ spatial directions the zero-point energy is $\frac{\hbar\kappa}{2}(k-l)$: $0$ on $\mathbb{R}^{3,3}$, $\pm\frac{3}{2}\hbar\kappa$ on the temporal and spatial triples, $\pm\frac{1}{2}\hbar\kappa$ on the $(2,1)$ and $(1,2)$ planes.*

*Proof.* The first identity uses $\eta(GX, GX) = -\eta(X, X)$; the spatial Hamiltonian is the Legendre transform of $-\frac{m}{2}(\dot{x}^2 - \kappa^2x^2)$. With $x = \sqrt{\hbar/2m\kappa}\,(\hat{a} + \hat{a}^\dagger)$, $p_x = i\sqrt{\hbar m\kappa/2}\,(\hat{a}^\dagger - \hat{a})$, $p_x^2/2m + m\kappa^2x^2/2 = \frac{\hbar\kappa}{2}(\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a})$, checked exactly on truncated ladder matrices. The cross term is a total derivative; it shifts the momenta by a canonical transformation and leaves the spectrum unchanged. □

**Corollary 3.11** (No ground state)**.** *With positive norm the spectrum of the full frame is $\hbar\kappa\mathbb{Z}$, every level is infinitely degenerate, and every state has one of lower energy (add a spatial quantum). Making the spatial energies positive instead requires negative norm (Section 18 of [1]).*

The proposition restates Theorem 18.6 of [1] (fixed there; the per-mode quantisation is the inherited indefinite oscillator): the zero-point energy vanishes because the signature is neutral, the condition under which factorisation and binding coexist (Theorem 16.2 of [1]). The corollary, which makes explicit for the full space the remark after Theorem 18.6 of [1], is an obstruction under the hypothesis of positive norm, the familiar fate of indefinite oscillators [41]. The two vacua differ in kind: the Dirac vacuum has energy $-2\varepsilon$ per momentum and is bounded below; the frame's zero-point energy cancels between positive temporal and negative spatial oscillators, and there is no ground state for it to be the energy of.

### 3.6 Goldstone bookkeeping across the tower

**Proposition 3.12** (Goldstone directions)**.** *(i) The $\mathfrak{l}_{\text{int}}$-invariant kernels supported on $\Gamma = +1$ form a complex line; each $M$ in it anticommutes with $J$, has rank four, $M^\top M \propto \frac{1}{2}(\mathbf{1}+\Gamma)$ and $\operatorname{Tr}(M^\top M)^2/(\operatorname{Tr} M^\top M)^2 = \frac{1}{4}$; its stabiliser in $\mathfrak{g}_{\text{massless}}$ is $\mathbb{R}(J - J\Gamma) = \mathbb{R}(J+\Omega)$ and its orbit is three-dimensional.*

| stage | order parameter | kept | broken directions | status here |
|---|---|---|---|---|
| $n = 1$, Dirac | $T = \gamma^0 J$ | $\mathfrak{u}(1)_J$ | 3: $T\Omega$, $Te_2\Omega$, $Te_3\Omega$ (global) | re-verified |
| $n = 1$, Majorana | $M = \Gamma M = M\Gamma$ | $\mathbb{R}(J - J\Gamma)$ | 3 (global) | re-verified |
| $n = 3$, Pati–Salam | $\Phi_0$ ($\nu_R\nu_R$) | 14 of 23 | 9, eaten by $Z'$, $W_R^{\pm}$, six $X$ | re-verified |
| $n = 3$, electroweak | $\Phi_0$ + bidoublet | 10 of 23 | 12 eaten, 1 axion-like $(1 - \Gamma)$ | re-derived (d2-24) |

Table 2: Broken directions attached to the specified generators: at $n = 1$ the four of $\mathfrak{g}_{\text{massless}}$, at $n = 3$ the 21 Pati–Salam generators and the central $J$, $J\Gamma$ (Proposition 22.1 of [1]). The rows restate Theorems 18.5, 22.2 and 22.3 of [1].

*(ii) At three constituents, with the generators $E_A$ of* $\mathrm{Cl}(9,9)$ *and $P_*$, $\Phi_0$, $t_3^R$ of Appendix A of [1] (its $\mathfrak{su}(2)$ generators normalised as in Section 8, as the corrected Appendix has them):* $[J, \Phi_0] = [J\Gamma, \Phi_0] = 2[t_3^R, \Phi_0] \neq 0$*, and the orbits of the* 21 *Pati–Salam generators and of all* 23 *have tangent spaces of the same dimension* 9 *at $\Phi_0$.*

*(iii) If a kernel $M$ commutes with $J$ and anticommutes with $\Gamma$ (Dirac type),* $[J - J\Gamma, M] = -2J\Gamma M \neq 0$*; if it lives on $\Gamma = +1$ and anticommutes with $J$,* $[J - J\Gamma, M] = 0$*. At one constituent $J - J\Gamma = J + \Omega$.*

*Proof.* (i) and (ii) are exact computations, at three constituents on $\mathbb{R}^{512}$ with dyadic matrices and ranks over $\mathbb{Q}$; the reconstruction reproduces Theorem 19.2(v) of [1]: $\Phi_0$ antisymmetric, $\{J, \Phi_0\} = 0$, $\Phi_0^{\top}\Phi_0 = P_\nu$, the rank-four $\nu_R$ projector of the block (on $\mathbb{R}^{512}$, $P_*$ times a rank-eight joint eigenprojector), $[t_3^R, \Phi_0] = J\Phi_0$. The identity in (ii) has a one-line reason: $\Phi_0$ lives on $\Gamma = +1$, so $J\Gamma\Phi_0 = J\Phi_0$ and $\Phi_0 J\Gamma = -J\Phi_0$. (iii) $[J, \Gamma] = 0$; if $\{M, \Gamma\} = 0 = [M, J]$ then $MJ\Gamma = -J\Gamma M$; if $M$ lives on $\Gamma = +1$ and $\{M, J\} = 0$ then $[J, M] = 2JM = [J\Gamma, M]$; and $J\Gamma = J^2\Omega = -\Omega$. The bidoublet kernel $P_*E_1E_7E_8P_*$ served as an explicit Dirac-type instance. □

Items (i) and (ii) restate Theorems 18.5(b) and 22.2 of [1] (fixed there), re-verified: the would-be Majoron is eaten. Item (iii) is inherited, the two-line fact that a charge vanishing on one chirality is kept by a Majorana mass of that chirality and broken by every mass pairing opposite chiralities; what the arena adds is the cross-scale identification: one element, $J - J\Gamma$, of charge $1 - \Gamma$, is kept by a Majorana mass on $\Gamma = +1$ and broken by every Dirac-type mass, at one constituent, where it is $J + \Omega$ (Theorem 18.5(b) of [1]), and at three. That at the generic electroweak vacuum the broken $1 - \Gamma$ direction escapes the gauge orbit restates Theorem 22.3 of [1]. We re-derived its whole table in integer arithmetic (d2-24: the 2176 kernel blades, the up- and down-type $(\mathbf{1}, \mathbf{2}, \mathbf{2})$ kernels $K_{1u}, K_{1d}$, and orbit dimensions $(12, 13)$, $(12, 12)$, $(5, 6)$, the uneaten tangent being that of $J - J\Gamma$); the adversarial check x2-24 re-derived it numerically.

### 3.7 What the hypothesis gets right, and where the $b$-field belongs

Table 3 collects the answer. *$m$ breaks the symmetry* is true for duality in the frame, in the orbit-space sense and for a configuration rather than a vacuum, and for the chiral and Pauli–Gürsey symmetries of Dirac matter; it is false for every spacetime symmetry, the six-dimensional one included: a mass changes which little group a state has and removes the duality generator from the stabiliser of a field.

*$c^2$ is the symmetric part* is an interpretive entry that needs correction. The constant $c$ is the Schur-forced conversion $c\,\mathbf{1}$ between the triples (Theorem 3.2 of [1]), and a constant generates nothing. It enters the interior in two places: in $\eta$ itself, through the conversion that makes $\eta$ dimensionally consistent (Theorem 3.2 and the opening of Section 3 of [1]), and in the shift $\kappa G$, $\kappa = c/\ell$ (eq. (7) of [1]), where $(\kappa G)^2 = -c^2/\ell^2$ is definitional ($G^2 = -\mathbf{1}$ is adopted in Theorem 16.2 of [1]). Of the two, the conversion inside $\eta$ (Lorentz class) and the shift $\kappa G$, the second generates duality. But $G$ is not an infinitesimal isometry of $\eta$ (Proposition 3.1): duality

| symmetry | broken by a mass? | class | source in [1] / here |
|---|---|---|---|
| Lorentz $\mathfrak{l}_{\text{int}}$ | no: $P^2$, $T$ invariant; little groups stay 3-dim | inherited | Theorem 3.4(iii) |
| Lorentz $\mathfrak{l}_G$ | no: $q$ invariant; stabilisers stay 2-dim | inherited | Sect. 14.3; Theorem 3.4(i) |
| Spin$(3,3)$ | no: $T$ invariant (the kinetic term keeps 7) | fixed | Cor. 18.4; Proposition 3.5(ii) new |
| duality $U(1)_G$ | yes, to $\{\pm\mathbf{1}\}$ on Lorentz orbits, iff $q \neq 0$ | inherited | Sect. 14.3 |
| axial $U(1)_\Omega$ | yes, to $\{\pm\mathbf{1}\}$ | inherited | Props. 4.6, 5.5 |
| Pauli–Gürsey $\mathfrak{su}(2)$ | yes, to $\mathfrak{u}(1)_J$ | inherited | Prop. 5.5, Thm. 18.5(a) |
| $U(1)_{1-\Gamma}$ | by Dirac type; not by Majorana on $\Gamma = +1$ | inherited | Thms. 18.5(b), 22.3 |

Table 3: What a mass breaks in the arena.

is a symmetry of the positive form $u$, while the interior Lagrangian is built on $\eta$. The split form and the positive form lie in the Lorentz and the duality class of Theorem 17.11 of [1], which exclude each other (remark after that theorem); $c$ sits in both, and the symmetric part and the Lorentz-invariant dynamics cannot be one object. In the Dirac reading the symmetric part is the massless kinetic algebra $\mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2)$, which the mass cuts to $\mathfrak{u}(1)_J$; for Spin$(3,3)$ the roles are reversed (Proposition 3.5). Which term is symmetric depends on which symmetry is meant, and the hypothesis has to say which.

Nor is the undivided energy the interior's conserved energy. Of the frame field, $u(X)$ depends on the Lorentz frame and is not conserved by the binding flow, and $E_0(X)$ oscillates; the Noether energy has duality charge two and lives on the overlap of the two senses; the conserved $u(\pi)$, $E_0(\pi)$ and $\mathbf{S}(\pi)$ belong to the momentum field (Proposition 3.3). Pointwise, the only listed quantity invariant under both Lorentz transformations and duality is $E_0 = \frac{1}{2}|q|$, the infimum of $u$ over Lorentz transformations (Proposition 3.2). If an undivided energy whose differentiation is the mass is wanted, the arena's candidate is $u$ of the momentum field, where it is conserved, with its split $u^2 = E_0^2 + |\mathbf{S}|^2$, the inherited field form of $E^2 = p^2c^2 + m^2c^4$; the differentiation it describes is the choice of a rest frame, not a breaking.

Finally the $b$-field. An admissible Yang–Mills potential takes compact values (Theorem 2.3 and Corollary 2.4 of [1]; Section 5). The broken generators at one constituent are compact, but none of them can carry a Yang–Mills field: duality is not a symmetry of the split-form Lagrangian (Proposition 3.1) and has no spin lift, $U(1)_\Omega$ is anomalous, and the Pauli–Gürsey $\mathfrak{su}(2)$ acts on a single doublet and mixes a field with its conjugate (Proposition 5.5 of [1], [60]). The one gaugeable generator there, $J$, is abelian and unbroken by the Dirac mass ($[T, J] = 0$); in the projector model of Section 26 of [1], where the charge is $J_N/2$, the normal-plane form of $J$, it acquires a Stückelberg mass exactly when the vacuum projector fails to commute with it (Theorem 26.1 and Corollary 26.2 of [1]). In the paper's vacuum analysis broken non-abelian generators are first gauged at three constituents, where the nine Goldstone directions of $\Phi_0$ are eaten by $Z'$, $W_R^\pm$ and six leptoquarks (Proposition 3.12; Theorems 21.1 and 22.2 of [1]). There the gauge quanta of Yang and Mills [63] acquire their masses from the breaking, by the inherited mechanism of [14, 23]. The same condensate gives the right-handed neutrino its Majorana mass $|y|r$ (Theorem 21.1(d) of [1], with the single Yukawa coupling $y$ that Assumption 19.4 of [1] adopts), while the charged-fermion and Dirac-neutrino masses require the bidoublet values of the electroweak stage, which are inputs (Proposition 21.2 of [1]); there the $SU(2)_L$ gauge fields, the closest analogue of the isotopic $b$-field of [63], eat three further directions ($W^\pm$, $Z$; Theorem 22.3 of [1]). In the arena the $b$-particle receives mass from the differentiation; it does not generate it. That is where the $b$-field belongs, and it reverses the order the hypothesis proposes.

| statement | class | source in [1] | checks |
|---|---|---|---|
| Proposition 3.1 | inherited | Sect. 14.3, Prop. 12.12 | d2-1, d2-2 |
| Proposition 3.2 | inherited | Sect. 14.3 | d2-3 |
| Proposition 3.3 (i), (ii); (iv) at $c_+ = 0$ | fixed there | Prop. 16.5 | d2-4, d2-4b |
| Proposition 3.3 (iii); (iv) at $c_- = 0$ | fixed (new) | — | d2-4b |
| Theorem 3.4 | inherited | Thm. 13.8, Ex. 13.11 | d2-5a, d2-5, d2-6, d2-7 |
| Proposition 3.5 (i) / (ii) | fixed there / fixed (new) | Cor. 18.4 / — | d2-8 / d2-8b |
| Proposition 3.6 (i), (ii) | fixed there; directions new | Prop. 5.5, Thm. 18.5(a) | d2-9 |
| Proposition 3.6 (iii), (iv) | inherited | Prop. 4.6 | d2-10 |
| Proposition 3.7 | inherited | Sect. 4 | d2-11 |
| Proposition 3.8 (i) / (ii), (iii) | inherited / fixed (new) | Thms. 5.1, 5.3 / 16.6 | d2-12 / d2-13, d2-14 |
| Proposition 3.9 | inherited | Thm. 5.7, Prop. 5.8 | d2-15a, d2-15b, d2-16, d2-17 |
| Proposition 3.10 / Corollary 3.11 | fixed there / obstructed | Thm. 18.6 | d2-18 / d2-19 |
| Proposition 3.12 (i), (ii) / (iii) | fixed there / inherited | Thms. 18.5(b), 22.2 | d2-20, d2-21a, d2-21 / d2-22 |
| electroweak count | fixed there | Thm. 22.3 | d2-24 (exact), x2-24 |
| $(\kappa G)^2 = -c^2/\ell^2$ | constructed (definitional) | Thm. 16.2, eq. (7) | d2-23 |
| readings of Section 3.7 | interpretive | — | — |

Table 4: Claims of this section, their class ("fixed there": a restated result of [1], fixed in the paper), and the labels of the checks printed by `d2_symmetry_breaking.py` (and, for x2-24, by `x2_check.py`). Check d2-0 is a consistency check of the constructed realisation (eq. (1) of [1]), not a claim.

# 4 The differentiation in motion: rotor, rate and duality angle

This section follows the undivided energy and its differentiation in motion inside the one dynamics that the arena closes by itself, the Lie–Poisson closure of the force law (Section 17 of [1]), in which the rate $\kappa = c/\ell$ of $\alpha_\pm = \partial_\tau \pm \kappa G$, where the hypothesis puts $c^2$, is the angular velocity of a rotor. No Yang–Mills field is used. As in Proposition 3.1, two fields are kept apart: the *momentum field* $\pi = \eta^{-1}p$ ($\pi = m\alpha_+ X$ in the clamped interior) and the *frame field* $X$. In brief: the closure freezes the undivided energy, rest energy and Poynting vector of the momentum field, so no Hamiltonian generates or changes the rest energy of that field, while the rest energy of the frame field oscillates in the clamped interior and grows without bound under the free-rotor drift (Theorem 4.1 and Corollary 4.2). The conserved total splits into a rate part $\frac{1}{2}(I_0/\ell^2)c^2$ and a Lorentz-invariant frame part with an amplitude set by the rest energy of the momentum field, and the duality angle swings with $\Omega^2$ proportional to it (Propositions 4.3, 4.4 and 4.7 and Corollary 4.5, restating Theorem 17.13 and Corollaries 17.14–17.15 of [1]). A constant rate is the heavy-rotor limit (Proposition 4.8); nothing in the algebra fixes $\ell$ (Proposition 4.12); and $\hbar\kappa = mc^2$, equivalently $\ell = \hbar/(mc)$, is an added identification, not a result, under which a quantised rotor has the first level $\frac{1}{2}mc^2$ (Propositions 4.10 and 4.16).

**Conventions.** As in Section 17 of [1]: $V = \mathbb{R}^{3,3}$, $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$, $Ge_i = e_{i+3}$, $Ge_{i+3} = -e_i$, so $G^2 = -\mathbf{1}$, $G^\top = -G$, $G^\dagger := \eta^{-1}G^\top\eta = G$, $\eta(GX, GY) = -\eta(X,Y)$ (Proposition 3.3, Theorem 16.2 of [1]), and $G = -\mathcal{K}\circ\mathrm{Ad}_{e_1e_2e_3}$, $\mathcal{K} = \frac{1}{2}\,\mathrm{ad}_{\mathsf{P}}$ (Lemma 3.4 of [1]). For $Y \in V$ with components $(\mathbf{B};\mathbf{E})$ along $(e_1,e_2,e_3;e_4,e_5,e_6)$, $z(Y) = \mathbf{B} + i\mathbf{E}$, so $G$ acts as $i$ (Theorem 12.3

of [1]), $q(z) = z \cdot z = |\mathbf{B}|^2 - |\mathbf{E}|^2 + 2i\,\mathbf{B}\cdot\mathbf{E}$, and

$$E_6(Y) := \tfrac{1}{2}|z(Y)|^2, \qquad E_0(Y) := \tfrac{1}{2}|q(z(Y))|, \qquad \mathbf{S}(Y) := \tfrac{i}{2}\,\overline{z(Y)} \times z(Y) = \mathbf{E} \times \mathbf{B}$$

are the undivided energy, the rest energy (the rest-frame energy density of Section 14.4 of [1]) and the frame Poynting vector of $Y$; as $\mathbf{B}_{\mathrm{t}} = -\mathbf{B}$ (Section 12 of [1]), $\mathbf{S} = -\mathbf{E} \times \mathbf{B}_{\mathrm{t}}$ is minus the textbook one, which the rest-frame velocity of Section 14.4 uses; only $|\mathbf{S}|$ enters below. A covector $p$ is a column, $C(p) = p^{\top}\eta^{-1}p$, $D(p) = p^{\top}G\eta^{-1}p$, $w(p) := C - iD = q(z(\pi))$, so $E_0(\pi) = \frac{1}{2}|w(p)|$, $E_6(\pi) = \frac{1}{2}|p|^2$. Lorentz-class inertia (Theorem 17.11(ii) of [1]) is written $N = (\nu_1\eta^{-1} + \nu_2 G\eta^{-1})/m$, $\nu := \nu_1 + i\nu_2$ (the $(a, b)$ of [1]), as in Theorem 2.13, keeping $b$ for the Yang–Mills field. Units are $\hbar = c = 1$ unless shown; $\tau$ is a time; $\mathrm{K}(\mu)$, $\mathrm{E}(\mu)$ are the complete elliptic integrals in the parameter $\mu$ [11]. d3-$n$ refers to `verify/d3_closure_dynamics.py` (integrations: eighth-order Runge–Kutta, relative tolerance $10^{-12}$; d3-0 checks the conventions), x3-$n$ to the independent `verify/x3_check.py`.

### 4.1 What the closure conserves

The only $\mathfrak{l}_G$-equivariant closure in which the rotor rotates the frame covector without dilating it is $\mathfrak{s} = \mathbb{R}\hat{g} \ltimes_G V$, $[\hat{g}, X] = GX$, $[X, Y] = 0$ (Theorem 17.7 of [1]). On $\mathfrak{s}^*$, with coordinates $(h, p)$, $F_h = \partial F/\partial h$ and $\nabla F = \partial F/\partial p$, the Lie–Poisson equations of an arbitrary Hamiltonian $H$, and the bracket for which they read $\dot{F} = \{F, H\}$, are

$$\dot{p} = H_h\, G^{\top}p, \qquad \dot{h} = -\,p^{\top}G\,\nabla H, \qquad \{F_1, F_2\} = p^{\top}\left(F_{2,h}\, G\nabla F_1 - F_{1,h}\, G\nabla F_2\right). \tag{7}$$

For $H = h^2/(2I_0) + \frac{1}{2}p^{\top}Np$ this is Proposition 17.8 of [1], $\dot{p} = (h/I_0)G^{\top}p$, $\dot{h} = -p^{\top}GNp$; re-derived from the structure constants, with $H$ conserved for every symmetric $N$ and $|p|^2$ for every $H$ (d3-1).

**Theorem 4.1** (The momentum field is frozen up to its duality phase)**.** *(i) A function $f(p)$ is a Casimir of $\mathfrak{s}^*$ iff it is invariant under $p \mapsto e^{tG}p$, i.e. under the duality rotations $z(\pi) \mapsto e^{-it}z(\pi)$ of the momentum field. So $E_6(\pi) = \frac{1}{2}|p|^2$, $E_0(\pi)$ and the vector $\mathbf{S}(\pi)$ are Casimirs; $C$, $D$ are not.*

*(ii) At $p \neq 0$ the symplectic leaves are two-dimensional, so there are five independent local Casimirs; the real and imaginary parts of the $\bar{z}_j z_k$, $z = z(\pi)$, provide them (Jacobian rank five), and $E_6(\pi)$, $E_0(\pi)$, $\mathbf{S}(\pi)$ account for four, with $E_6(\pi)^2 = E_0(\pi)^2 + |\mathbf{S}(\pi)|^2$.*

*(iii) Along the flow of any Hamiltonian, $p(\tau) = e^{-\theta(\tau)G}p_0$ with $\dot{\theta} = H_h$, $\theta(0) = 0$, so $z(\pi(\tau)) = e^{i\theta(\tau)}z(\pi_0)$ and $w(p(\tau)) = e^{2i\theta(\tau)}w(p_0)$.*

*Proof.* (i) For $f = f(p)$, (7) gives $\{f, F\} = F_h\, p^{\top}G\nabla f = -F_h\,(Gp)\cdot\nabla f$, zero for every $F$ iff $f$ is constant along $t \mapsto e^{tG}p$. As $\eta G = -G\eta$, $\eta^{-1}e^{tG}p = e^{-tG}\pi$, on which $G$ acts as $i$. $|z|^2$, $|z \cdot z|$ and $\bar{z} \times z$ are phase invariant, while $C = \operatorname{Re} w$ and $D = -\operatorname{Im} w$ rotate into each other; $|\pi| = |p|$ as $\eta$ is orthogonal. (ii) $\{p_i, p_j\} = 0$ and $\{h, p_j\} = (Gp)_j$, so the Poisson tensor has rank two at $p \neq 0$ and $\dim \mathfrak{s}^* - 2 = 5$. The $\bar{z}_j z_k$ are phase invariant with Jacobian rank five at a generic point; the relation is the Lagrange identity $u^2 - |\mathbf{S}|^2 = \frac{1}{4}|q|^2$ of Section 14.3 of [1]. (iii) $\dot{p} = -H_h Gp$ integrates to $p = e^{-\theta G}p_0$, so $\pi = e^{\theta G}\pi_0$, and $w$ is quadratic in $z$. □

The theorem is fixed, and in substance a corollary of Proposition 17.8 of [1], which identifies the coadjoint orbit through $(h, p_0)$ with the cylinder $\mathbb{R} \times \{e^{\theta G}p_0\}$, so that every function of $p$ constant on these circles is a Casimir; (iii) is the phase law of Theorem 17.13 of [1] and the relation in (ii) is inherited. What is added is the naming and the count of the Casimirs. Already in the clamped interior $p = e^{-\kappa\tau G}p_0$, so these functions of the momentum field were constant before the closure (Theorem 2.13). All items were checked exactly (d3-2); along five random Lorentz-class flows $H$, $|p|^2$, $E_6(\pi)$, $E_0(\pi)$, $\mathbf{S}(\pi)$ and (iii) hold to $7 \times 10^{-12}$ (d3-3).

**Corollary 4.2** (What the closure can and cannot change)**.** *(i) For every Hamiltonian on $\mathfrak{s}^*$, quadratic or not, of Lorentz class or not, $E_0(\pi)$, $E_6(\pi)$ and $\mathbf{S}(\pi)$ of the momentum field are constant along every solution: a null momentum field stays null, and one of rest energy $E_0$ keeps it.*

*(ii) The same functions of the frame field are not constants of motion. In the clamped split-form interior, $z(X) = c_- e^{-i\kappa\tau} + c_+ e^{i\kappa\tau}$ (Proposition 16.5(ii) of [1]) gives*

$$E_6(X) = \tfrac{1}{2}(|c_-|^2+|c_+|^2)+\mathrm{Re}(\bar{c}_-\cdot c_+\, e^{2i\kappa\tau}), \quad q(z(X)) = c_-\cdot c_-\, e^{-2i\kappa\tau}+2\,c_-\cdot c_+ + c_+\cdot c_+\, e^{2i\kappa\tau},$$

*so $E_6(X)$ is constant iff $\bar{c}_-\cdot c_+ = 0$, and $c_- = (1,i,0)$, $c_+ = (1,0,0)$ give $|q(z(X))|^2 = 9 - 8\sin^2\kappa\tau$. Under the free-rotor drift of Example 4.13 with $x_0 = 0$, the frame field $X = e^{-\theta G}x$ has $E_0(X) = 0$ at $\tau = 0$ and $\frac{1}{2}(3\,\Delta x)^2 \approx 1.807$ after three full turns; on every rotating solution with $R_0 > 0$, $E_0(X)$ and $E_6(X)$ grow without bound.*

*Proof.* (i) Casimirs Poisson-commute with every Hamiltonian (Theorem 4.1(i)). (ii) Expand $|z|^2$ and $z\cdot z$; in the example $c_-\cdot c_- = 0$, $c_-\cdot c_+ = c_+\cdot c_+ = 1$, so $|q|^2 = |2+e^{2i\kappa\tau}|^2$. In Example 4.13 the motion is $2\pi$-periodic in $\theta$, so at $\theta = 2\pi k$, $e^{-\theta G} = \mathbf{1}$ and $x = k\,\Delta x$, a real multiple of $e_1$, whence $q(z(X)) = k^2\Delta x^2$. The growth is Theorem 2.13, item 4, from Theorem 17.16 of [1]. □

Item (i) is an obstruction for the momentum field, checked with the generic Hamiltonian $h^2/(2I_0) + \frac{1}{10}h^4 + \sin(p^\top M_1 p) + \frac{1}{5}h\,p^\top M_2 p$, $M_{1,2}$ random symmetric: the Casimirs and $H$ stay constant to $5\times10^{-12}$ while the phase of $w$ moves through 6.3 radians (d3-4; x3-3). Item (ii) is fixed, from Proposition 16.5(ii) and Theorem 17.16 of [1] (d3-22; x3-4, x3-23). This first verdict cuts both ways. For the momentum field, $E_6(\pi)$ and its split $E_6(\pi)^2 = E_0(\pi)^2+|\mathbf{S}(\pi)|^2$ are labels of the leaf: its rest energy is carried by the dynamics, not produced by it. For the frame field, to which the dictionary's $u = \frac{1}{2}|z|^2$ and $m \leftrightarrow \frac{1}{2}|q|$ refer, the closure does change the rest energy, by an oscillation or an unbounded drift, not by a mechanism that selects a value; and $E_6(X)$ is not a function of $\frac{1}{2}|p|^2 = E_6(\pi)$ (on zero-momentum orbits with $X \neq 0$, $|p|^2 = 0 < E_6(X)$, Proposition 3.1). What moves is the duality angle $\theta$ and the rate $h/I_0$.

## 4.2 The duality pendulum

From now on $N = (\nu_1\eta^{-1} + \nu_2 G\eta^{-1})/m$ is of Lorentz class and $\nu\,w(p_0) = R_0 e^{i\varphi_0}$, $R_0 \geq 0$. On a leaf $(\theta, h)$ are canonical and $H = h^2/(2I_0) + U(\theta)$ (Proposition 17.8 of [1]).

**Proposition 4.3** (The potential of the duality angle)**.** $\frac{1}{2}p^\top Np = \mathrm{Re}(\nu\,w(p))/(2m)$*, and with* $\psi := 2\theta + \varphi_0$*,*

$$U(\theta) = \frac{R_0}{2m}\cos\psi, \qquad R_0 = |\nu|\,|w(p_0)| = 2|\nu|\,E_0(\pi_0), \qquad I_0\ddot{\theta} = \dot{h} = \frac{R_0}{m}\sin\psi.$$

*Proof.* $\frac{1}{2}p^\top Np = (\nu_1 C + \nu_2 D)/(2m) = \mathrm{Re}(\nu(C - iD))/(2m)$, and $w(p(\theta)) = e^{2i\theta}w(p_0)$ by Theorem 4.1(iii). For $p(\theta) = e^{-\theta G}p_0$, $dU/d\theta = -p^\top NGp = p^\top GNp$ ($N$ symmetric, $G$ antisymmetric), so $\dot{h} = -dU/d\theta$; and $\dot{\theta} = h/I_0$. □

This restates Theorem 17.13 of [1] (fixed there), re-verified exactly and along three random flows (d3-5); $R_0 = 2|\nu|E_0(\pi_0)$ only renames $\frac{1}{2}|w|$, and $E_6(\pi)$, $\mathbf{S}(\pi)$ do not enter.

**Proposition 4.4** (Small oscillations)**.** *The stable equilibrium is $\psi = \pi$, $U = -R_0/(2m)$. With $\psi = \pi + \xi$, $\ddot{\xi} = -\Omega^2\sin\xi$ and*

$$\Omega^2 = \frac{2R_0}{mI_0} = \frac{4|\nu|\,E_0(\pi_0)}{mI_0};$$

*$\theta$ oscillates at the same frequency with half the amplitude, and at amplitude $0 < \xi_0 < \pi$ the period is* $4\mathrm{K}(\sin^2(\xi_0/2))/\Omega$.

*Proof.* $\ddot{\psi} = 2\ddot{\theta} = (2R_0/mI_0)\sin\psi$ and $\sin(\pi+\xi) = -\sin\xi$. The period follows from $\frac{1}{2}\dot{\xi}^2 = \Omega^2(\cos\xi - \cos\xi_0)$ with $\sin(\xi/2) = \sin(\xi_0/2)\sin t$. □

Fixed, as a corollary of Theorem 17.13 of [1], whose pendulum it linearises; the period is the textbook one. At amplitude $10^{-3}$ the measured period exceeds $2\pi/\Omega$ by $6.25 \times 10^{-8}$, the shift $\xi_0^2/16$; at amplitude 2 it matches the elliptic formula to $5 \times 10^{-14}$ (d3-6).

**Corollary 4.5** (When the rate is constant)**.** *(i) If $\nu \neq 0$, the rate $h/I_0$ is constant and nonzero iff $R_0 = 0$, that is $E_0(\pi_0) = 0$ (a null momentum field, $|\mathbf{B}| = |\mathbf{E}|$, $\mathbf{B}\cdot\mathbf{E} = 0$ [54]), and $h(0) \neq 0$; then $U \equiv 0$ and $\theta = h\tau/I_0$, an exact flat direction. For $E_0(\pi_0) > 0$ the rate is constant only at the equilibria $h(0) = 0$, $\psi(0) \in \pi\mathbb{Z}$, where it vanishes and the frame moves at the constant velocity $Np_0$, $|Np_0| = |\nu||p_0|/m \neq 0$.*

*(ii) If $\nu = 0$, then $U \equiv 0$ and the rate is constant for all data.*

*(iii) For duality-class inertia, $[N, G] = 0$ (nine dimensions), the rate is constant for all data and $U = \frac{1}{2}p_0^\top N p_0$ is constant, in general nonzero.*

*Proof.* (i) $\dot{h} = (R_0/m)\sin\psi$. If $R_0 > 0$ and $h$ is constant, $\sin\psi \equiv 0$, so $\theta$ is constant and $h = I_0\dot{\theta} = 0$; conversely these points are equilibria, where Theorem 17.12 of [1] ($N_c = 0$, $\theta(0) = 0$) gives $\dot{x} = Np_0$, and $N^2 = |\nu|^2/m^2$. (ii) $N = 0$. (iii) $GN$ is antisymmetric, so $\dot{h} = -p^\top GNp = 0$, and $U(\theta) = \frac{1}{2}p_0^\top e^{\theta G} N e^{-\theta G} p_0 = \frac{1}{2}p_0^\top N p_0$. □

Items (i)–(ii) restate Corollary 17.14 of [1] (fixed there), with the equilibria explicit as in Theorem 2.13; (iii) restates Theorems 17.11(i),(iii) and 17.12 of [1]. Numerically $|h - h_0| < 10^{-14}$ over $\tau \in [0, 30]$ on null data and for a random duality-class $N$, and at the equilibria $h$ stays 0 while $x = \tau Np_0$ (d3-7).

**Remark 4.6** (A pseudo-Goldstone reading, and its limits)**.** On a leaf $\theta$ is the angle of the duality group $U(1)_G$ of the momentum field, broken by a nonzero rest energy (Proposition 3.1): in the duality class it is a symmetry of $H$, in the Lorentz class it is broken on each leaf in proportion to $E_0(\pi)$. Exactly (d3-8),

$$I_0\,\Omega^2 = U''(\theta_{\min}) = \frac{2R_0}{m} = \frac{4|\nu|\,E_0(\pi_0)}{m}.$$

The first equality defines $\Omega$ for every one-degree-of-freedom oscillator; the closure-specific content is the second, $U'' = 2R_0/m$, which is Theorem 17.13 of [1]. With $I_0$ for $f_\pi^2$ and $|\nu|E_0(\pi_0)/m$ for the explicit breaking it has the shape of the Gell-Mann–Oakes–Renner relation [17, 56], as every pendulum does; the reading is interpretive. The breaking is explicit, by the inertia, not spontaneous: there is no vacuum manifold, $E_0(\pi)$ is a label fixed by the data (Corollary 4.2), not an order parameter (Section 14.3 of [1]), and with one degree of freedom "mass" means a frequency. What survives: on a null momentum field the angle is an exact zero mode, and a rest energy of the momentum field gives it the frequency $\Omega \propto \sqrt{E_0(\pi_0)}$.

### 4.3 The energy budget: a rate part and a frame part

**Proposition 4.7** (Rate part and frame part)**.** *With Lorentz-class inertia, along every solution:*

*(i) $H = K_{\mathrm{rot}} + U$, $K_{\mathrm{rot}} = \frac{1}{2}I_0\kappa^2 \geq 0$, $\kappa = h/I_0$, $U = \frac{R_0}{2m}\cos\psi = \mathrm{Re}(\nu\,w(p))/(2m)$, and $\dot{K}_{\mathrm{rot}} = -\dot{U} = \frac{R_0}{m}\,\kappa\sin\psi$;*

*(ii) in the rotating regime $K_{\mathrm{rot}}$ runs through exactly $[H - \frac{R_0}{2m}, H + \frac{R_0}{2m}]$ in every period of the rate, so the energy exchanged is $R_0/m = 2|\nu|E_0(\pi_0)/m$; in the librating regime it runs through $[0, H + \frac{R_0}{2m}]$;*

*(iii)* $K_{\mathrm{rot}}$ *and* $U$ *are separately invariant under* $\mathfrak{l}_G$ *acting by* $(h,p)\mapsto(h,\Lambda^{-\top}p)$, $\Lambda=e^r$, $r\in\mathfrak{l}_G$, *and the flow commutes with it;*

*(iv)* $(\theta,h)(\tau)$ *depends on* $p_0$ *only through* $\nu\,w(p_0)$*:* $E_6(\pi_0)$ *and* $\mathbf{S}(\pi_0)$ *do not enter the motion.*

*Proof.* (i) By Proposition 4.3, $\dot K_{\mathrm{rot}}=h\dot h/I_0$ and $\dot U=-(R_0/2m)\,2\kappa\sin\psi$. (ii) Rotating, $\dot\psi=2\kappa$ never vanishes and $\cos\psi$ covers $[-1,1]$ in each period; librating, $K_{\mathrm{rot}}=0$ at the turning points and is largest at $\psi=\pi$. (iii) The Lorentz requirement $rN+Nr^{\top}=0$ (Definition 17.10 of [1]) integrates to $\Lambda^{-1}N\Lambda^{-\top}=N$, so $U(\Lambda^{-\top}p)=U(p)$; $\Lambda^{-\top}$ commutes with $G^{\top}$ and maps solutions to solutions. (iv) $\pi\mapsto\Lambda\pi$ as $\Lambda\in O(3,3)$, and $\mathfrak{l}_G\cong\mathfrak{so}(3,\mathbb{C})$ preserves $q$ (Theorem 12.1 of [1]), so $w(\Lambda^{-\top}p)=w(p)$; the equations for $(\theta,h)$ contain only $R_0$ and $\varphi_0$. □

Fixed; (i) restates the energy $E_\theta=H$ of Theorem 17.13 of [1], (ii)–(iv) are added. Numerically $H=K_{\mathrm{rot}}+U$ to $3\times10^{-11}$, the range of $K_{\mathrm{rot}}$ equals $R_0/m$ to $3\times10^{-10}$, and a Lorentz transformation of $p_0$ changed $E_6(\pi_0)$ by 5.0 and $\theta(\tau)$ by less than $10^{-11}$ (d3-9). So the conserved total differentiates into two Lorentz-invariant parts whose exchange is fixed by the rest energy of the momentum field alone; neither is the undivided $E_6(\pi)$, which is inert.

**Units, and the factor one half.** With $[\tau]=\mathsf{T}$, $[\kappa]=\mathsf{T}^{-1}$ (Section 3 of [1]) and $[X]=\mathsf{L}$, a convention by Proposition 4.11(ii) that makes $m$ a mass (d3-10): $[I_0]=\mathsf{ML}^2$, $[h]=\mathsf{ML}^2\mathsf{T}^{-1}$ (an action), $[p]=\mathsf{MLT}^{-1}$, $[N]=\mathsf{M}^{-1}$, $[H]=[K_{\mathrm{rot}}]=[U]=\mathsf{ML}^2\mathsf{T}^{-2}$, $[M_{\mathrm{rot}}]=\mathsf{M}$ for $M_{\mathrm{rot}}:=I_0/\ell^2$; and $E_0(\pi)$ is a momentum squared, $|\nu|E_0(\pi)/m$ an energy. With the clamped value $\kappa=c/\ell$, $K_{\mathrm{rot}}=\frac12 I_0\kappa^2=\frac12 M_{\mathrm{rot}}c^2$, the "$c^2$ part" of the conserved total (interpretive): literally the Newtonian kinetic energy of a mass $M_{\mathrm{rot}}$ on a circle of radius $\ell$ at rim speed $c$, outside Newtonian kinetics. It is not $E=mc^2$: the $\frac12$ comes from an energy quadratic in the rate, where Planck's relation is linear; $M_{\mathrm{rot}}$ is not tied to $m$ (Proposition 4.11(iv)); and $\kappa=c/\ell$ holds only for the clamped rotor.

## 4.4 When the rate is constant

**Proposition 4.8** (The multiscale criterion)**.** *Let* $R_0>0$, $\kappa_0=\kappa(0)\neq0$, $\psi_0=\psi(0)$ *and* $\varepsilon:=R_0/(mI_0\kappa_0^2)=U_{\max}/K_{\mathrm{rot}}(0)$, $U_{\max}=R_0/(2m)$. *Then* $\kappa^2=\kappa_0^2[1+\varepsilon(\cos\psi_0-\cos\psi)]$, *so* $|\kappa^2/\kappa_0^2-1|\le2\varepsilon$*; the rotor turns for all time iff* $\varepsilon(1-\cos\psi_0)<1$, *which holds for every phase when* $\varepsilon<\frac12$*; for* $\varepsilon(1-\cos\psi_0)>1$ *it librates, and equality is the separatrix.*

*Proof.* Divide the conservation of $H=\frac12 I_0\kappa^2+\frac{R_0}{2m}\cos\psi$ by $\frac12 I_0\kappa_0^2$. The least value of $\kappa^2$, at $\cos\psi=1$, is $\kappa_0^2[1-\varepsilon(1-\cos\psi_0)]$; if it is negative, $\kappa$ vanishes where $\sin\psi\neq0$, hence $\ddot\theta\neq0$, and changes sign. □

This restates Corollary 17.15 of [1] (fixed there), checked along four flows to $8\times10^{-12}$ and in five cases on both sides of the criterion (d3-11); the form $\varepsilon=U_{\max}/K_{\mathrm{rot}}(0)$ is ours. The reading is interpretive (d3-12): if $c$ is the fixed conversion of Theorem 3.2 of [1], a varying $\kappa=c/\ell$ is a varying length $\ell(\tau)=c/\kappa(\tau)$, with relative spread $\varepsilon+O(\varepsilon^2)$ along a rotating solution (measured at $\varepsilon=10^{-1},10^{-2},10^{-3}$). A constant "$c/\ell$" requires $\frac12 M_{\mathrm{rot}}c^2\gg|\nu|E_0(\pi_0)/m$: the rotor energy must dominate the rest-energy amplitude of the momentum field. The clamped split-form interior of Section 16 of [1] is the limit $\varepsilon\to0$ (Proposition 17.9 and Corollary 17.15 of [1]); the positive-form interior of Section 15 has duality-class inertia and an exactly constant rate (Theorem 17.11(iii) of [1]). The "symmetric" $c^2/\ell^2$ is constant only insofar as the differentiated part is negligible beside the rotor, and $\varepsilon$ is an input.

## 4.5 The rest energy as a rate of rotation

**Proposition 4.9** (Two rotations at rest)**.** *(i) For the real first-order Dirac form of Section 4.4 of [1] at* $\mathbf{p}=0$, $H(0)=me_1$, $Je_1=T=e_1e_2e_3$, $T^2=-\mathbf{1}$, *and* $\Psi(\lambda)=\cos(m\lambda)\Psi_0-$

$\sin(m\lambda)Je_1\Psi_0 = e^{-m\lambda T}\Psi_0$*: angular frequency* $m$*, that is* $mc^2/\hbar$*. (ii) The zero-momentum orbits* $X(\tau) = e^{-\kappa G\tau}X_0$ *satisfy* $\alpha_+X = 0$*,* $\ddot{X} = -\kappa^2 X$*: angular frequency* $\kappa = c/\ell$*,* $z \mapsto e^{-i\kappa\tau}z$*,* $q \mapsto e^{-2i\kappa\tau}q$*. The orbits* $X(\tau) = e^{\kappa G\tau}X_0$*,* $z = e^{i\kappa\tau}c_+$*, are duality rotations of the other sense at the same rate, with momentum* $\alpha_+X = 2\kappa GX \neq 0$*.*

*Proof.* (i) $\dot{\Psi} = -JH\Psi$ with $JH(0) = mT$, since $e_1$ commutes with $J = e_2e_3$. (ii) $\frac{d}{d\tau}e^{\mp\kappa G\tau} = \mp\kappa Ge^{\mp\kappa G\tau}$, $G^2 = -\mathbf{1}$. □

Item (i) is inherited, the realification of $\psi = e^{-imc^2t/\hbar}\psi_0$, de Broglie's internal clock [9] (compare Proposition 3.7); (ii) restates Proposition 16.5(i)–(ii) of [1] (fixed there); both checked exactly (d3-13). Duality acts along both senses at the same rate: the zero-momentum sense is singled out by the momentum, not by duality.

**Proposition 4.10** (No operator relates the two rates)**.** *(i)* $\mathrm{Ad}_{e^{\vartheta T}}$ *fixes* $e_1, e_2, e_3$ *and maps a spatial* $e_a$ *to* $\cos 2\vartheta\, e_a + \sin 2\vartheta\, Te_a$*, a vector only for* $\vartheta \in \frac{\pi}{2}\mathbb{Z}$*: the Dirac rest flow is not a motion of* $V$*, and* $T$*, of grade three, is not in* $\mathfrak{so}(3,3)$*. (ii)* $(e^{\theta G})^\top \eta\, e^{\theta G} = \cos 2\theta\, \eta + \sin 2\theta\, \eta G$*, so* $e^{\theta G} \in O(3,3)$*, and lifts to* $\mathrm{Pin}(3,3)$*, only for* $\theta \in \pi\mathbb{Z}$*. Hence neither one-parameter group is the image of the other under the spin homomorphism, so this route supplies no relation between* $m$ *and* $\kappa$*, which remain independent inputs.*

*Proof.* (i) $T$ commutes with the temporal and anticommutes with the spatial generators (Lemma 2.1 of [1]), so $e^{\vartheta T}e_ae^{-\vartheta T} = e^{2\vartheta T}e_a$, and $Te_a$ has grade four. (ii) Expand with $e^{\theta G} = \cos\theta + \sin\theta\, G$, $G^\top\eta = \eta G$ symmetric, $G^\top\eta G = -\eta$. □

Item (ii) is the remark of Sections 14.3 and 17.6 of [1]. The proposition obstructs one route only, a map of the arena from one rate to the other (d3-15); $m$ and $\kappa$ are independent because both are free inputs. What remains is an added identification, interpretive (d3-14): with $E = \hbar\times$rate, $\hbar\kappa = mc^2$ holds iff $\ell = \hbar/(mc)$, the reduced Compton wavelength. The dynamics does not supply it. In the interior $\ddot{X} = -\kappa^2 X$ does not contain $m$; and the rescaling $(\tau, \kappa, I_0, m) \mapsto (s\tau, \kappa/s, sI_0, sm)$ of Proposition 4.11, which maps solutions to solutions and preserves the action (hence any quantisation of $h$), keeps $m\kappa$ and multiplies $m/\kappa$ by $s^2$, so $m = \hbar\kappa/c^2$ picks one point of that orbit by fiat. The rate is also ambiguous by a factor two ($q$ turns at $2\kappa$, $z$ at $\kappa$, the Dirac bilinears at 0 or $2m$): $2\hbar\kappa = mc^2$ gives $\ell = 2\hbar/(mc)$, and the algebra decides neither.

## 4.6 Scale freedom

**Proposition 4.11** (Scalings)**.** *(i)* $\eta$*,* $G$*,* $G\eta^{-1}$*,* $\eta G$*,* $\mathcal{K}$ *and the structure constants of the complex cross product* $\beta_0$ *are integer matrices.*

*(ii)* $\ddot{X} = -\kappa^2 X$ *is invariant under* $X \mapsto \sigma X$*, which scales* $\frac{m}{2}\eta(\alpha_+X, \alpha_+X)$ *by* $\sigma^2$*, absorbed by* $m \mapsto m/\sigma^2$*, and under* $(\tau, \kappa) \mapsto (s\tau, \kappa/s)$*, which with* $m \mapsto sm$ *preserves the action.*

*(iii) The closure is invariant under* $(\tau, I_0, m) \mapsto (s\tau, sI_0, sm)$ *with* $(\theta, h, p)$ *unchanged as functions of the rescaled time;* $\kappa \mapsto \kappa/s$*, and* $m\kappa$*,* $\varepsilon$ *and the action are invariant.*

*(iv) For* $s, t > 0$*,* $(I_0, m) \mapsto (sI_0, tm)$ *at fixed initial rate* $\kappa_0$ *(*$h_0 \mapsto sh_0$*) keeps* $I_0 > 0$ *and* $N$ *of Lorentz class and multiplies* $\varepsilon$ *by* $1/(st)$ *(by* $s/t$ *at fixed* $h_0$*).*

*Proof.* (i) $\beta_0(e_i, e_j) = \epsilon_{ijk}e_k$, $\beta_0(e_i, e_{j+3}) = \epsilon_{ijk}e_{k+3}$, $\beta_0(e_{i+3}, e_{j+3}) = -\epsilon_{ijk}e_k$; the rest is Lemma 3.4 of [1]. (ii) Linearity; $Y(\tau') = X(\tau'/s)$ solves $Y'' = -(\kappa/s)^2 Y$, the Lagrangian scales by $s^{-1}$ and $d\tau' = s\, d\tau$. (iii) With $d/d\tau' = s^{-1}d/d\tau$, $I_0' = sI_0$, $N' = N/s$, both equations of Proposition 17.8 of [1] are reproduced, $H' = H/s$, and $\kappa_0' = \kappa_0/s$ gives $\varepsilon' = \varepsilon$. (iv) $R_0$ does not depend on $(I_0, m)$, and $\varepsilon = R_0/(mI_0\kappa_0^2)$ with $\kappa_0$ held fixed; at fixed $h_0$, $\kappa_0 \mapsto \kappa_0/s$. □

Items (i) and (iv) restate the remark closing Section 17.5 of [1] (fixed there; the fixed-$\kappa_0$ qualifier is item 12 of `investigations/errata_2026-10-17`); (ii) and (iii) are added; all are fixed. The closure rescaled by $s = 2.5$ reproduces $(\theta, h, p)$ to $7 \times 10^{-12}$, and $\varepsilon$ measured from the range of $\kappa^2$ goes from 0.27209155 to $0.04534858 = \varepsilon/6$ under $(I_0, m) \mapsto (2I_0, 3m)$ at fixed $\kappa_0$, and to $\frac{2}{3}\varepsilon$ at fixed $h_0$ (d3-16, d3-17).

**Proposition 4.12** (Nothing in the algebra fixes $\ell$)**.** *(i) Schur's lemma [51] fixes only the form $\mathcal{C} = c\,\mathbf{1}$ of the conversion operator (Theorem 3.2 of [1]), and $c$ converts units of time into units of length. (ii) No monomial $c^{n_1}\hbar^{n_2}$ is a length, a mass or a time; with a mass $m$ adjoined, the only length is $\hbar/(mc)$ up to a number. (iii) The interior systems with different $\kappa > 0$ are carried into each other by a rescaling of $\tau$ that fixes every Clifford datum (Proposition 4.11(ii)); so no condition formulated in the Clifford data distinguishes one value of $\kappa$, or of $\ell = c/\kappa$, and in the closure none fixes $\varepsilon$ (Proposition 4.11(iv)).*

*Proof.* (i) The commutant of the vector representation of $\mathfrak{so}(3)$ in $M_3(\mathbb{R})$ is $\mathbb{R}\,\mathbf{1}$. (ii) In the basis $(\mathsf{M}, \mathsf{L}, \mathsf{T})$, $[c] = (0, 1, -1)$ and $[\hbar] = (1, 2, -1)$ span a plane containing none of $(0, 1, 0)$, $(1, 0, 0)$, $(0, 0, 1)$; adjoining $[m] = (1, 0, 0)$ gives an invertible matrix, with $(0, 1, 0) = -[c] + [\hbar] - [m]$. (iii) The data are integer matrices, and the rescaling acts on $\tau$ and $\kappa$ only. □

This is inherited dimensional analysis, already implicit in equation (kappa) of Section 3 of [1] ("their sum requires a scale") and in the remark closing Section 17.5 (d3-18). It is the one exact form of "$m$ breaks the symmetry of $c^2$" available here: $c$ carries no scale, the shift $\kappa G$ needs a length $\ell$ that the algebra cannot supply, and through $\ell = \hbar/(mc)$ that length is a mass. What it "breaks" is the change of the unit of $\tau$ (Proposition 4.11(iii)), not a symmetry of a vacuum; the reading is interpretive.

## 4.7 The frame drifts

**Example 4.13** (Example 17.17 of [1], re-verified)**.** Take $p_0 = e^1$, $m = I_0 = \kappa_0 = 1$, $\nu = 1$: $w = 1$, $\varphi_0 = 0$, $\varepsilon = 1$, $\dot\theta^2 = 1 + 2\sin^2\theta$, and by Theorem 17.12 of [1] ($N_c = 0$) $\dot x = e^{2\theta G} N p_0 = \cos 2\theta\, e_1 + \sin 2\theta\, e_4$. The $e_4$ part integrates to zero ($\theta \mapsto \pi - \theta$), and with $(1 - 2\sin^2\theta)/\sqrt{1 + 2\sin^2\theta} = 2/\sqrt{1 + 2\sin^2\theta} - \sqrt{1 + 2\sin^2\theta}$,

$$\Delta x\big|_{\theta:0\to\pi} = \int_0^\pi \frac{\cos 2\theta\, d\theta}{\sqrt{1 + 2\sin^2\theta}}\, e_1 = (4\mathrm{K}(-2) - 2\mathrm{E}(-2))e_1 \approx 0.3168040511\, e_1.$$

Over a full turn, $\theta : 0 \to 2\pi$, $\Delta x = (8\mathrm{K}(-2) - 4\mathrm{E}(-2))e_1 \approx 0.6336081022\, e_1$, the value of Example 17.17 of [1], whose corrected wording (erratum item 13) makes the turn $\theta \mapsto \theta + 2\pi$, two periods of the rate. After half a turn $h$ has returned and $p = -p_0$ ($e^{\pi G} = -\mathbf{1}$); after a full turn both have returned.

Fixed, as a re-verification (d3-19): quadratures agree with the elliptic expressions to $10^{-25}$, and integrating $(\theta, h, p, x)$ to the events $\theta = \pi, 2\pi$ reproduces both values and the returns of $h$, $p$ to $3 \times 10^{-14}$. By Theorem 17.16 of [1] every rotating solution with a nonzero rest energy of the momentum field has an unbounded frame; with a free rotor, a constant nonzero rate and a bounded frame coexist exactly on null data (end of Section 17.5 of [1]; at the equilibria of Corollary 4.5 the rate is zero and the frame moves uniformly), or, for several constituents, on the balance surface (Proposition 17.18 of [1]). In motion the differentiation unbinds the frame, and the frame field's rest energy grows with it (Corollary 4.2(ii)).

## 4.8 A quantum rotor, conditionally

**Proposition 4.14** (The half turn)**.** *(i) $e^{\pi G} = -\mathbf{1}$, $e^{2\pi G} = \mathbf{1}$, and for $p_0 \neq 0$ the leaf circle $\{e^{\theta G} p_0\}$ has period exactly $2\pi$. (ii) For every symmetric $N$, $U(\theta+\pi) = U(\theta)$. (iii) $\Gamma = e_1 e_2 \cdots e_6$ anticommutes with every $e_a$ and $\Gamma^2 = \mathbf{1}$, so $\mathrm{Ad}_\Gamma = -\mathbf{1}$ on $V$: $\Gamma$ implements $e^{\pi G} = -\mathbf{1}$ on*

*spinors; by Proposition 4.10(ii) the multiples of the half turn are the only rotor elements that lift to* $\mathrm{Pin}(3,3)$.

*Proof.* (i) $(e^{\theta G}-\mathbf{1})p_0 = (\cos\theta-1)p_0+\sin\theta\, Gp_0$ with $p_0 \perp Gp_0$, $|Gp_0| = |p_0|$. (ii) $p(\theta+\pi) = -p(\theta)$ and $U$ is quadratic. (iii) A product of all six anticommuting generators anticommutes with each, and $(e_1\cdots e_6)^2 = (-1)^{15}\prod_a \eta_{aa} = \mathbf{1}$. □

This restates the remark of Section 17.6 of [1] (fixed there; in the realisation $\Gamma = J\,e_1e_4e_5e_6 = +e_1\cdots e_6$), re-verified exactly (d3-20). The quantum statement it permits is conditional on an assumption.

**Assumption 4.15** (A quantised leaf)**.** The leaf $T^*S^1$ of Theorem 4.1(ii) is quantised by wave functions with $\Phi(\theta+2\pi) = e^{2\pi i\delta}\Phi(\theta)$, $h = -i\hbar\,\partial_\theta$, and the Casimirs of the leaf are treated as numbers.

**Proposition 4.16** (A conditional rate ladder)**.** *Under Assumption 4.15: (i) $h$ has spectrum $\hbar(\mathbb{Z}+\delta)$; on flat leaves (Corollary 4.5) the levels are $E_n^{\mathrm{rot}} = \hbar^2(n+\delta)^2/(2I_0)$ (plus the constant $\frac{1}{2}p_0^{\top}Np_0$ on duality-class leaves), $\kappa_n = (n+\delta)\hbar/I_0$ and $E_{n+1}^{\mathrm{rot}} - E_n^{\mathrm{rot}} = \frac{\hbar}{2}(\kappa_n+\kappa_{n+1})$. (ii) If $R_0 > 0$ the Hamiltonian is a Mathieu operator [36, 11] whose potential couples $e^{i(n+\delta)\theta}$ only to $e^{i(n\pm2+\delta)\theta}$, so for every $\delta$ the parity of $n$ is conserved; it labels the eigenvalues $\pm e^{i\pi\delta}$ of the half-turn translation $\theta\mapsto\theta+\pi$ ($\pm1$ for $\delta = 0$). For $q_{\mathrm{M}} := I_0R_0/(2m\hbar^2) \gg 1$ the lowest gap in either parity sector is $\hbar\Omega - \hbar^2/(2I_0) + O(q_{\mathrm{M}}^{-1/2}\hbar^2/I_0)$, and the two sectors are degenerate up to tunnelling. (iii) If moreover $\delta = 0$, and, as a separate assumption, the first level $h = \hbar$ is matched by $\hbar\kappa_1 = mc^2$ with $\ell = \hbar/(mc)$, then $I_0 = \hbar^2/(mc^2)$, $M_{\mathrm{rot}} = m$, and $E_1^{\mathrm{rot}} = \frac{1}{2}mc^2$.*

*Proof.* (i) Spectrum of $-i\hbar\partial_\theta$ on twisted periodic functions; for $[N,G]=0$, $U$ is the constant of Corollary 4.5(iii). (ii) $\cos(2\theta+\varphi_0)$ shifts $n$ by $\pm2$, and $e^{i(n+\delta)(\theta+\pi)} = (-1)^n e^{i\pi\delta}e^{i(n+\delta)\theta}$. Dividing the Schrödinger equation by $-\hbar^2/(2I_0)$ gives $\Phi'' + (a_{\mathrm{M}} - 2q_{\mathrm{M}}\cos(2\theta+\varphi_0))\Phi = 0$, $a_{\mathrm{M}} = 2I_0E/\hbar^2$, with $a_{\mathrm{M}} \sim -2q_{\mathrm{M}} + 2(2r_{\mathrm{M}}+1)\sqrt{q_{\mathrm{M}}} - \frac{1}{8}((2r_{\mathrm{M}}+1)^2+1)$ as $q_{\mathrm{M}}\to\infty$ (DLMF 28.8.1); the levels $r_{\mathrm{M}} = 0, 1$ lie in the same sector and differ by $4\sqrt{q_{\mathrm{M}}} - 1$, and $\hbar\Omega = (\hbar^2/2I_0)\,4\sqrt{q_{\mathrm{M}}}$. (iii) $\hbar/I_0 = c/\ell = mc^2/\hbar$ and $E_1^{\mathrm{rot}} = \hbar^2/(2I_0)$. □

Checked (d3-21; independently, x3-21): at $R_0 = 0$ the ladder is exact for $\delta = 0$ and 0.3; at $R_0/m = 2000$, $\hbar = I_0 = 1$, the lowest gap in each sector is 62.7415 for both $\delta$, against $\hbar\Omega - \hbar^2/(2I_0) = 62.7456$ (62.7416 with the next term of DLMF 28.8.1), and the sectors' ground states agree to $3\times10^{-12}$. The proposition is constructed: the quantisation of the leaf exists (the orbit method, $\delta$ a character of the stabiliser $2\pi\mathbb{Z}$ in the universal cover of the rotor group), and nothing selects $\delta$, $I_0$ or the matchings; whether it is physical is open. Missing are a reason for the assumption (a quantum theory of the momentum field, which the closure does not give and the split-form frame obstructs, having no ground state, Corollary 3.11), a value of $I_0$, a coupling to matter (only the $\mathbb{Z}_2$ of the half turn reaches spinors, as $\Gamma$, and nothing ties the parity of $n$ to chirality), and any derivation of the two matchings of (iii). $E_1^{\mathrm{rot}} = \frac{1}{2}mc^2$ is $K_{\mathrm{rot}} = \frac{1}{2}h\kappa$ at $h = \hbar$: the $\frac{1}{2}$ is built in, and $mc^2$ appears only as $\hbar$ times the rate put in, neither as a level nor as a transition energy.

## 4.9 Assessment

Against the working dictionary (Table 5) the closure gives four answers. *$E$ as the Casimir $|p|^2$:* $E_6(\pi) = \frac{1}{2}|p|^2$ is a definition; the identification of $u = \frac{1}{2}|z|^2$ of the frame with $\frac{1}{2}|p|^2$ fails, since the frame drifts and its $u$ is not conserved (Corollary 4.2(ii)); and the Casimir does not enter the rotor's motion. What is conserved and differentiates in motion is $H = K_{\mathrm{rot}} + U$. *$c^2$ as the factorising shift $\kappa G$:* in the closure $c^2$ appears as the rotor term $\frac{1}{2}(I_0/\ell^2)c^2$ and, clamped, as $(\kappa G)^2 = -c^2/\ell^2$; $c$ is fixed up to units by Schur's lemma, $\ell$ is free, $\kappa$ is dynamical and constant only when $\varepsilon \ll 1$, and the factor $\frac{1}{2}$ is that of a rotor. *$m$ as the rest energy $\frac{1}{2}|q|$:* confirmed in

| claim | statement | class | check |
|---|---|---|---|
| Theorem 4.1 | $E_6(\pi)$, $E_0(\pi)$, $\mathbf{S}(\pi)$ are Casimirs; only the duality phase moves | fixed (cf. Prop. 17.8) | d3-1–3 |
| Corollary 4.2 | (i) no Hamiltonian changes $E_0(\pi)$; (ii) $E_0(X)$ oscillates or grows | (i) obstructed, (ii) fixed | d3-4, 22 |
| Proposition 4.3 | $U = \frac{R_0}{2m}\cos(2\theta+\varphi_0)$, $R_0 = 2\lvert\nu\rvert E_0(\pi_0)$ | fixed (restated) | d3-5 |
| Proposition 4.4 | $\Omega^2 = 4\lvert\nu\rvert E_0(\pi_0)/(mI_0)$; elliptic period | fixed (corollary) | d3-6 |
| Corollary 4.5 | constant rate: null data, $\nu = 0$, equilibria, duality class | fixed (restated) | d3-7 |
| Remark 4.6 | $U'' = 2R_0/m$ / $\theta$ as a pseudo-Goldstone angle | restated / interpretive | d3-8 |
| Proposition 4.7 | $H = K_{\rm rot} + U$, exchange $R_0/m$, Lorentz invariance, $E_6(\pi)$ inert | fixed | d3-9 |
| units | $K_{\rm rot} = \frac{1}{2}M_{\rm rot}c^2$ is the "$c^2$ part"; not $E = mc^2$ | interpretive | d3-10 |
| Proposition 4.8 | $\kappa^2$ law, bound $2\varepsilon$, rotation criterion | fixed (restated) | d3-11 |
| heavy rotor | constant $c/\ell$ needs $\frac{1}{2}M_{\rm rot}c^2 \gg \lvert\nu\rvert E_0(\pi_0)/m$ | interpretive | d3-12 |
| Proposition 4.9 | rest energy as a rotation rate (Dirac; frame, both senses) | inherited / restated | d3-13 |
| matching | $\hbar\kappa = mc^2 \iff \ell = \hbar/(mc)$, an added identification | interpretive | d3-14 |
| Proposition 4.10 | no Spin or $O(3,3)$ map relates the two rates | obstructed (one route) | d3-15 |
| Proposition 4.11 | integer data; scalings of $X$, $\tau$, $I_0$, $m$ | fixed | d3-16, 17 |
| Proposition 4.12 | nothing in the algebra fixes $\ell$ (a mass scale) | inherited | d3-18 |
| Example 4.13 | $\Delta x = (8\mathrm{K}(-2) - 4\mathrm{E}(-2))e_1$ per full turn | fixed (restated) | d3-19 |
| Proposition 4.14 | $e^{\pi G} = -\mathbf{1}$, implemented by $\Gamma$ | fixed (restated) | d3-20 |
| Proposition 4.16 | conditional ladder $\kappa_n = (n+\delta)\hbar/I_0$; $E_1^{\rm rot} = \frac{1}{2}mc^2$ | constructed | d3-21 |
| $b$-field | compact transports (Sections 2 and 5): Zeeman, Jordan, screening; $\mathfrak{so}(4)_\Delta$ | fixed (restated) | d3-23 |

Table 5: The claims of Section 4, their classes and the checks of `d3_closure_dynamics.py` (d3-0 verifies the conventions). "Fixed (restated)" and "restated": fixed in [1] (or in another section of this note), restated and re-verified here.

a precise dynamical sense for the momentum field, whose $E_0(\pi)$ is the amplitude of the duality potential, sets $\Omega^2$ and the energy exchanged, and vanishes on the flat direction; but there it is a Casimir, carried and not generated. For the frame field $\frac{1}{2}|q(X)|$ is not conserved even in the clamped interior (Proposition 16.5(ii) of [1]) and grows without bound under the free-rotor drift, which changes it without selecting a value; and $mc^2 = \hbar\kappa$, which would tie $m$ to the rotor, is an added identification (Proposition 4.10). *b, the Yang–Mills field*: the closure contains none and needs none (its frame is abelian, Theorems 17.6–17.7 of [1]); a Yang–Mills field can only be added as a transport. An admissible one is compact (Theorem 2.3 and Corollary 2.4 of [1]): $b = (A_{\rm d} + A_{\rm a}, A_{\rm d} - A_{\rm a})$ on the temporal and spatial triples, rotation vectors $\mathbf{a}_{\rm d}, \mathbf{a}_{\rm a}$, diagonal part in $\mathfrak{l}_G$, anti-diagonal part in $\mathfrak{p}_G$. In the interior with the shift $b + \kappa G$ (Section 5) the diagonal part alone gives the bounded lines $\kappa$, $\kappa \pm |\mathbf{a}_{\rm d}|$, removable along a worldline [33] and so not a gauge-invariant spectrum; the anti-diagonal part alone gives Jordan blocks at the rate $\sqrt{\kappa^2 + |\mathbf{a}_{\rm a}|^2}$ and secular growth; $\mathbf{a}_{\rm d}\cdot\mathbf{a}_{\rm a} = 0$, $\mathbf{a}_{\rm d} \neq 0$ screens it, every orbit bounded although $[b, G] \neq 0$, except on the ratio $|\mathbf{a}_{\rm d}|^2 = \kappa^2 + |\mathbf{a}_{\rm a}|^2$, where a static mode produces a Jordan block (Theorem 5.10); a generic compact $b$ is unbounded (d3-23). At three constituents $\mathfrak{so}(4)_\Delta$ is the part of $\mathfrak{ps}_W$ commuting with $G_3$ (Theorem 17.21 of [1]), not the only bounded part ($b = R_{7,13} \pm R_{10,11}$ is bounded with $[b, G_3] \neq 0$, Section 5); it preserves the total $\frac{1}{2}|Z|^2$ and $q_3$, hence $\frac{1}{2}|q_3|$, but changes the constituents' rest energies, their sum included ($\frac{1}{2} \to \frac{3}{2}$), and the Lorentz length of the summed density (Proposition 2.14; d3-23, x3-22). The conclusion, interpretive and drawn from fixed statements: the undivided energy and its differentiation in

motion are complete before any $b$-field appears, and where the hypothesis needs the closure to produce a rest energy the reading fails: for the momentum field it cannot, and the frame field's rest energy changes only by an oscillation or an unbounded drift, with no value selected.

# 5 The $b$-field: the antisymmetric part of the shift

The hypothesis places the $b$-field of Yang and Mills [63] at the origin of the mass. This section asks where a connection enters the interior dynamics of Sections 15–17 of [1], and what it can do there. A constant shift of the interior Lagrangian splits into the $\eta$-self-adjoint binding term $\kappa G$ and an $\eta$-antisymmetric part $b \in \mathfrak{so}(3,3)$, and only $b$ is a connection (Theorem 5.2). A Yang–Mills $b$-field is moreover compact (Theorem 2.3 and Corollary 2.4 of [1]), $b = (A + C, A - C)$ on the two triples. Its diagonal part splits the binding frequency into Zeeman lines that can be gauged away along any worldline (Theorem 5.6 and Proposition 5.7); its anti-diagonal part alone tears the binding (Theorem 5.8); together they keep every orbit bounded exactly when the triples rotate at equal rates about axes that are not opposite, outside one exceptional ratio (Theorem 5.10). On its own $b$ does not bind, and nothing in the interior dynamics ties the mass $m$ to the rate $\kappa$ (Proposition 5.12). Units are $\hbar = c = 1$ unless a symbol is shown; $\tau$ has dimension $T$, $\kappa = c/\ell$ (equation (7) of [1]) and every eigenvalue of $b$ are rates, and $\hbar\kappa$ is a quantum of interior energy. Every finite-dimensional statement was checked by `verify/d4_bfield.py` and independently by `verify/x4_check.py` (Table 6).

**Notation.** For $a < b$, $R_{ab} \in \mathfrak{so}(3,3)$ is defined by $R_{ab}e_b = \eta_{bb}e_a$ and $R_{ab}e_a = -\eta_{aa}e_b$; on vectors it equals $\frac{1}{2}\operatorname{ad}(e_a e_b)$. The generators of $\mathfrak{l}_G$ (Theorems 12.1 and 12.3 of [1]) are $L_i = \frac{1}{2}\operatorname{ad}(e_{j+3}e_{k+3} - e_j e_k)$ and $K_i = \operatorname{ad}(\frac{1}{2}(e_k e_{j+3} - e_j e_{k+3}))$ for cyclic $(i,j,k)$. On $z = \mathbf{B} + i\mathbf{E}$ they act as $z \mapsto \mathbf{e}_i \times z$ and $z \mapsto i\,\mathbf{e}_i \times z$, so $K_i = GL_i$; the $L_i$ are rotations and the $K_i$ boosts, and

$$b = \textstyle\sum_i (w_i^{\mathrm{r}} L_i + w_i^{\mathrm{b}} K_i) \in \mathfrak{l}_G \quad \text{acts as} \quad z \mapsto w \times z, \qquad w = w^{\mathrm{r}} + i\,w^{\mathrm{b}} \in \mathbb{C}^3. \tag{8}$$

In particular $R_{12} - R_{45} = -L_3$. The grading root is $\mathcal{K} = \frac{1}{2}\operatorname{ad}\big(\sum_i e_i e_{i+3}\big)$, with $G = -\mathcal{K}\eta$ (Lemma 3.4 of [1]). We write $\sigma_j$ for the boost exchanging $e_j$ and $e_{j+3}$, $F^\dagger = \eta^{-1}F^\top\eta$, $[\mathbf{v}]_\times$ for the matrix of $\mathbf{x} \mapsto \mathbf{v} \times \mathbf{x}$ on $\mathbb{R}^3$, $\mathcal{E}_\mu(T) = \ker(T - \mu)$, and $h(y, y') = \bar{y}^\top \eta y'$. The interior Lagrangian with a constant shift is

$$\mathcal{L} = \frac{m}{2}\,\eta(u,u), \qquad u = \dot{X} + SX, \qquad S = b + \kappa G, \qquad b \in \mathfrak{so}(3,3), \tag{9}$$

with $p = m\eta u$ and $\pi = \eta^{-1}p = mu$. For $b = 0$ this is the split-form Lagrangian of Theorem 16.2 of [1].

## 5.1 Two summands, and which one is a connection

**Proposition 5.1** (Two summands)**.** $\operatorname{End}(V) = \mathfrak{so}(3,3) \oplus \{F : F = F^\dagger\}$*, of dimensions* $15 + 21$*. The part of* $S = b + \kappa G$ *in the first summand is* $b$*, and the part in the second is* $\kappa G$*. The binding root lies in the second summand (*$G^\dagger = G$*) and the grading root in the first (*$\mathcal{K}^\dagger = -\mathcal{K}$*).*

*Proof.* $F \mapsto F^\dagger$ is an involution, and $F$ lies in its $\mp 1$ eigenspace exactly when $\eta F$ is antisymmetric (respectively symmetric). □

This restates Theorem 17.3 of [1] at $n = 1$, and the placement of the two roots is Proposition 3.3 and Theorem 17.3 of [1] (fixed there); the decomposition itself is linear algebra valid for any nondegenerate form. The next theorem gives it dynamical meaning.

**Theorem 5.2** (Covariant factorisation)**.** *Let* $b \in \mathfrak{so}(3,3)$ *be constant and* $D := \frac{d}{d\tau} + b$*.*

*(i) The Euler–Lagrange equations of (9) are $(D-\kappa G)\pi = 0$ with $\pi = m(D+\kappa G)X$, that is*

$$(D-\kappa G)(D+\kappa G)X = 0 \qquad \Longleftrightarrow \qquad D^2X + \kappa^2 X + \kappa\,(D_\tau G)\,X = 0, \qquad D_\tau G := [b, G]. \tag{10}$$

*(ii) Let $X = g(\tau)Y$ with $g(\tau) \in O(3,3)$. Then $(\frac{d}{d\tau} + S)(gY) = g(\frac{d}{d\tau} + S^g)Y$, where $S^g = g^{-1}Sg + g^{-1}\dot{g}$. The $\mathfrak{so}(3,3)$ part $b^g = g^{-1}bg + g^{-1}\dot{g}$ transforms as a connection, the self-adjoint part $\kappa g^{-1}Gg$ as a tensor, and $(D_\tau G)^g = g^{-1}(D_\tau G)g$.*

*Proof.* (i) $\partial\mathcal{L}/\partial\dot{X} = p$ and $\partial\mathcal{L}/\partial X = S^\top p$, so $\dot{p} = S^\top p$, that is $\dot{\pi} = S^\dagger\pi = (-b + \kappa G)\pi$, which is $(D - \kappa G)\pi = 0$. Moreover $\pi = m(D+\kappa G)X$, $D(GX) - GDX = [b, G]X$ and $G^2 = -\mathbf{1}$. (ii) Differentiating $g^\top\eta g = \eta$ gives $(g^{-1}\dot{g})^\dagger = -g^{-1}\dot{g}$, and $(g^{-1}Fg)^\dagger = g^{-1}F^\dagger g$ preserves both summands; finally $\frac{d}{d\tau}(g^{-1}Gg) + [b^g, g^{-1}Gg] = g^{-1}[b, G]g$. Checked with fifteen symbolic coordinates for $b$ and a Cayley curve in $SO(3,3)$. □

Part (ii) is inherited, the transformation law of a connection and of an endomorphism field in any gauge theory. Part (i) is fixed: the split-form, internal-connection version of the curved extension at the end of Section 15 of [1], with the single new term $\kappa(D_\tau G)X$. That the binding part is a tensor restates Theorem 17.3 of [1] ($G = G^\dagger$): no gauge transformation creates, removes or rescales it. The dictionary entry "$b \leftrightarrow$ transport part of the shift" is correct, and forced.

**Proposition 5.3** (Commuting and anticommuting parts)**.** *$\mathrm{Ad}_G : b \mapsto GbG^{-1}$ is an involution of $\mathfrak{so}(3,3)$, so $\mathfrak{so}(3,3) = \mathfrak{l}_G \oplus \mathfrak{p}_G$, where $\mathfrak{l}_G = \{[b, G] = 0\}$ has dimension 6 and $\mathfrak{p}_G = \{bG = -Gb\}$ has dimension 9. In the variable $z$,*

$$\mathfrak{l}_G = \{z \mapsto (P + iQ)z\} \cong \mathfrak{so}(3, \mathbb{C}), \qquad \mathfrak{p}_G = \{z \mapsto (C + i\Sigma)\bar{z}\},$$

*with $P, Q, C$ real antisymmetric and $\Sigma$ real symmetric. The Killing form $4\,\mathrm{Tr}$ has signature $(3,3)$ on $\mathfrak{l}_G$. On $\mathfrak{p}_G$ it is negative on the three directions $\Sigma = 0$, compact counter-rotations of the two triples such as $R_{12} + R_{45}$, and positive on the six boosts $C = 0$. Both $\mathcal{K}$ ($z \mapsto -i\bar{z}$) and the pairing boosts $\sigma_j$ lie in $\mathfrak{p}_G$.*

*Proof.* $G^{-1} = -G$ and $G^\dagger = G$ give $(GbG^{-1})^\dagger = -GbG^{-1}$ and $\mathrm{Ad}_G^2 = \mathbf{1}$. In $(\mathbf{B}, \mathbf{E})$ blocks, commuting with $G$ means the form $\left(\begin{smallmatrix} P & -Q \\ Q & P \end{smallmatrix}\right)$ and anticommuting means $\left(\begin{smallmatrix} C & \Sigma \\ \Sigma & -C \end{smallmatrix}\right)$; $b^\dagger = -b$ gives the symmetry types. The Killing form was computed in the adjoint representation. □

This is fixed; the description of $\mathfrak{l}_G$ restates Theorem 12.1 of [1]. The pairing boosts that tear the binding at a threshold (Proposition 16.7 of [1]) are anticommuting, and so is the grading root.

**Gauge admissibility.** By Theorem 2.3 and Corollary 2.4 of [1] ("no gauge boson is a boost"), a Yang–Mills $b$-field with a positive kinetic term takes values in the compact part $\mathfrak{so}(3) \oplus \mathfrak{so}(3)$ of $\mathfrak{so}(3,3)$, spanned by the $R_{ab}$ with $a, b$ in the same triple. On the two triples,

$$b = \mathrm{diag}(A + C,\, A - C), \qquad A = [\mathbf{a}]_\times, \quad C = [\mathbf{c}]_\times, \tag{11}$$

so the triples rotate about $\mathbf{a} + \mathbf{c}$ and $\mathbf{a} - \mathbf{c}$. The diagonal part $(A, A)$ lies in $\mathfrak{l}_G$, with $w = \mathbf{a}$ real in (8); the anti-diagonal part $(C, -C)$ lies in $\mathfrak{p}_G$, with $\Sigma = 0$ ($z \mapsto \mathbf{c} \times \bar{z}$). Boost-valued $b$ below ($K_i$, $\sigma_j$, $\mathcal{K}$, the $\Sigma$-part of $\mathfrak{p}_G$) are metric connections in the sense of Proposition 16.7 of [1], legitimate in (9), but not gauge fields.

### 5.2 The phase-space generator and a criterion for boundedness

**Lemma 5.4** (Bookkeeping). *On $(X,\pi)$ the flow is generated by $M = \left(\begin{smallmatrix} -S & \mathbf{1}/m \\ 0 & S^\dagger \end{smallmatrix}\right)$, with $S^\dagger = -b + \kappa G$. It is the Hamiltonian flow of the Noether energy $H(X,p) = \frac{1}{2m} p^\top \eta^{-1} p - p^\top S X$. $H$ vanishes on the invariant Lagrangian subspace $p = 0$, and $\det(\lambda - M) = \chi_S(\lambda)\chi_S(-\lambda)$, where $\chi_S(\lambda) = \det(\lambda - S)$ (since $\dot X = \pi/m - SX$, $\dot\pi = S^\dagger \pi$, and $S^\dagger$ is similar to $S^\top$).*

This is Hamiltonian bookkeeping, inherited. The classification rests on the following criterion.

**Theorem 5.5** (When the interior motion stays bounded). *Let $S$ be any constant real $6 \times 6$ matrix.*

*(i) $M$ is semisimple if and only if $S$ is semisimple and, for every $\mu \in \mathbb{C}$, $\mathcal{E}_\mu(S^\dagger)$ and $\mathcal{E}_{-\mu}(S^\dagger)$ are orthogonal for the bilinear extension of $\eta$.*

*(ii) Every solution is bounded on $\mathbb{R}$ if and only if, in addition, $\operatorname{spec} S \subset i\mathbb{R}$. Then (i) says that every eigenspace of $S^\dagger$ is totally isotropic for $h$. Equivalently, every momentum normal mode $\pi = \operatorname{Re}(y e^{i\omega\tau})$ with $\omega \neq 0$, superpositions within a degenerate frequency included, has period mean $\langle \eta(\pi,\pi) \rangle = \frac{1}{2} h(y,y) = 0$, and the static momenta ($\omega = 0$) span a totally $\eta$-null space.*

*Proof.* $M$ is semisimple if and only if $-S$ (its restriction to $\pi = 0$) and $S^\dagger$ (the quotient map) are, and $M$ is similar to $\operatorname{diag}(-S, S^\dagger)$, which by Roth's theorem [50] means that $-S\Xi - \Xi S^\dagger = -\mathbf{1}/m$ is solvable. With $-S = F_1 \Lambda_1 F_1^{-1}$ and $S^\dagger = F_2 \Lambda_2 F_2^{-1}$ diagonal, this holds if and only if $(F_1^{-1} F_2)_{ij} = 0$ whenever $(\Lambda_1)_i = (\Lambda_2)_j =: \mu$. A row $\ell^\top$ of $F_1^{-1}$ has $S^\top \ell = -\mu \ell$, and $S^\top = \eta S^\dagger \eta^{-1}$ puts $\eta^{-1}\ell$ in $\mathcal{E}_{-\mu}(S^\dagger)$; the condition is $\ell^\top y = \eta(\eta^{-1}\ell, y) = 0$ for $y \in \mathcal{E}_\mu(S^\dagger)$. (ii): a linear flow is bounded on $\mathbb{R}$ exactly when it is semisimple with imaginary spectrum, and then $\mathcal{E}_{-\mu}(S^\dagger) = \overline{\mathcal{E}_\mu(S^\dagger)}$. □

The theorem is fixed, though valid for any nondegenerate $\eta$ and any $S$. Its necessity half is the split virial identity (Theorem 16.1 of [1]: a bound orbit has null momentum on average); the new content is that the averaged virial condition, imposed mode by mode, is also sufficient. For $S = \kappa G$ it is the isotropy of Proposition 16.3 of [1]. On ten test connections, exact semisimplicity, exact Sylvester solvability over $\mathbb{Q}$ and numerical isotropy (to $10^{-8}$) agree.

### 5.3 Connections commuting with $G$: a Zeeman pattern without a scale

**Theorem 5.6** (Connections commuting with $G$). *Let $b \in \mathfrak{l}_G$ act as in (8).*

*(i) $\Xi = -G/(2\kappa m)$ solves $-S\Xi - \Xi S^\dagger = -\mathbf{1}/m$, so $M$ is semisimple if and only if $S$ is.*

*(ii) $\chi_S(\lambda) = (\lambda^2 + \kappa^2)\big((\lambda - i\kappa)^2 + w \cdot w\big)\big((\lambda + i\kappa)^2 + \overline{w \cdot w}\big)$.*

*(iii) In the frame in which $b$ is constant, every solution is bounded if and only if $e^{-b\tau}$ is bounded, that is, if and only if $w = 0$ or $w \cdot w > 0$. The frequencies are then $\kappa$ and $|\kappa \pm s|$, where $s = \sqrt{w \cdot w}$. For $w \cdot w = 0 \neq w$ the zero-momentum flow grows like $\tau^2$; for $w \cdot w \notin [0,\infty)$ it grows exponentially. For a gauge-admissible $b$, $w = \mathbf{a}$ is real and every solution is bounded.*

*(iv) For the rotation $b = \beta(R_{12} - R_{45}) = -\beta L_3$, the spectrum of $M$ is $\pm i\kappa$, $\pm i(\kappa + \beta)$, $\pm i(\kappa - \beta)$, each double. For the boost $b = \beta K_3$, the exponents are $\pm\beta \pm i\kappa$: complex rather than real, so the orbits spiral outward at the binding frequency. The growth is the boost factor $e^{-\beta K_3 \tau}$ of Proposition 5.7, and $K_3$ is not a gauge generator (Corollary 2.4 of [1]).*

*Proof.* (i) With $[b, G] = 0$: $-(b + \kappa G)\Xi - \Xi(-b + \kappa G) = \frac{1}{2\kappa m}(bG - Gb + 2\kappa G^2) = -\frac{1}{m}$. (ii) On $(V, G) \cong \mathbb{C}^3$, $S$ is $z \mapsto i\kappa z + w \times z$, and $(w\times)^3 = -(w \cdot w)(w\times)$ gives $w\times$ the eigenvalues $0, \pm is$;

the real characteristic polynomial of a complex-linear map is its complex one times the conjugate. (iii) By Proposition 5.7, $X = e^{-b\tau}Y$ with $Y$ a solution at $b = 0$, bounded by (i)–(ii); conversely the zero-momentum solutions are $e^{-b\tau}e^{-\kappa G\tau}X_0$ with $e^{\kappa G\tau}$ bounded. And $e^{-b\tau}$ is bounded exactly when $w\times$ is semisimple with imaginary eigenvalues: $s$ real, and not $w\cdot w = 0 \neq w$, where $N = w\times$ has $N^2\bar{w} = |w|^2 w \neq 0 = N^3$. For real $w$, $w\times$ is real antisymmetric. (iv) $s = \pm\beta$, respectively $\pm i\beta$; semisimplicity was checked exactly at $(\kappa, \beta) = (1, \frac{3}{7})$ and $(1, \frac{3}{2})$. □

The theorem is fixed, but its content deflates. Criterion (iii) is a property of the frame in which $b$ is constant, not of the binding: in the co-moving frame of Proposition 5.7 the binding survives every commuting connection, as the paper warns after Proposition 16.7 of [1] ("growth on one worldline proves nothing"), and for admissible $b$ it holds automatically. In that frame a rotation lifts the degeneracy noted there ("the single frequency is degenerate across the three pairs") into the triplet $\kappa - \beta$, $\kappa$, $\kappa + \beta$. Read as a frame vector through the Lie isomorphism of the frame with $\mathfrak{l}_G$ (Theorem 17.5 of [1]), the connection has invariant $q = w \cdot w$; the interpretive reading is that the lab-frame motion stays bounded exactly for a connection of magnetic type, $\mathbf{E}_b \cdot \mathbf{B}_b = 0$ and $|\mathbf{B}_b| > |\mathbf{E}_b|$, while the binding survives every commuting connection, and an admissible one has $\mathbf{E}_b = 0$.

**Proposition 5.7** (The commuting part is a moving frame)**.** *Let $b \in \mathfrak{l}_G$. Then $(\frac{d}{d\tau} + b + \kappa G)(e^{-b\tau}Y) = e^{-b\tau}(\dot{Y} + \kappa G Y)$. The exponents are therefore $\pm i\kappa$ shifted by the eigenvalues of $-b$, and with $p_X = e^{b^\top \tau}p_Y$,*

$$H_{\text{lab}} = H_{\text{co}} - p_Y^\top b Y.$$

*Proof.* $e^{-b\tau}$ is an isometry that commutes with $b$ and $G$; moreover $e^{b\tau}\eta^{-1}e^{b^\top\tau} = \eta^{-1}$ and $e^{b\tau}(b + \kappa G)e^{-b\tau} = b + \kappa G$. □

For compact $b$ this is Larmor's theorem [33], inherited (a rotating frame trades the field for an angular velocity), and the first sentence of Proposition 16.7 of [1]; for an $\mathfrak{l}_G$ boost, $e^{-b\tau}$ is a uniformly boosted frame instead. $Q_b = p_Y^\top b Y$ is the Noether charge of the $\mathfrak{l}_G$ generator, so $H_{\text{lab}}$ carries a chemical potential for it. The reading is interpretive but sharp: the Zeeman pattern labels the levels of the binding frequency by their $\mathfrak{l}_G$ charge, has no gauge-invariant content on one worldline, and is observable only against a reference that $b$ does not rotate, as the Zeeman effect is against the laboratory. A commuting $b$ supplies a spectrum but no invariant scale; $\beta/\kappa$ is free.

## 5.4 Connections anticommuting with $G$, and screening

**Theorem 5.8** (Connections anticommuting with $G$)**.** *Let $b \in \mathfrak{p}_G$ with $b \neq 0$.*

*(i) $(\pm b + \kappa G)^2 = b^2 - \kappa^2$, and $M^2 = \operatorname{diag}(b^2 - \kappa^2, b^2 - \kappa^2) + N$ with $N = \left(\begin{smallmatrix} 0 & -2b/m \\ 0 & 0 \end{smallmatrix}\right) \neq 0$. Hence $M$ is never semisimple, and some solutions grow at least linearly.*

*(ii) For the admissible counter-rotation $b = \beta(R_{12} + R_{45})$, which is $(C, -C)$ with $\mathbf{c} = -\beta\mathbf{e}_3$, $\det(\lambda - M) = (\lambda^2 + \kappa^2)^2(\lambda^2 + \kappa^2 + \beta^2)^4$. The exponents $\pm i\sqrt{\kappa^2 + \beta^2}$ have algebraic multiplicity four and geometric multiplicity two, and $\pm i\kappa$ are semisimple. The bounded initial data are exactly those with $\pi_1 = \pi_2 = \pi_4 = \pi_5 = 0$, that is, with no momentum on the support of $b$. The zero-momentum orbits are bounded and turn at $\sqrt{\kappa^2 + \beta^2}$ on the support.*

*(iii) For a pairing-plane boost $b = \nu\,\sigma_j$, the exponents in that plane are $\pm\sqrt{\nu^2 - \kappa^2}$, and there $M^2 - (\nu^2 - \kappa^2) = \left(\begin{smallmatrix} 0 & -2\nu\sigma_j/m \\ 0 & 0 \end{smallmatrix}\right)$: Jordan blocks for every $\nu \neq 0$, below, at and above the threshold (Proposition 16.7 of [1] and the computation after it, where $\nu$ is called $c$).*

*(iv) The grading root as a connection satisfies $(\nu\mathcal{K} + \kappa G)^2 = (\nu^2 - \kappa^2)\mathbf{1}$ on all of $V$.*

*Proof.* (i) $\{b, G\} = 0$ gives both squares, and $M^2$ has off-diagonal block $-2b/m$. With $b^2 - \kappa^2 = Q_s + Q_n$ (Jordan–Chevalley), $M^2 = \mathrm{diag}(Q_s, Q_s) + (\mathrm{diag}(Q_n, Q_n) + N)$ with commuting summands ($Q_n$ is a polynomial in $b$), so by uniqueness $M^2$, and hence $M$, is semisimple only if $\mathrm{diag}(Q_n, Q_n) + N = 0$, impossible for off-diagonal $N \neq 0$. (ii)–(iv): exact ranks and null spaces over $\mathbb{Q}(i)$ at $(\kappa, \beta) = (4, 3)$ and at $\nu = \frac{1}{2}, 1, 2$, $\kappa = m = 1$ ($\mathrm{rank}(M^2 + \frac{3}{4}) = 2$ on the plane at $\nu = \frac{1}{2}$), with symbolic characteristic polynomials; (i) with nine symbolic coordinates on $\mathfrak{p}_G$. □

Part (i) is an obstruction: no purely anticommuting connection, admissible or not, is compatible with bounded normal modes. Parts (ii) and (iv) are fixed; (iii) re-verifies Proposition 16.7 of [1] and the computation following it. The rate changes to $\sqrt{\kappa^2 + |\mathbf{c}|^2}$ for an admissible element and $\sqrt{\kappa^2 - \nu^2}$ for a pairing boost (a metric connection, not a gauge field), and the momentum sector behaves as under the positive form of Theorem 15.5 of [1]: exponents carried by Jordan blocks, imaginary for a compact element and for a boost below the threshold.

**Proposition 5.9** (What a gauge transformation removes)**.** *Let $b = b_l + b_p \in \mathfrak{l}_G \oplus \mathfrak{p}_G$ be constant. Some $g(\tau) \in O(3,3)$ carries $(b, G)$ to $(0, G')$ with $G'$ constant if and only if $[b, G] = 0$. For $b \in \mathfrak{p}_G$, $e^{b\tau} G e^{-b\tau} = e^{2b\tau} G$. The gauge-covariant remainder is $D_\tau G = [b, G] = 2b_p G$: along a worldline, $b_l$ is pure gauge and $b_p$ is not.*

*Proof.* $b^g = 0$ forces $g = e^{-b\tau} g_0$, and then $G' = g_0^{-1} e^{b\tau} G e^{-b\tau} g_0$ has derivative $g_0^{-1} e^{b\tau} [b, G] e^{-b\tau} g_0$. If $b$ anticommutes with $G$, then $Gf(b) = f(-b)G$ for every power series $f$. □

This is fixed. A connection on a line has no curvature (the non-abelian $[b_\mu, b_\nu]$ needs two directions; inherited); once the binding root is present, the invariant is $D_\tau G$, exactly the anticommuting part. One might expect boundedness to force $b \in \mathfrak{l}_G$. It does not, and the admissible case shows it.

**Theorem 5.10** (Compact screening)**.** *Let $b$ be admissible, as in (11), and $\kappa > 0$.*

*(i) If every solution is bounded, then $\mathbf{a} \cdot \mathbf{c} = 0$: the two triples rotate at equal rates, $|\mathbf{a} + \mathbf{c}| = |\mathbf{a} - \mathbf{c}|$.*

*(ii) Let $\mathbf{a} \cdot \mathbf{c} = 0$ and $\mathbf{a} \neq 0$. Then $\operatorname{spec} S = \{\pm i\omega_1, \pm i\omega_2, \pm i\omega_3\}$, where the $\omega_j$ are the roots of*

$$f(\omega) = \omega^3 + \kappa\omega^2 - (\kappa^2 + |\mathbf{a}|^2 + |\mathbf{c}|^2)\,\omega - \kappa(\kappa^2 + |\mathbf{c}|^2 - |\mathbf{a}|^2),$$

*and if $|\mathbf{a}|^2 \neq \kappa^2 + |\mathbf{c}|^2$, every solution is bounded, although $[b, G] = 2b_p G \neq 0$ when $\mathbf{c} \neq 0$.*

*(iii) If $\mathbf{a} \cdot \mathbf{c} = 0$, $\mathbf{c} \neq 0$ and $|\mathbf{a}|^2 = \kappa^2 + |\mathbf{c}|^2$, the vector $X_0 = (\mathbf{n}, (\mathbf{a} + \mathbf{c}) \times \mathbf{n}/\kappa)$, $\mathbf{n} = \mathbf{a} \times \mathbf{c}$, is static ($SX_0 = 0$) with $\eta(X_0, X_0) = -2|\mathbf{c}|^2|\mathbf{n}|^2/\kappa^2 \neq 0$, and some solutions drift linearly. For $\mathbf{a} = 0$ see Theorem 5.8.*

*Proof.* $S = b + \kappa G$ is real antisymmetric, since $b$ and $G$ are; so $S$ and $S^\dagger = -\eta S \eta = -b + \kappa G$ are normal with imaginary spectrum, and by Theorem 5.5 the motion is bounded if and only if $P_\mu \eta P_\mu = 0$ for the orthogonal spectral projectors $P_\mu$ of $S^\dagger$. (i) Then $\mathrm{Tr}(\eta (S^\dagger)^2) = \sum_\mu \mu^2 \,\mathrm{Tr}(P_\mu \eta P_\mu) = 0$. But $\{b, G\}$ is block-off-diagonal, $\mathrm{Tr}\,\eta = 0$ and $\mathrm{Tr}[\mathbf{v}]_\times^2 = -2|\mathbf{v}|^2$, so $\mathrm{Tr}(\eta (S^\dagger)^2) = \mathrm{Tr}(A + C)^2 - \mathrm{Tr}(A - C)^2 = -8\,\mathbf{a} \cdot \mathbf{c}$. (ii) Let $\rho$ be the half-turn about $\mathbf{a}$ and $\hat{\rho} = \mathrm{diag}(\rho, \rho)$. Since $\rho\mathbf{a} = \mathbf{a}$ and $\rho\mathbf{c} = -\mathbf{c}$, $\hat{\rho} b \hat{\rho} = \mathrm{diag}(A - C, A + C) = G^{-1} b G$, so $G' := G\hat{\rho}$ commutes with $b$ and $G$, hence with $S$ and $S^\dagger$. Moreover $G'^2 = -\mathbf{1}$ and $G'^\dagger = G'$, so, as for $G$ in Proposition 16.3 of [1], each eigenspace $\mathcal{V}_\pm$ of $G'$ (eigenvalue $\pm i$) is totally $h$-isotropic: $-i\,h(y, y') = h(G'y, y') = h(y, G'y') = i\,h(y, y')$ on $\mathcal{V}_+$. In the $(\mathbf{B}, \mathbf{E})$ blocks, $\mathcal{V}_+ = \{(\mathbf{x}, -i\rho\mathbf{x})\}$, and $S$ acts there as $\mathbf{x} \mapsto (\mathbf{a} + \mathbf{c}) \times \mathbf{x} + i\kappa\rho\mathbf{x}$; after a common rotation of both triples (which commutes with $G$ and $\eta$) to $\mathbf{a} = |\mathbf{a}|\mathbf{e}_3$, $\mathbf{c} = |\mathbf{c}|\mathbf{e}_1$, one finds $\det(i\omega - [\mathbf{a} + \mathbf{c}]_\times - i\kappa\rho) = -if(\omega)$. So $S$ has eigenvalues $i\omega_j$ on $\mathcal{V}_+$ and $-i\omega_j$ on $\mathcal{V}_- = \overline{\mathcal{V}_+}$; since $\eta$ exchanges $\mathcal{V}_\pm$, $S^\dagger$ has the same eigenvalues there. Every eigenspace of $S^\dagger$ splits along $\mathcal{V}_\pm$, and lies in one of them unless

$f(\omega) = f(-\omega) = 0$ for some $\omega$. But $\frac{1}{2}[f(\omega) + f(-\omega)] = \kappa(\omega^2 - \kappa^2 - |\mathbf{c}|^2 + |\mathbf{a}|^2)$ and $\frac{1}{2}[f(\omega) - f(-\omega)] = \omega(\omega^2 - \kappa^2 - |\mathbf{a}|^2 - |\mathbf{c}|^2)$, so a common root forces $\mathbf{a} = 0$, or $\omega = 0$ and $|\mathbf{a}|^2 = \kappa^2 + |\mathbf{c}|^2$. (iii) With $\mathbf{u} = \mathbf{a}+\mathbf{c}$, $\mathbf{v} = \mathbf{a}-\mathbf{c}$: $SX_0 = \big(\mathbf{u}\times\mathbf{n} - \mathbf{u}\times\mathbf{n},\ \mathbf{v}\times(\mathbf{u}\times\mathbf{n})/\kappa + \kappa\mathbf{n}\big) = 0$, because $\mathbf{v}\cdot\mathbf{n} = 0$ and $\mathbf{u}\cdot\mathbf{v} = |\mathbf{a}|^2 - |\mathbf{c}|^2 = \kappa^2$; and $\eta(X_0, X_0) = |\mathbf{n}|^2 - |\mathbf{u}|^2|\mathbf{n}|^2/\kappa^2$. So $\mathcal{E}_0(S^\dagger) = \eta\,\mathcal{E}_0(S)$ is not $\eta$-null, and Theorem 5.5 fails at $\mu = 0$. Exact checks: the identities symbolically, boundedness at four rational points (one with $\mathbf{a}, \mathbf{c}$ in general position), and the exceptional case at $(\kappa, |\mathbf{c}|, |\mathbf{a}|) = (3, 4, 5)$ and $(\frac{3}{5}, \frac{4}{5}, 1)$. □

The theorem is fixed, and it is the case that matters for a gauge field. Its mechanism is a second binding root $G' = G\hat{\rho}$ commuting with $S$. The family was found by the adversarial check (x4-N1: 500 of 500 random $\mathbf{a} \perp \mathbf{c}$ bounded, none of 500 generic compact $b$); part (i) explains the second count, and the exceptional ratio, of measure zero, was found here. At $\mathbf{c} = 0$ it gives the Zeeman lines of Theorem 5.6. For $\mathbf{c} \neq 0$ the screened frequencies cannot be removed along a worldline (Proposition 5.9), but they are $\kappa$ times functions of $|\mathbf{a}|/\kappa$ and $|\mathbf{c}|/\kappa$: screening supplies no scale.

**Proposition 5.11** (Gyroscopic screening by a boost)**.** *Let $b = \alpha L_3 + b_\Sigma$, where $b_\Sigma \in \mathfrak{p}_G$ is the boost $z \mapsto i\Sigma\bar{z}$ with $\Sigma = \left(\begin{smallmatrix} y_1 & y_2 & 0 \\ y_2 & -y_1 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, and let $r^2 = y_1^2 + y_2^2 > 0$. Then $[b, G] \neq 0$ and*

$$\chi_S(\lambda) = (\lambda^2 + \kappa^2)\big(\lambda^2 + (\kappa - \alpha)^2 - r^2\big)\big(\lambda^2 + (\kappa + \alpha)^2 - r^2\big).$$

*If $\alpha \neq 0$ and $r < \min(|\kappa - \alpha|, |\kappa + \alpha|)$, every solution is bounded; for $\alpha = 0$ (Theorem 5.8) and for $|\kappa - \alpha| < r < |\kappa + \alpha|$ some are not. Numerically, near bounded points at which the five conditions of the proof are independent, the set of $b \in \mathfrak{so}(3, 3)$ with all motion bounded is a 10-dimensional submanifold; at the tested points of $\mathfrak{l}_G$ and of this family the conditions are degenerate (Jacobian rank four), so these lie on its singular locus.*

*Proof.* On $\langle e_1, e_2, e_4, e_5\rangle$, $K_3 = GL_3$ is an $\eta$-antisymmetric involution commuting with $G$, $L_3$ and $b_\Sigma$, so its eigenplanes $\mathcal{U}_\pm$ are totally $\eta$-null; there $L_3 = \mp G$, $S = (\kappa \mp \alpha)G + b_\Sigma$ and $S^\dagger = (\kappa \pm \alpha)G - b_\Sigma$, each squaring to a negative multiple of $\mathbf{1}$ ($b_\Sigma^2 = r^2$, $\{b_\Sigma, G\} = 0$). For $\alpha \neq 0$ the planes carry different frequencies, so every eigenvector of $S^\dagger$ lies in $\mathcal{U}_+$, $\mathcal{U}_-$ or the $b$-free plane $\langle e_3, e_6\rangle$ and is $h$-isotropic; Theorem 5.5 applies (exact checks at $(\alpha, y_1, y_2) = (\frac{1}{2}, \frac{1}{4}, 0)$, $(\frac{1}{3}, \frac{1}{5}, \frac{1}{7})$, $\kappa = 1$). For simple frequencies Theorem 5.5(ii) imposes five real conditions (two for imaginary spectrum after the trace relation, three for isotropy); their finite-difference Jacobian has rank five at four least-squares solutions and at an exact compact screened point, and rank four (residual $\sim t^2$ along kernel directions) at three points of this family and two elliptic points of $\mathfrak{l}_G$. □

The family is fixed; the dimension statements are numerical. This is gyroscopic stabilisation of a saddle [55, 29]: the rotation detunes the two light-like planes of $K_3$ in which the boost acts, so $\eta(\pi, \pi) = 2\eta(\pi_+, \pi_-)$ averages to zero. Since $b_\Sigma$ is a boost, the family is not a gauge field. With Theorem 5.10 it corrects the working dictionary: an anticommuting part need not cause secular growth, and boundedness does not single out $\mathfrak{l}_G$; gauge removability does (Proposition 5.9).

**Proposition 5.12** (A connection alone does not bind)**.** *(i) Let $\kappa = 0$ and let $b \in \mathfrak{so}(3, 3)$ be any constant connection, $b = \nu\mathcal{K}$ included. Then $X = e^{-b\tau}Y$ gives $\mathcal{L} = \frac{m}{2}\eta(\dot{Y}, \dot{Y})$ and $\ddot{Y} = 0$. The generator $M = \left(\begin{smallmatrix} -b & \mathbf{1}/m \\ 0 & -b \end{smallmatrix}\right)$ has $\det(\lambda - M) = \chi_{-b}(\lambda)^2$ and is never semisimple: the motion is free drift, and there is no binding frequency.*

*(ii) The rescaling $(\tau, \kappa, b, m) \mapsto (\vartheta\tau, \kappa/\vartheta, b/\vartheta, \vartheta m)$ maps $M$ to $M/\vartheta$, solutions to solutions and the action $\int \mathcal{L}\,d\tau$ to itself, and leaves $\eta$, $G$, $\mathfrak{l}_G$, $\mathfrak{p}_G$ and $\mathfrak{so}(4)_\Delta$ unchanged. No value of $\kappa$, hence of $\ell = c/\kappa$, is selected, with or without a constant $b$.*

*(iii) $M(m) = TM(1)T^{-1}$ with $T = \mathrm{diag}(\mathbf{1}, m\mathbf{1})$: the spectrum of $M$ does not involve $m$, and the rescaling of (ii) keeps $m\kappa$, not $m/\kappa$, fixed.*

*Proof.* (i) $\dot{X} + bX = e^{-b\tau}\dot{Y}$. Write $-\operatorname{diag}(b,b) = S' + N'$ (Jordan–Chevalley). Then $M = S' + (N' + N)$ with $N = \left(\begin{smallmatrix} 0 & \mathbf{1}/m \\ 0 & 0 \end{smallmatrix}\right)$, and $N' + N \neq 0$ because $N$ is off-diagonal and $N'$ is not. (ii) and (iii) are direct. □

Part (i) is inherited (every connection on a line is pure gauge); its consequence is an obstruction: a Yang–Mills potential on a worldline, with no self-adjoint shift, produces free motion. Part (ii) is dimensional analysis, inherited; it restates [1] ("$\kappa$ is not fixed", end of Section 16; no ratio among the inertia constants and $m$ is fixed by the Clifford structure, Section 17.5). Part (iii) is fixed and elementary: nothing in the interior dynamics ties $m$ to $\kappa$, and $mc^2 = \hbar\kappa$, that is $\ell = \hbar/(mc)$, is a matching added from outside, never a result.

### 5.5 The split virial identity and the vacuum

**Proposition 5.13** (Split virial with a connection)**.** *For every constant $S$, and in particular for every $b \in \mathfrak{so}(3,3)$, $\frac{d}{d\tau}\eta(X,\pi) = \eta(\pi,\pi)/m$ along every solution.*

*Proof.* $\eta(\pi/m - SX, \pi) + \eta(X, S^\dagger\pi) = \eta(\pi,\pi)/m$; also checked in integral form along numerical solutions (DOP853, error below $10^{-8}$). □

This restates and re-verifies Theorem 16.1 of [1] (fixed there), whose metric-compatible connection is our $b$.

**Theorem 5.14** (Krein signatures and the zero-point energy)**.** *Suppose every solution is bounded, and let $\omega > 0$ have multiplicity $k_\omega$ in* spec $S$*. On the eigenspace $\mathcal{E}_{i\omega}(M)$, the energy form $v \mapsto \bar{v}^\top \mathcal{H} v$, where $\mathcal{H}$ is the Hessian of $H$, has signature $(k_\omega, k_\omega)$. Hence, with positive-norm quantisation of the normal modes, the zero-point energy $\sum_j (\pm)\frac{1}{2}\hbar\omega_j$ cancels frequency by frequency, and every level has a lower one.*

*Proof.* On $\mathcal{E}_{i\omega}(M)$, $\mathcal{H}v = -i\omega\mathbb{J}v$, so the energy form equals $\omega$ times the Krein form $-i\bar{v}^\top\mathbb{J}v$, which is nondegenerate there because eigenspaces of distinct imaginary eigenvalues are orthogonal for it [62]. The zero-momentum part of $\mathcal{E}_{i\omega}(M)$ has dimension $k_\omega$ (the multiplicity of $i\omega$ in $-S$), and $H$ vanishes on it (Lemma 5.4). Since $M$ is semisimple, $\dim \mathcal{E}_{i\omega}(M) = 2k_\omega$. A nondegenerate Hermitian form with a half-dimensional isotropic subspace has signature $(k,k)$, and a mode of energy sign $\pm$ quantises to $\pm\hbar\omega(n + \frac{1}{2})$. □

The theorem is fixed, though structural (any nondegenerate $\eta$); it was checked numerically (to $10^{-9}$) on the Zeeman triplet, on $w = (2, i, 0)$, on the gyroscopic family, on a compact screened connection and at three constituents. It extends Theorem 18.6 of [1] (zero-point energy 0 on $\mathbb{R}^{3,3}$ at $b = 0$) to every bounded connection: the cancellation, forced by the perfect-square form (9), survives the splitting line by line, and so does the absence of a ground state. On the coordinate subfields of that theorem (zero-point energy $\pm\frac{3}{2}\hbar\kappa$ or $\pm\frac{1}{2}\hbar\kappa$), $G$ does not preserve the subfield (the zero-momentum flow stays in it only on its intersection with its preimage under $G$, of dimension 0 or 2), and the theorem does not apply.

**Corollary 5.15** (No Fock vacuum with Jordan blocks)**.** *Suppose $M$ is not semisimple at a frequency $\omega \neq 0$, as for every nonzero $b \in \mathfrak{p}_G$. Then no linear symplectic change of variables writes the Hamiltonian on that block as a sum of signed oscillators, and there is no Fock vacuum annihilated by normal-mode lowering operators.*

*Proof.* A sum of signed oscillators generates a semisimple flow, and symplectic changes of variables act by similarity. □

This is an obstruction. A model block with the same Jordan structure (checked) is $H = \omega(q_1 p_2 - q_2 p_1) + \frac{1}{2}(q_1^2 + q_2^2)$, whose terms Poisson-commute [59]; that its quantisation has continuous spectrum is inherited and not re-derived.

### 5.6 Three constituents: the part of $\mathfrak{ps}_W$ that commutes with $G_3$

We use the generators $E_{6(k-1)+a} = \omega_6^{\otimes(k-1)} \otimes e_a \otimes \mathbf{1}^{\otimes(3-k)}$, $\omega_6 = e_1e_2\cdots e_6$, of equation (2) of [1] on $\mathbb{R}^{512}$, with the adjoint action of bivectors on the 18 vectors computed exactly, and set $G_3 = G \oplus G \oplus G$. We follow the numbering of Section 17 of [1]: $W = \{7, 13, 14, 15\}$, the colour directions are $\{10, 11, 12, 16, 17, 18\}$, $E = \mathrm{span}\{7, 8, 9, 13, 14, 15\}$, and $\mathrm{rot}(i, j) = e_{i'}e_{j'} - e_ie_j$ acts by $\frac{1}{2}\,\mathrm{ad}$.

**Proposition 5.16** (The binding-compatible algebra)**.** *The $E_A$ satisfy the relations of* $\mathrm{Cl}(9, 9)$*, $G_3$ is an anti-isometry with $G_3^2 = -\mathbf{1}$, and* $\dim(\mathrm{Cent}(G_3) \cap \mathfrak{so}(9, 9)) = 72$*. For four-dimensional $U \subset \mathrm{span}\{2, 3, 7, 8, 9, 13, 14, 15\}$, neither $\mathfrak{so}(U)$ nor $\mathfrak{so}(6)_C$ meets* $\mathrm{Cent}(G_3)$*, and their sum $\mathfrak{ps}_U$ meets it in dimension $\binom{d}{2}$, $d = \dim(U \cap E)$; for $U = W$ in $\mathfrak{so}(4)_\Delta = \mathrm{span}\{\mathrm{rot}(i, j) : i < j \in W\}$. The elements $T_1^\pm = \frac{1}{2}(\mathrm{rot}(13, 14) \pm \mathrm{rot}(7, 15))$, $T_2^\pm = \frac{1}{2}(\mathrm{rot}(7, 14) \mp \mathrm{rot}(13, 15))$ and $T_3^\pm = \frac{1}{2}(\mathrm{rot}(7, 13) \pm \mathrm{rot}(14, 15))$ satisfy $[T_i^+, T_j^-] = 0$ and $[T_1^\pm, T_2^\pm] = -T_3^\pm$ cyclically. The trace form is negative definite, so $\mathfrak{so}(4)_\Delta = \mathfrak{su}(2)_+ \oplus \mathfrak{su}(2)_-$ is compact. Each $T_3^\pm$ has eigenvalues $\pm\frac{i}{2}$, four times each, and 0, ten times.*

This restates and re-verifies Theorem 17.21 of [1] (fixed there), here for $U = W$, $\{3, 7, 8, 9\}$, $\{2, 3, 7, 8\}$ (dimensions $6, 3, 1$) and in `x4_check.py` for all 70 coordinate $U$. Each $\mathfrak{su}(2)$ factor locks a weak rotation of $W$ to the colour rotation of the partners $\{10, 16, 17, 18\}$; reading $\mathfrak{so}(4)_\Delta$ as "Yang and Mills' isospin, doubled" is interpretive, as after Theorem 17.22 of [1].

**Proposition 5.17** (Spectra on the eighteen-dimensional frame)**.** *Let $\kappa = 5$ and $m = 1$.*

*(i) For $b = \beta T_3^+$ with $\beta = 2$, $\chi_S = (\lambda^2 + 25)^5(\lambda^2 + 16)^2(\lambda^2 + 36)^2$: frequency $\kappa$ on the five complex singlet directions and $\kappa \mp \beta/2$ on the two $\mathfrak{su}(2)_+$ doublets, of weights $\pm\frac{1}{2}$.*

*(ii) For $b = \beta_+T_3^+ + \beta_-T_3^-$ with $(\beta_+, \beta_-) = (2, 4)$, the lines are $\kappa \pm \frac{1}{2}(\beta_+ \pm \beta_-) = 8, 6, 4, 2$, together with $\kappa$.*

*(iii) In both cases $\Xi = -G_3/(2\kappa m)$ solves the Sylvester equation, $M$ is semisimple, the Krein signatures are $(k, k)$, and the zero-point energy vanishes.*

*(iv) The weak-only rotation $R_{7,13}$ and the colour-only rotation $R_{10,16}$ give a non-semisimple $M$, hence secular growth.*

*(v) $b = R_{7,13} \pm R_{10,11} \in \mathfrak{ps}_W$ has $[b, G_3] \neq 0$, and every solution is bounded, for every $\kappa > 0$. At $\kappa = 5$ the frequencies are $\kappa$ (six pairs) and $|\omega_j| \approx 4.319, 5.050, 5.731$, with $\omega_j$ the roots of $\omega^3 + 5\omega^2 - 26\omega - 125$.*

*Proof.* (i)–(iv): exact characteristic polynomials and semisimplicity tests on $36 \times 36$ integer matrices; Krein signatures numerical. (v) $G_3$ maps the six-plane $\langle e_7, e_8, e_{13}; e_{10}, e_{11}, e_{16}\rangle$ as $G$ maps $\langle e_1, e_2, e_3; e_4, e_5, e_6\rangle$, and $b$ preserves it and vanishes on its $\eta$-orthogonal complement, where the motion is the $b = 0$ binding. On the six-plane $b$ is admissible, with unit rotation vectors $\mathbf{e}_2$ and $\pm\mathbf{e}_3$ for the two triples: equal rates, perpendicular axes, so $\mathbf{a} = \frac{1}{2}(\mathbf{e}_2 \pm \mathbf{e}_3) \perp \mathbf{c} = \frac{1}{2}(\mathbf{e}_2 \mp \mathbf{e}_3)$ and $|\mathbf{a}|^2 = |\mathbf{c}|^2 = \frac{1}{2} \neq \kappa^2 + \frac{1}{2}$. Theorem 5.10(ii) applies with $f$ as stated; $M$ is also exactly semisimple at $\kappa = 5$. □

This is fixed. Every constant $\mathfrak{so}(4)_\Delta$-valued connection is compact and commutes with $G_3$, so by Theorem 5.6(i) it keeps the interior bounded; the splitting follows the $\mathfrak{su}(2)$ weights, $\beta/\kappa$ is free, and by Proposition 5.7 the pattern can be removed along one worldline. The single-factor rotations tear the binding, but boundedness does not single out the locked diagonal: part (v), found by the adversarial check (x4-N2) and explained by Theorem 5.10, is bounded with $[b, G_3] \neq 0$ (the anti-locked $R_{7,13} + R_{10,16}$, $\mathbf{a} = 0$ on its six-plane, is not). What singles out $\mathfrak{so}(4)_\Delta$ is commutation with $G_3$ (Theorem 17.21 of [1]), that is, removability along a worldline (Proposition 5.9).

### 5.7 Quantum Yang–Mills facts, and the channels of the paper's vacuum

The only place left for $b$ to generate a scale is the quantum theory. We re-verify the facts used by the earlier session.

**Proposition 5.18** (Inherited facts)**.** *(i) For* $\mathfrak{su}(4)$ *with* $\mathrm{tr}(T^A T^B) = \frac{1}{2}\delta^{AB}$, $\sum_A T^A Z T^A = \frac{1}{2}(\mathrm{tr}(Z)\mathbf{1} - \frac{1}{4}Z)$ *for every* $4\times 4$ *matrix* $Z$*:* $\frac{15}{8} = C_F$ *on* $\mathbf{1}$, $-\frac{1}{8}$ *on traceless* $Z$*.*

*(ii) The channel factors* $T_1 \cdot T_2 = \frac{1}{2}[C_2(pair) - C_2(R_1) - C_2(R_2)]$ *are* $+\frac{3}{8}, -\frac{5}{8}$ *on* $\mathbf{10}, \mathbf{6} \subset \mathbf{4}\otimes\mathbf{4}$*;* $-\frac{15}{8}, +\frac{1}{8}$ *on* $\mathbf{1}, \mathbf{15} \subset \bar{\mathbf{4}}\otimes\mathbf{4}$*;* $+\frac{1}{4}, -\frac{3}{4}$ *on* $\mathbf{3}, \mathbf{1} \subset \mathbf{2}\otimes\mathbf{2}$ *of* $\mathfrak{su}(2)$ *(*$C_2 = \frac{15}{8}, \frac{9}{2}, \frac{5}{2}, 4$ *on* $\mathbf{4}, \mathbf{10}, \mathbf{6}, \mathbf{15}$*). Hence the Dirac bilinear* $\overline{(\mathbf{4},\mathbf{2},\mathbf{1})}\otimes(\mathbf{4},\mathbf{1},\mathbf{2}) = (\mathbf{1},\mathbf{2},\mathbf{2})\oplus(\mathbf{15},\mathbf{2},\mathbf{2})$ *has* $\mathfrak{su}(4)$ *factors* $-\frac{15}{8}, +\frac{1}{8}$*, and a Lorentz-scalar Majorana bilinear of* $(\mathbf{4},\mathbf{1},\mathbf{2})$*, symmetric in its internal indices, lies in* $\mathrm{Sym}^2(\mathbb{C}^4\otimes\mathbb{C}^2) = (\mathbf{10},\mathbf{1},\mathbf{3})\oplus(\mathbf{6},\mathbf{1},\mathbf{1})$ *(*$30+6$*), with factors* $(+\frac{3}{8}, +\frac{1}{4})$ *and* $(-\frac{5}{8}, -\frac{3}{4})$*.*

*(iii)* $\int_{k^2\le k^2_{\max}} \frac{d^4k}{(2\pi)^4}\frac{1}{k^2+M^2_{\mathrm{dyn}}} = \frac{1}{16\pi^2}\left[k^2_{\max} - M^2_{\mathrm{dyn}}\ln(1+k^2_{\max}/M^2_{\mathrm{dyn}})\right]$ *(Euclidean);* $f_{\mathrm{gap}}(x) = 1 - x^2\ln(1+x^{-2})$ *decreases strictly from* 1 *to* 0 *(*$f'_{\mathrm{gap}} < 0$ *since* $\ln(1+u) > u/(1+u)$*), so* $1 = \lambda_c f_{\mathrm{gap}}(x)$ *has a root* $x = M_{\mathrm{dyn}}/k_{\max} > 0$ *if and only if* $\lambda_c > 1$*; at* $\lambda_c = \frac{7}{4}$ *it is* $x = 0.5330234964\ldots$*, confirming the earlier* 0.533*.*

*(iv) The ladder critical coupling* $\pi/(3C_F)$ *[15, 38] is* $8\pi/45 \approx 0.5585$ *at* $C_F = \frac{15}{8}$*. For* $SU(2)$ *with* $N_f$ *Dirac doublets* $\beta_0 = (22-2N_f)/3$*,* 6 *at* $N_f = 2$ *[20, 46], and* $\Lambda = \mu e^{-8\pi^2/(\beta_0 g^2)}$ *is independent of* $\mu$*.* $SU(4)\to Sp(4)$ *(*$N_f = 2$*) leaves* $15-10 = 5$ *Goldstone directions [45].*

All of this is inherited [40, 53], and was computed exactly (generalised Gell-Mann basis, joint Casimir eigenspaces on $\mathrm{Sym}^2(\mathbb{C}^8)$), the root of (iii) with mpmath to 25 digits. The reading of (ii) is interpretive. Under the most-attractive-channel heuristic of one-boson exchange [48] (negative factor: attraction), with a common coupling for $\mathfrak{su}(4)$ and $\mathfrak{su}(2)_R$, which the paper itself adopts only as an assumption (Assumption 19.4 and the remark after Proposition 19.5 of [1]), the exchange favours $(\mathbf{1},\mathbf{2},\mathbf{2})$ and the colour-breaking $(\mathbf{6},\mathbf{1},\mathbf{1})$, and disfavours $(\mathbf{15},\mathbf{2},\mathbf{2})$, which the Georgi–Jarlskog factor $-3$ needs (Proposition 21.2 of [1]; [19]), and $(\mathbf{10},\mathbf{1},\mathbf{3})$, which carries the paper's Standard-Model vacuum (Section 20 of [1]). That vacuum is a minimum of a scalar potential, not a gauge-exchange condensate, so the heuristic addresses a different, dynamical scenario, in which a gauge-generated vacuum would point away from the paper's.

### 5.8 Synthesis: where the $b$-field enters

(a) *The binding frequency comes from the self-adjoint part, never from a connection* (fixed; obstructed for $b$ alone): $\kappa$, with $(\kappa G)^2 = -c^2/\ell^2$, sits in the tensor part; $b$ alone gives free drift (Theorem 5.2 and Proposition 5.12).

(b) *A commuting* $b$ *splits the binding frequency without supplying a scale* (fixed; Zeeman reading interpretive): lines $\kappa$, $\kappa \pm \sqrt{w\cdot\bar{w}}$, bounded for admissible $b$, removable along any worldline, $\beta/\kappa$ free (Theorem 5.6 and Proposition 5.7); at three constituents $\mathfrak{so}(4)_\Delta$, the part of the Pati–Salam algebra commuting with the binding root, gives $\mathfrak{su}(2)$-weight lines and is not the only bounded part (Proposition 5.17); the zero-point cancellation survives line by line (Theorem 5.14).

(c) *A purely anticommuting* $b$ *tears the binding* (obstructed; spectra fixed), with rates $\sqrt{\kappa^2+|\mathbf{c}|^2}$ (admissible) or $\sqrt{\kappa^2-\nu^2}$ (pairing boost) carried by Jordan blocks (Theorem 5.8 and Corollary 5.15). A mixed one may or may not: an admissible $b$ keeps every orbit bounded exactly at equal rates of the triples about non-opposite axes, outside $|\mathbf{a}|^2 = \kappa^2 + |\mathbf{c}|^2$ (Theorem 5.10); a boost can be screened gyroscopically (Proposition 5.11); neither gives a scale.

| statement | class | checks |
|---|---|---|
| Proposition 5.1: decomposition; roots on opposite sides | inherited; fixed (restated: Prop. 3.3, Thm 17.3) | d4-0, d4-1 |
| Theorem 5.2: (i) covariant factorisation; (ii) $b$ connection, $\kappa G$ tensor | fixed; inherited | d4-1, d4-2 |
| Proposition 5.3, (11): $\mathfrak{l}_G \oplus \mathfrak{p}_G$; admissible split; Lemma 5.4 | fixed; inherited (Lemma 5.4) | d4-3, d4-4 |
| Theorem 5.5: Roth–virial criterion | fixed | d4-11 |
| Theorem 5.6: Zeeman lines; bounded iff $e^{-b\tau}$ bounded; boost spirals | fixed | d4-5 to d4-7 |
| Proposition 5.7: moving frame; Zeeman reading | inherited; interpretive | d4-13 |
| Theorem 5.8: (i) never semisimple; (ii), (iv) spectra; (iii) pairing boost | obstructed; fixed; fixed (restated: Prop. 16.7) | d4-8 to d4-10 |
| Proposition 5.9: removable part $\mathfrak{l}_G$, invariant $D_\tau G$ | fixed | d4-13 |
| Theorem 5.10: admissible $b$ bounded iff $\mathbf{a} \perp \mathbf{c}$, $\mathbf{a} \neq 0$, $\lvert\mathbf{a}\rvert^2 \neq \kappa^2 + \lvert\mathbf{c}\rvert^2$ | fixed | d4-26 |
| Proposition 5.11: screening by a boost; dimension 10 | fixed (dimension numerical, regular points) | d4-12 |
| Proposition 5.12: (i) free drift; (ii) no scale; (iii) $m$ drops out | inherited; inherited (restated); fixed | d4-25 |
| Proposition 5.13; Theorem 5.14; Corollary 5.15 | fixed (restated: Thm 16.1); fixed; obstructed | d4-14 to d4-16 |
| Proposition 5.16: $\mathfrak{so}(4)_\Delta = \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ | fixed (restated: Thm 17.21) | d4-17, d4-18 |
| Proposition 5.17: (i)–(iv) spectra, single-factor tearing; (v) unlocked bounded $b$ ("only the locked diagonal" withdrawn) | fixed | d4-19, d4-27 |
| Proposition 5.18: Yang–Mills facts; channel, isospin and magnetic-type readings | inherited; interpretive | d4-20 to d4-24 |
| Conjecture 5.19: $\hbar\kappa = \Lambda_b$ ($mc^2 = \hbar\kappa$ a separate assumption) | open | — |

Table 6: Claims of Section 5, their classes, and the checks printed by `d4_bfield.py`; each is re-checked independently by `x4_check.py`, whose findings x4-N1 to x4-N4 are incorporated in Theorems 5.6 and 5.10 and Propositions 5.11 and 5.17. "Restated": fixed in [1], restated and re-verified here.

(d) *The only role left for $b$ in generating a scale is quantum* (open). The arena leaves $\ell$ free (Proposition 5.12) and classical four-dimensional Yang–Mills theory has no dimensionful parameter; a scale arises only by dimensional transmutation, $\Lambda_b = \mu\, e^{-8\pi^2/(\beta_0 g^2)}$ [8, 20].

**Conjecture 5.19** (A transmuted binding length)**.** *$\ell = \hbar c/\Lambda_b$ for the scale $\Lambda_b$ of a Yang–Mills $b$-field, that is, $\hbar\kappa = \Lambda_b$.*

Reading the rest energy $mc^2$ as $\hbar\kappa$ is a separate assumption, not part of the conjecture: the spectrum of $M$ does not involve $m$, and the rescaling of Proposition 5.12 keeps $m\kappa$, not $m/\kappa$, fixed. The conjecture is open: nothing in the arena derives it. A mechanism would need $\kappa$ to be dynamical, as the rotor rate of Section 17 of [1] is, and coupled to a quantum gauge sector with a fixed $\beta_0$ and matching coupling; even then, by Proposition 5.18(ii), the most attractive channels are not those of the paper's vacuum.

Corrected so that it is true, the user's sentence reads as follows. In $E = mc^2$, read through the added identification $mc^2 = \hbar\kappa$, the symmetric part is the self-adjoint binding shift $\kappa G$, with square $-c^2/\ell^2$. The Yang–Mills $b$-field is the antisymmetric, transport part of the same shift, and it is compact. Classically it generates neither $m$ nor $\kappa$: when it commutes with $G$ it splits the interior rate $\kappa$ into charge-labelled lines; a purely anticommuting $b$ tears the binding, and a mixed one may or may not (Theorem 5.10 and Proposition 5.11). Whether its quantum scale is the $\ell$ that the arena leaves free is open.

# 6 Synthesis: the hypothesis, sentence by sentence

We now return to the hypothesis of Section 1 and test each of its clauses against what Sections 2 to 5 establish. The verdicts are summarised in Table 7. Throughout, the relation $E = mc^2$ itself is inherited; what is at stake is only how the arena realises its three ingredients.

## 6.1 "The energy is one undivided entity, and it is six-dimensional"

This clause survives, in the field-strength reading of $\mathbb{R}^{3,3}$. Among quadratic forms on the six frame components, exactly one is invariant under the diagonal rotations and under duality, and it is positive: $E_6 = \frac{1}{2}|z|^2 = \frac{1}{2}(|\mathbf{B}|^2 + |\mathbf{E}|^2)$ (Section 2.1; this restates Theorem 17.11(iii) of [1]). Its symmetry group is $U(3)$, so the energy is "unitary" in a literal sense, but only as a pointwise quadratic form: $U(3)$ is not a symmetry of Maxwell's equations on the frame, nor of the split-form interior. In the closure of the force law the undivided energy of the *momentum* field, $E_6(\pi) = \frac{1}{2}|p|^2$, is a Casimir, conserved by every Hamiltonian, together with its rest energy $E_0(\pi)$ and its Poynting vector $\mathbf{S}(\pi)$ (Section 4.1). The energy of the *frame* field is not conserved. The other reading of "six-dimensional", three temporal energies forming a vector in $V^+$, fails: splitting that vector lowers the interior mass, $m_4^2c^4 = \mu^2c^4 - |\mathcal{E}_\perp|^2$ (Section 2.7).

## 6.2 "Mass emerges through an internal differentiation of that unity"

This clause survives as an identity, not as a process. With no interaction and no second field, the undivided energy of a field point splits as

$$E_6^2 = E_0^2 + |\mathbf{S}|^2, \qquad E_0 = \tfrac{1}{2}|q| = \tfrac{1}{2}|z \cdot z|,$$

the complex Lagrange identity (Section 2.3). In SI units it reads $u^2 = c^2|\mathbf{g}|^2 + u_0^2$, the field-point form of $E^2 = (pc)^2 + (mc^2)^2$, with rest-mass density $u_0/c^2$. The rest part $E_0$ is the complete invariant of Lorentz transformations together with duality, the Lorentz scalar of the stress tensor whose 00 component is $E_6$ (Rainich), and the minimum of $E_6$ over boosts, attained exactly in the rest frame (Section 2.4). All of this is inherited classical electrodynamics written in the arena's variables. The arena adds which forms play which role: $E_6$ belongs to the duality class of Theorem 17.11 of [1], $E_0$ to the Lorentz class, and $E = mc^2$ is the locus, the rest frame, where the two coincide.

The differentiation does not happen in time. In the closure the split $E_6^2 = E_0^2 + |\mathbf{S}|^2$ of the momentum field is fixed by the initial data, because all three terms are Casimirs. What moves is the duality angle and the rotor rate, and the conserved total $H = \frac{1}{2}I_0\kappa^2 + U(\theta)$ exchanges energy between a rate part and a Lorentz-invariant frame part (Section 4.3). The mass of a whole is not produced by interference either: two disjoint counter-propagating null pulses have $q \equiv 0$ everywhere and $M = U > 0$ (Section 2.6).

## 6.3 "$c^2$ is the symmetric part"

This clause survives in one precise form. The constant $c$ is the unique isotropic conversion between the temporal and the spatial triple, forced by Schur's lemma (Theorem 3.2 of [1]). It is a conversion, not a symmetry, and its weight in $E_6$ is a unit convention (Section 2.5). The form in which the clause is exact is the decomposition of a shift operator into its two summands (Theorem 17.3 of [1]):

$$\mathrm{End}(V) = \mathfrak{so}(3,3) \oplus \{X : X^\dagger = X\}, \qquad \text{shift} = b + \kappa G, \qquad (\kappa G)^2 = -\frac{c^2}{\ell^2}\,\mathbf{1}.$$

The $\eta$-self-adjoint summand is the symmetric part: it is the summand in which factorisation holds, the one that binds in neutral signature, and the one that carries $c$, through $\kappa = c/\ell$.

The antisymmetric summand is a connection. Reading the first as "$c^2$" and the second as "$b$" is interpretive, but the decomposition, and the fact that only the first summand binds, are fixed by the paper.

### 6.4 "$m$ is what breaks the symmetry"

This clause is true for the right symmetries and false for the ones it is usually read about (Section 3).

1. A nonzero rest energy breaks duality. The duality $U(1)$ generated by $G$ acts on the Lorentz invariant by $q \mapsto e^{2i\theta}q$, so a field with $E_0 \neq 0$ keeps only $\{\pm\mathbf{1}\}$, and $\arg q$ labels the broken directions. At a null field point, duality joins the little group, which becomes $\mathfrak{e}(2)$ with helicity as its compact generator. This is inherited, pulled back to the frame.

2. For Dirac matter the mass breaks the chiral and Pauli–Gürsey symmetries, $\mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2) \to \mathfrak{u}(1)_J$, with three Goldstone directions, one of them the pseudoscalar mass angle (Theorem 18.5 of [1], re-verified).

3. The mass breaks no spacetime symmetry, not even the six-dimensional one. The Dirac mass form $T = e_1e_2e_3$ is invariant under all of $\mathrm{Spin}(3,3)$ and is the only such kernel (Corollary 18.4 of [1]). Within $\mathfrak{so}(3,3)$ it is the kinetic term, where $c$ enters, that cuts the symmetry down to the interior Lorentz algebra and $J$. In this sense the hypothesis is inverted.

4. In the closure the rest energy of the momentum field is exactly the amplitude of the potential for the duality angle, $R_0 = 2|\nu_1 + i\nu_2|\, E_0(\pi)$, with $(\nu_1, \nu_2)$ the Lorentz-class inertia coefficients of Section 2. A null (massless) momentum field leaves the angle flat and the rate constant; a massive one makes it a pendulum whose squared small-oscillation frequency is $4|\nu_1 + i\nu_2|E_0(\pi)/(mI_0)$ (Section 4.2). This restates Theorem 17.13 and Corollary 17.14 of [1]. Reading the angle as a pseudo-Goldstone boson of duality is interpretive: the breaking is explicit, carried by the Lorentz-class inertia, and every pendulum has this shape.

### 6.5 "$m$ is generated from the $b$-particle of Yang–Mills"

Classically, this clause fails, and the failure is sharp (Section 5).

1. The binding rate is never a gauge potential. For every connection $b$ the field equation factorises covariantly, $(D - \kappa G)(D + \kappa G)X = 0$ with $D = \partial_\tau + b$; $\kappa G$ transforms as a tensor, $b$ as a connection. Without $\kappa$, a constant $b$ is pure gauge along the worldline: no rest frequency, no mass scale.

2. An admissible Yang–Mills field is compact, since no gauge boson is a boost (Corollary 2.4 of [1]). Its part commuting with $G$ splits the rest frequency into charge-labelled lines $\kappa$, $\kappa \pm \beta$, removable along a single worldline by a rotating frame (Larmor's theorem). This is a spectrum without a scale, since $\beta/\kappa$ is free. A purely anticommuting part produces Jordan blocks and secular growth. A mixed connection keeps every orbit bounded exactly when it rotates the temporal and the spatial triple at equal rates about axes that are not opposite, outside one exceptional ratio of the rates to $\kappa$ (Theorem 5.10).

3. At three constituents the part of the Pati–Salam algebra commuting with the binding root is $\mathfrak{so}(4)_\Delta \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ (Theorem 17.21 of [1]): two copies of Yang and Mills' isospin algebra. It splits the rest frequency by $\mathfrak{su}(2)$ weights and supplies no scale.

4. In the selected vacuum the order is the reverse of the hypothesis. The Pati–Salam vector bosons acquire mass by eating nine Goldstone directions (Theorem 22.2 of [1]), rather than generating the mass of anything.

| clause | what the arena gives | verdict |
|---|---|---|
| energy undivided, six-dimensional | $E_6 = \frac{1}{2}(\|\mathbf{B}\|^2 + \|\mathbf{E}\|^2)$, unique, positive, symmetry $U(3)$ pointwise; Casimir of the momentum field | holds (inherited / restates [1]) |
| mass from internal differentiation | $E_6^2 = E_0^2 + \|\mathbf{S}\|^2$, $E_0 = \min_{\text{boosts}} E_6$; split fixed by initial data in the closure | holds as an identity, not as a process |
| $c^2$ the symmetric part | $c$ a Schur conversion; $\kappa G$ the $\eta$-self-adjoint summand of the shift, the only one that binds | holds in the shift decomposition (interpretive reading of a fixed result) |
| $m$ breaks the symmetry | breaks duality and the chiral algebra; breaks neither Lorentz nor $\mathrm{Spin}(3,3)$ | true for internal symmetries, false for spacetime ones |
| $m$ generated from the $b$-field | $b$ is the connection summand: splits or tears the binding, never creates it; vector bosons are massive by eating Goldstones | fails classically; scale by transmutation open |

Table 7: The hypothesis against the arena.

5. At the quantum level, one-gauge-boson exchange favours the Dirac singlet channel $(\mathbf{1},\mathbf{2},\mathbf{2})$ and the colour-breaking $(\mathbf{6},\mathbf{1},\mathbf{1})$, and disfavours both the $(\mathbf{15},\mathbf{2},\mathbf{2})$ needed for the Georgi–Jarlskog factor and the $(\mathbf{10},\mathbf{1},\mathbf{3})$ that the paper's vacuum uses. This is inherited group theory applied to the arena's representations.

The one role left open for the $b$-field is the scale. The algebra fixes $c$ and leaves $\ell$, hence $\kappa$, free; classical Yang–Mills theory in four dimensions has no scale either, but its quantum theory generates one by dimensional transmutation, $\Lambda_b = \mu \exp[-8\pi^2/(\beta_0 g^2(\mu))]$. The identification $\hbar\kappa = \Lambda_b$, that is $\ell = \hbar c/\Lambda_b$, is a conjecture, not derived. Even granted, reading the rate as a mass, $mc^2 = \hbar\kappa$, is a further assumption: in the interior dynamics $m$ and $\kappa$ are independent inputs, the spectrum does not depend on $m$, and the classical rescaling invariant is $m\kappa$.

### 6.6 Particles, antiparticles, the vacuum and the Goldstone bosons

The original questions of this investigation have the following answers in the arena (Sections 3.4 to 3.6). For Dirac matter at rest the flow is a rotation generated by the six-dimensional mass form itself, $\Psi(\lambda) = e^{-m\lambda T}\Psi_0$, with rate equal to the mass. Particle and antiparticle are its two senses, and charge conjugation, the surplus plane, exchanges them. The binding rotation of the frame also has two senses, but parity exchanges them, not charge conjugation, so its rate $\kappa = c/\ell$ is a conversion rate and its two senses are not a particle–antiparticle pair. The Fock vacuum of one momentum is unique and neutral, with energy $-2E$ ($-2mc^2$ at rest). The frame has no ground state, its spectrum being $\hbar\kappa\,\mathbb{Z}$, but its zero-point energy cancels in neutral signature, and this cancellation survives, frequency by frequency, every connection whose motion stays bounded (Section 5). Goldstone directions: three at one constituent in the Dirac phase; nine at three constituents, all eaten, with no Majoron because $[J,\Phi_0] = [J\Gamma,\Phi_0] = 2[t_3^R,\Phi_0]$; and one axion-like direction of charge $1-\Gamma$ at the electroweak stage, which a realistic potential must lift explicitly.

### 6.7 The hypothesis, restated so that the algebra supports it

Collecting the verdicts, the strongest statement of the hypothesis that the arena supports is the following. *The undivided energy is the duality-invariant, unitary energy* $\frac{1}{2}(|\mathbf{B}|^2+|\mathbf{E}|^2)$ *of the six field components. The constant c is the unique isotropic conversion between times and lengths, and it enters the dynamics through the η-self-adjoint part of the shift, the binding root. The mass is the Lorentz-and-duality invariant* $\frac{1}{2}|q|$*, the minimum of the energy over boosts, and* $E = mc^2$

*is the rest frame, where the two coincide. What the mass breaks is duality and, for matter, chirality, never Lorentz invariance or* $\mathrm{Spin}(3,3)$*. The Yang–Mills b-field is the antisymmetric part of the same shift: it can split or destabilise the binding, but it cannot create it, and the only role left for it in generating mass is to supply, by dimensional transmutation, the one scale* $\ell$ *that the algebra leaves free.*

### 6.8 Open problems

1. A mechanism fixing $\ell$, and with it $\kappa$, from the arena or from the quantum dynamics of a $b$-field, so that $\hbar\kappa = \Lambda_b$ becomes a derivation; and a reason, absent so far, to identify the interior rate with a particle mass.

2. A dynamical process in which rest energy is produced rather than carried: in the closure the rest energy of the momentum field is a Casimir, while that of the frame field drifts without bound.

3. A common Lorentz action for the two readings of $\mathbb{R}^{3,3}$: the field-strength reading, where $E_6$ and $E_0$ live, and the interior reading, where the Dirac mass form lives, exclude each other on the same six directions (Proposition 12.6 of [1]).

4. The classification of compact connections that keep the binding bounded, beyond the commuting ones and the equal-rate family found here.

## 7 Duplication: from $\mathrm{Cl}(3,3)$ to $\mathrm{Cl}(6,6)$

The author's second remark asks which physical and mathematical process makes a unitary energy "result in a duplication", from $\mathrm{Cl}(3,3)$ to $\mathrm{Cl}(6,6)$. A duplication acts on four levels at once: on the vector space, $V \to V_2 := V \oplus V$; on the algebra, $\mathrm{Cl}(V) \to \mathrm{Cl}(V_2) = \mathrm{Cl}(6,6)$; on the spinor module, $S \to S_2 = S \otimes S$; and on the energy, $E_6 \to E_6 \oplus E_6$. This section takes the four steps in turn, asks which are forced and which are chosen, and records what the duplicate can do that one copy cannot. The answers, with their classes, are these. The step on the algebra is forced: it is the graded tensor product, its Klein factor is the Jordan–Wigner sign, and $S_2$ is the Fock space of six fermionic modes (inherited; Section 7.1). The step on the energy is a choice within a three-sphere of energy-preserving, Lorentz-covariant duplications, which is the algebra of a lossless beam splitter. Only a circle in it, the copy rotation, respects $\eta$, and the two notions of rest energy that coincide on one copy separate on two (fixed; Section 7.3). The duplicate carries structures one copy cannot carry: an $\eta$-isometric complex structure, a compact, Lorentz-invariant copy charge, the copy algebra $\mathfrak{so}(2,\mathbb{C})$, a closure that runs at constant rate with massive copies, and a non-rigid, Lorentz-invariant vacuum (fixed, partly restated from [1]; Sections 7.4 to 7.7). Finally, the copy rotation is the only admissible $b$-field between two binding frames that commutes with the binding root and the rotations. When the two rates differ it cannot be removed, and it confines a frame that has no rate of its own; it creates no rate where both frames are free (fixed, with the mixing formulas inherited; Section 7.8). Every finite-dimensional statement was checked by `verify/d5_duplication_math.py`; identifiers d5-$n$ refer to its `PASS` lines (Table 10).

**Notation.** $V_2 = V \oplus V$ has the basis $e_1, \dots, e_6$ of copy 1 followed by that of copy 2, the metric $\eta_2 = \mathbf{1}_2 \otimes \eta$, and the binding root $G_2 = \mathbf{1}_2 \otimes G$ of both copies (as $G_3$ in Theorem 17.19 of [1]). $\varepsilon = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $\sigma_x$, $\sigma_z$ act on the copy index, and $C_v := \varepsilon \otimes \mathbf{1}_6$ is the copy rotation on vectors. A pair of field strengths is $(z_1, z_2) \in \mathbb{C}^3 \oplus \mathbb{C}^3$, with $q_2 := q(z_1) + q(z_2)$ and duplicated energy $E_6(z_1) + E_6(z_2)$. On $S_2 = \mathbb{R}^8 \otimes \mathbb{R}^8$ the tower of equation (2) of [1] gives $e_a^{(1)} = e_a \otimes \mathbf{1}$, $e_a^{(2)} = \omega \otimes e_a$, $\omega = e_1 \cdots e_6$, and $E_{6(k-1)+a} = e_a^{(k)}$. Statements about $V_2$ are pointwise linear

algebra and apply to whichever field is placed there. Where the field matters we name it: the frame field $X$, or the momentum field $\pi = \eta^{-1}p$ of the closure, to which the Casimirs belong (Section 17 of [1], Section 4). Rates $(\kappa, \omega)$ are rates, and reading a rate as a rest energy through $mc^2 = \hbar\kappa$ is an added identification.

### 7.1 The duplicated algebra: graded tensor product and Klein factor

**Proposition 7.1** (The duplicated algebra)**.** *(i)* $\omega^2 = \mathbf{1}$, $\omega^\top = \omega$, $\operatorname{Tr}\omega = 0$, *and* $\omega$ *anticommutes with every* $e_a$. *The twelve matrices* $e_a^{(k)}$ *satisfy the Clifford relations of* $\mathbb{R}^{6,6}$. *Their* 4096 *blades are signed permutation matrices, and all except* $\mathbf{1}$ *are traceless, so* $\mathrm{Cl}(6,6) \cong M_{64}(\mathbb{R})$.

*(ii) For blades* $e_I, e_J$ *of* $\mathrm{Cl}(3,3)$, $e_I^{(1)}e_J^{(2)} = (e_I\,\omega^{|J|}) \otimes e_J$ *and* $e_J^{(2)}e_I^{(1)} = (-1)^{|I||J|}e_I^{(1)}e_J^{(2)}$. *Hence* $a \hat{\otimes} b \mapsto a\,\omega^{|b|} \otimes b$ *is an algebra isomorphism* $\mathrm{Cl}(3,3)\hat{\otimes}\mathrm{Cl}(3,3) \to \mathrm{Cl}(6,6)$, *where* $\hat{\otimes}$ *is the graded tensor product,* $(a\hat{\otimes}b)(c\hat{\otimes}d) = (-1)^{|b||c|}ac \hat{\otimes} bd$.

*(iii) The Klein factor is forced. In* $M_{64}(\mathbb{R})$ *the elements commuting with every* $e_a \otimes \mathbf{1}$ *form* $\mathbf{1} \otimes M_8(\mathbb{R})$, *and those anticommuting with every* $e_a \otimes \mathbf{1}$ *form* $\omega \otimes M_8(\mathbb{R})$. *Any realisation of* $\mathrm{Cl}(V_2)$ *on* $\mathbb{R}^{64}$ *that extends* $e_a \mapsto e_a \otimes \mathbf{1}$ *is therefore conjugate to the tower by an element of* $\mathbf{1} \otimes GL(8,\mathbb{R})$.

*Proof.* (i) $\omega^2 = (-1)^{15}(-1)^3\mathbf{1}$ by the transpose rule (4) of [1], and $\omega$ has even grade in an even dimension. The relations follow from $(\omega \otimes e_a)(\omega \otimes e_b) = \mathbf{1} \otimes e_a e_b$ and $(e_a \otimes \mathbf{1})(\omega \otimes e_b) = -(\omega \otimes e_b)(e_a \otimes \mathbf{1})$. A product of signed permutation matrices is one, and if $I \neq J$ then $e_I^\top e_J = \pm e_{I\triangle J}$ is traceless, so the blades are orthogonal for the trace form; $4096 = 64^2$. (ii) $e_J^{(2)} = \omega^{|J|} \otimes e_J$, and moving the $|J|$ copy-2 generators past the $|I|$ copy-1 generators costs $(-1)^{|I||J|}$. (iii) By (3) of [1] a blade $e_I^{(1)}e_J^{(2)}$ commutes with $e_a^{(1)}$ exactly when $|I| + |J| - [a \in I]$ is even. For all six $a$ this forces either $I = \emptyset$ and $|J|$ even, or $I = \{1,\dots,6\}$ and $|J|$ odd: 64 blades, each $\pm\mathbf{1} \otimes e_J$. The anticommuting case is the same with the parities reversed. If $e'_a$ anticommutes with all $e_b \otimes \mathbf{1}$, then $e'_a = \omega \otimes f_a$, and $(\omega \otimes f_a)(\omega \otimes f_b) = \mathbf{1} \otimes f_a f_b$, so the $f_a$ satisfy the relations of $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$. By the Skolem–Noether theorem $f_a = u e_a u^{-1}$, and conjugation by $\mathbf{1} \otimes u$ fixes $e_a \otimes \mathbf{1}$ and $\omega \otimes \mathbf{1}$. □

Everything here is inherited: it is the exponential law $\mathrm{Cl}(V \oplus W) \cong \mathrm{Cl}(V)\hat{\otimes}\mathrm{Cl}(W)$ [7, 35], and (i) restates equation (2) of [1]. The arena contributes only the fixed realisation (d5-1a to d5-1d). The duplication of the algebra is not a choice; given copy 1, copy 2 is determined up to conjugation.

**Proposition 7.2** (Fock space and Jordan–Wigner strings)**.** *Put* $n_i^\pm = \frac{1}{2}(e_i \pm e_{i+3})$, $i = 1,2,3$. *Then* $\{n_i^\pm, n_j^\pm\} = 0$ *and* $\{n_i^+, n_j^-\} = \delta_{ij}$, *so* $\mathrm{Cl}(3,3)$ *is the real algebra of canonical anticommutation relations of three fermionic modes and* $S = \mathbb{R}^8$ *is their Fock space. The occupation numbers are* $N_i = n_i^+ n_i^- = \frac{1}{2}(\mathbf{1} - \omega_i)$, *with* $\omega_i = e_i e_{i+3}$ *the pole operators. The eight eigenlines of Proposition 2.5(i) of [1] are therefore the eight Fock states, and* $(-1)^{N_1+N_2+N_3} = -\omega$. *At* $n = 2$ *the tower supplies six modes. The modes of copy* 2 *are* $\omega \otimes n_i^\pm$, *each carrying the string* $-(-1)^{N^{(1)}}$ *of copy* 1. *The total number has spectrum* $0,\dots,6$ *with multiplicities* $1, 6, 15, 20, 15, 6, 1$, *and* $(-1)^N$ *is the volume element of* $\mathrm{Cl}(6,6)$.

Inherited. In split signature the vectors $e_i \pm e_{i+3}$ span two maximal isotropic subspaces, and the spinor module is the exterior algebra of one of them [7]. The string is the Jordan–Wigner transformation [26], and the Klein factor is the device that makes the fields of two subsystems anticommute [30]. On the level of states, the mathematical process behind "$\mathrm{Cl}(3,3) \to \mathrm{Cl}(6,6)$" is the second-quantisation rule $\mathrm{Fock}(\mathcal{H} \oplus \mathcal{H}) = \mathrm{Fock}(\mathcal{H}) \otimes \mathrm{Fock}(\mathcal{H})$ (d5-1e).

## 7.2 Centre-of-mass and relative halves

**Proposition 7.3** (Two halves, and the exchange)**.** *Let* $f_a = (e_a^{(1)} + e_a^{(2)})/\sqrt{2}$ *and* $g_a = (e_a^{(1)} - e_a^{(2)})/\sqrt{2}$.

*(i)* $\{f_a, f_b\} = \{g_a, g_b\} = 2\eta_{ab}$ *and* $\{f_a, g_b\} = 0$ *for all* $a, b$, *so* $\mathrm{Cl}(6,6) = \mathrm{Cl}(f)\hat{\otimes}\mathrm{Cl}(g)$*: two mutually anticommuting copies of* $\mathrm{Cl}(3,3)$.

*(ii) The copy rotation* $C_b = \frac{1}{2}\sum_a \eta_{aa} e_a^{(1)} e_a^{(2)}$ *acts on vectors by* $e_a^{(1)} \mapsto -e_a^{(2)}$, $e_a^{(2)} \mapsto e_a^{(1)}$*; it is a sum of six commuting terms of square* $-\frac{1}{4}$, *so* $\exp(2\pi C_b) = \mathbf{1}$. *Conjugation by* $\exp(\frac{\pi}{4}C_b)$ *carries* $e_a^{(1)}$ *to* $g_a$ *and* $e_a^{(2)}$ *to* $f_a$, *and* $C_b = -\frac{1}{2}\sum_a \eta_{aa} f_a g_a$. *The split into halves is the split into constituents turned by an eighth of a turn of the copy rotation, which rotates one half into the other.*

*(iii) Let* $\omega = (-1)^{\mathsf{p}}$ *on the basis of equation (1) of [1], in which* $\omega$ *is diagonal, let* $\tau(x\otimes y) = y\otimes x$, *and let* $D = (-1)^{\mathsf{p}\otimes\mathsf{p}}$. *The Koszul swap* $\tau_K = D\tau$ *satisfies* $\tau_K e_a^{(1)} \tau_K = e_a^{(2)}$, $\tau_K^2 = \mathbf{1}$, $\mathrm{Tr}\,\tau_K = 0$, *and* $\tau_K = g_1 g_2 \cdots g_6$. *So the exchange commutes with every* $f_a$ *and anticommutes with every* $g_a$. *The naive swap* $\tau$ *does not implement the exchange.*

*(iv) The summed clock is* $e_1^{(1)} + e_1^{(2)} = \sqrt{2} f_1$. *Every diagonal bivector satisfies* $\sum_k e_a^{(k)} e_b^{(k)} = f_a f_b + g_a g_b$*; this covers the diagonal interior and binding Lorentz generators and the summed charge* $J^{(1)} + J^{(2)} = f_2 f_3 + g_2 g_3$, *which act on both halves alike with no cross term.* $G_2$ *preserves both halves. The copy rotations* $C_b$ *and* $M_{12}$ *(the copy rotation of the interior four-planes, Section 10.1 of [1]) are pure cross terms and anticommute with* $\tau_K$.

*Proof.* (i) and (iv) are expansions, using $e_a^{(1)} e_a^{(2)} = -f_a g_a$. (ii) By (3) of [1], $[C_b, e_b^{(1)}] = -e_b^{(2)}$. The six terms $\frac{1}{2}\eta_{aa} e_a^{(1)} e_a^{(2)}$ share no index, so they commute, and each squares to $-\frac{1}{4}$; hence $\exp(\theta C_b) = \prod_a (\cos\frac{\theta}{2} + \sin\frac{\theta}{2}\,\eta_{aa} e_a^{(1)} e_a^{(2)})$, which is $(-1)^6\mathbf{1}$ at $\theta = 2\pi$. On vectors $\exp(\theta\,\mathrm{ad}\,C_b)$ turns $e_a^{(1)}$ into $\cos\theta\, e_a^{(1)} - \sin\theta\, e_a^{(2)}$, which is $g_a$ at $\theta = \frac{\pi}{4}$. (iii) The swap of the copies is the reflection of $V_2$ in the span of the $g_a$, and the product $g_1 \cdots g_6$ of an even number of unit vectors implements it by conjugation. The Koszul swap implements it too, by the sign rule. The commutant of $\mathrm{Cl}(6,6) = M_{64}(\mathbb{R})$ is $\mathbb{R}$, so the two agree up to a scalar, which is $+1$. $\mathrm{Tr}\,\tau_K$ counts $\pm 1$ over the diagonal states $x_i \otimes x_i$ with sign $(-1)^{\mathsf{p}_i}$, and $\omega$ is traceless. □

Inherited: centre-of-mass and relative coordinates, and the graded swap, which is the exchange operator of two fermionic subsystems. The identity $\tau_K = g_1 \cdots g_6$, with its sign, and the list in (iv) are facts of the realisation (d5-1f, d5-2a to d5-2c). Because $\tau_K$ anticommutes with the copy rotation, exchanging the constituents reverses the copy charge, as exchanging two real vector fields $W^1, W^2$ exchanges $W^{\pm}$ (inherited). The halves are not invariant under the copy rotation. Only the exchange selects them, as its two eigenspaces on vectors.

## 7.3 Duplicating the undivided energy

**Proposition 7.4** (Symmetry of the duplicated energy)**.** *(i) The linear maps of* $V_2$ *that commute with the diagonal rotations and with* $G_2$ *form* $\mathfrak{gl}(2)\otimes \mathrm{span}\{\mathbf{1}, G\} \cong \mathfrak{gl}(2,\mathbb{C})$. *The infinitesimal symmetries of* $E_6 \oplus E_6$ *among them form* $\mathfrak{u}(2) = \mathrm{span}\{C_v,\ \mathbf{1}\otimes G,\ \sigma_z\otimes G,\ \sigma_x\otimes G\}$. *Those in* $\mathfrak{so}(6,6)$ *form* $\mathfrak{so}(2,\mathbb{C}) = \mathrm{span}\{C_v,\ \varepsilon\otimes G\}$, *and those in both form* $\mathfrak{so}(2) = \mathbb{R}\,C_v$. *The overall duality* $\mathbf{1}\otimes G = G_2$, *the relative duality* $\sigma_z \otimes G$ *(the binding rotation in opposite senses on the two copies) and* $\sigma_x \otimes G$ *are* $\eta_2$*-self-adjoint, like* $G$.

*(ii) Inside* $\mathfrak{u}(6) = \mathfrak{so}(12) \cap \mathrm{Cent}(G_2)$, *of dimension* 36, *the diagonal* $\mathfrak{u}(3)$ *and* $\mathfrak{u}(2)\otimes\mathbf{1}$ *are each other's commutants:* $U(3)\times U(2)$ *is a dual pair in* $U(6)$.

Part (i) restates rows four and five of Proposition 13.14 of [1] (fixed there), with the generators named. The four generators are those of a lossless beam splitter on two modes [6]: a real splitter $C_v$, an overall phase, a relative phase and a symmetric splitter. Their group $U(2)$ is the energy-preserving duality group of two free Maxwell fields [16], as Section 13.5 of [1] notes. Part (ii) is inherited from Howe duality for $\mathbb{C}^3 \otimes \mathbb{C}^2$ [24] (d5-3a, d5-3b).

**Theorem 7.5** (Duplication maps)**.** *(i) The $G$-linear, rotation-equivariant maps $V \to V_2$ are $\iota_{\alpha,\beta}(z) = (\alpha z, \beta z)$ with $\alpha, \beta \in \mathbb{C}$. Such a map preserves the energy, $E_6(\alpha z) + E_6(\beta z) = E_6(z)$, if and only if $|\alpha|^2 + |\beta|^2 = 1$, which gives a three-sphere $U(2)/U(1)$: the first column of a beam splitter. It is an isometry $(V, \eta) \to (V_2, \eta_2)$ if and only if $\alpha, \beta$ are real with $\alpha^2 + \beta^2 = 1$, which gives a circle, the orbit of the copy rotation.*

*(ii) On the image, $q_2 = (\alpha^2 + \beta^2)\, q$, $E_0^{(1)} + E_0^{(2)} = (|\alpha|^2 + |\beta|^2)E_0$, and the summed Maxwell stress tensor is $T(\alpha z) + T(\beta z) = (|\alpha|^2 + |\beta|^2)\, T(z)$. Every energy-preserving duplication therefore preserves the summed stress tensor, the sum of the copies' rest energies, and the pair invariant $E^2_{\text{pair}} := \frac{1}{4} T_{\text{tot}}{}^{\mu}{}_{\nu} T_{\text{tot}}{}^{\nu}{}_{\mu}$, which equals $E_0^2$ by Rainich's identity $(T\eta)^2 = E_0^2\, \mathbf{1}$ [49]. By contrast $|q_2| = |\alpha^2 + \beta^2|\, |q|$ sweeps $[0, |q|]$. It is preserved exactly on the circle; at the equal split with relative duality angle $\theta$ ($\beta = e^{i\theta}\alpha$), $|q_2| = |\cos\theta|\, |q|$, which vanishes at $\theta = \pi/2$.*

*(iii) Without equivariance, consider $X \mapsto (\cos\phi\, X,\ \sin\phi\, RX)$ with $R \in U(3)$ and $0 < \phi < \pi/2$. It preserves the summed stress tensor for all $X$ iff $R = e^{\vartheta G}$ is a duality rotation, preserves $q_2$ iff $R \in O(3)$, and preserves both iff $R = \pm\mathbf{1}$. More generally, a $G$-linear embedding $W = U + iV : \mathbb{C}^3 \to \mathbb{C}^6$ that preserves both $E_6 \oplus E_6$ and $q_2$ is real, since $W^*W - W^\top W = 2V^\top V - 2iV^\top U$.*

*Proof.* (i) The rotations act on $V = \mathbb{R}^2 \otimes \mathbb{R}^3$ through the absolutely irreducible second factor, so by Schur's lemma [51] the equivariant maps are $(A, B) \otimes \mathbf{1}_3$ with $A, B \in M_2(\mathbb{R})$, and $G$-linearity puts $A, B$ in $\text{span}\{\mathbf{1}, G\} \cong \mathbb{C}$. With $G^\top G = \mathbf{1}$, $G^\top \eta G = -\eta$ and $G^\top \eta = \eta G$, the map $L = \iota_{\alpha,\beta}$ has $L^\top L = (|\alpha|^2 + |\beta|^2)\mathbf{1}$ and $L^\top \eta_2 L = \text{Re}(\alpha^2 + \beta^2)\, \eta + \text{Im}(\alpha^2 + \beta^2)\, \eta G$. Since $|\alpha|^2 + |\beta|^2 - \text{Re}(\alpha^2 + \beta^2) = 2[(\text{Im}\,\alpha)^2 + (\text{Im}\,\beta)^2]$, the two conditions together force real coefficients. (ii) $q$ is holomorphic and quadratic. The components of $T$ are $\frac{1}{2}|z|^2$, the textbook Poynting vector $-\mathbf{S} = -\frac{i}{2}\bar{z} \times z$, and $\frac{1}{2}\delta_{ij}|z|^2 - \text{Re}(\bar{z}_i z_j)$, all sesquilinear, so $T(\alpha z) = |\alpha|^2 T(z)$. Rainich's identity is the tensor form of the identity $E_6^2 = E_0^2 + |\mathbf{S}|^2$ of Section 2.3. (iii) $T$ determines the Hermitian matrix $zz^*$, the real part from $T^{ij}$ and the imaginary part from $T^{0i}$, and conversely. So $T(Rz) = T(z)$ for all $z$ iff $R$ commutes with every rank-one Hermitian matrix, iff $R$ is a phase. $q(Rz) = q(z)$ for all $z$ iff $R^\top R = \mathbf{1}$, which for unitary $R$ means $\bar{R} = R$. If $W^*W = W^\top W = \mathbf{1}$ the identity gives $V^\top V = 0$. □

All three parts are inherited: they are the linear algebra of a beam splitter acting on $\mathbb{C}^3 \otimes \mathbb{C}^2$ (d5-3c to d5-3e). What is not inherited is the reading.

**Remark 7.6** (Two rest energies of a duplicate)**.** On one copy $E_0 = \frac{1}{2}|q|$ is at once the holomorphic invariant and the stress-energy invariant; this is Rainich's identity, the field-point form of $E^2 = (pc)^2 + (mc^2)^2$ (Section 2.3). On two copies the two separate. The stress-energy content $(T_{\text{tot}}, \sum_k E_0^{(k)}, E_{\text{pair}})$ is preserved by the whole three-sphere, the holomorphic invariant $q_2$ only by the circle (fixed: this is the point where duplication changes the meaning of "rest energy"). Which one deserves the name depends on the group. The invariants of a pair under the diagonal $\mathfrak{so}(3, \mathbb{C})$ are the entries of the complex Gram matrix $Q_{kl} = z_k \cdot z_l$; the copy algebra $\mathfrak{so}(2, \mathbb{C})$ preserves $\text{Tr}\, Q = q_2$ and $\det Q$ but not $Q_{11}$ (d5-3f); and the closure sees $q_2$ of the momentum field (Proposition 7.9). Two counter-propagating null fields, $z_1 = (1, i, 0)$ and $z_2 = (1, -i, 0)$, have $E_0^{(k)} = 0$ and $q_2 = 0$, yet $E^2_{\text{pair}} = 2$ (d5-3i). This is the field-point analogue of the invariant mass of two photons, which permits $\gamma\gamma \to e^+e^-$ [4]. A single null field duplicated by any $\iota_{\alpha,\beta}$ stays null in each copy, since $q(\alpha z) = \alpha^2 q(z)$; that is the analogue of the absence of

$\gamma \to e^+e^-$ in vacuum (both inherited). No isometric linear map copies the energy, $z \mapsto (z, z)$ has $E_6 \oplus E_6 = 2E_6$, and the genuine duplication of a quantum, $z \mapsto z \otimes z$, is quadratic; that is the no-cloning theorem [61] (inherited). A linear duplication divides the undivided energy; it does not multiply it.

**Proposition 7.7** (An isometric complex structure exists only on the duplicate)**.** *(i) No $J \in O(3,3)$ satisfies $J^2 = -\mathbf{1}$.*

*(ii) On $\mathbb{R}^{6,6}$, $C_v$ satisfies $C_v^2 = -\mathbf{1}$, $C_v^\top C_v = \mathbf{1}$ and $C_v^\top \eta_2 C_v = \eta_2$, and commutes with $G_2$ and with the diagonal $\mathfrak{l}_G$. It is a compact charge, invariant under the Lorentz group that acts on both frames alike.*

*(iii) $\mathfrak{u}(6) \cap \mathfrak{so}(6,6) = \mathfrak{so}(6)$ has dimension 15; it consists of the paired rotations and is the compact part of $\mathrm{Cent}(G_2)$. At $n = 1$ the corresponding algebra is $\mathfrak{u}(3) \cap \mathfrak{so}(3,3) = \mathfrak{so}(3)$. Inside $\mathfrak{so}(6)$ the centraliser of $C_v$ is $\mathfrak{u}(3)' = \{\mathbf{1} \otimes \mathrm{diag}(A, A) + \varepsilon \otimes \mathrm{diag}(B, B) : A^\top = -A,\ B^\top = B\}$, of dimension 9, and its part commuting with the diagonal boosts is $\mathbb{R}\, C_v$.*

*Proof.* (i) If $J^\top \eta J = \eta$ and $J^2 = -\mathbf{1}$, then $h(x, y) = \eta(x, y) + i\,\eta(x, Jy)$ is a Hermitian form on $(\mathbb{R}^6, J) \cong \mathbb{C}^3$ with $\mathrm{Re}\, h = \eta$. A Hermitian form of signature $(p, q)$ has real signature $(2p, 2q)$, never $(3, 3)$. (ii) and (iii) are computed (d5-3g, d5-3h). □

Part (i) is inherited, the signature parity of Hermitian forms. It bites because the arena's triples are odd, and that much is fixed. Read physically, $w = X^{(1)} + iX^{(2)}$ combines two neutral copies into one field charged under $C_v$, as two real fields form a charged one (inherited). The pointwise unitary symmetry $U(3)$ of $E_6$ is cut to $O(3)$ by Lorentz compatibility on one copy (Section 2). It reappears inside $SO(6,6)$ as $U(3)'$, with the copy index as imaginary unit and not the binding root (interpretive), and $U(3)'$ is not Lorentz invariant.

### 7.4 The duplicated closure symmetry

**Proposition 7.8** ($\mathrm{Cent}(G_2) = \mathfrak{so}(6, \mathbb{C})$ and the copy algebra)**.** *(i) The centraliser of $G_n = G \oplus \cdots \oplus G$ in $\mathfrak{so}(3n, 3n)$ has dimension $3n(3n-1)$: 6, 30 and 72 at $n = 1, 2, 3$. It is $\mathfrak{so}(3n, \mathbb{C})$, with compact part $\mathfrak{so}(3n)$ of dimension 3, 15 and 36. At $n = 2$ every element is $\left(\begin{smallmatrix} R & S \\ -S & R \end{smallmatrix}\right)$ in the basis of the six temporal directions followed by their partners, with $R, S$ antisymmetric.*

*(ii) $\mathfrak{so}(6, \mathbb{C}) = \mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)} \oplus (\mathbb{C}^3 \otimes \mathbb{C}^3)$, of dimensions $6 + 6 + 18$. Under the diagonal rotations the cross part is $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{5}$ (complex), with Casimir 0, 2, 6 on 2, 6, 10 real dimensions.*

*(iii) The commutant of the diagonal $\mathfrak{l}_G$ in $\mathfrak{so}(6,6)$ is the singlet of (ii), the copy algebra $\mathfrak{so}(2, \mathbb{C}) = \mathrm{span}\{C_v, \varepsilon \otimes G\}$. Here $C_v^2 = -\mathbf{1}$ is compact, while $(\varepsilon \otimes G)^2 = +\mathbf{1}$ is a copy boost, acting on $(z_1, z_2)$ by $\left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right)$; it preserves $q_2$ and not $E_6 \oplus E_6$.*

*(iv) $\mathrm{Cent}(G_2)$ meets $\mathfrak{l}_{\mathrm{int}}^{(1)}$ in 0, and the elements of $\mathfrak{so}(6,6)$ commuting with both $G_2$ and $\mathfrak{l}_{\mathrm{int}}^{(1)}$ form $\mathfrak{l}_G^{(2)} \cong \mathfrak{so}(3, \mathbb{C})$.*

Part (i) is the $n = 2$ member of Theorem 17.19 of [1], with the same proof, and the dimension 72 at $n = 3$ re-verifies that theorem. Part (iii) restates row three of Proposition 13.14 and Theorem 10.8 with $k_B = 2$ of [1] (fixed there). Parts (ii) and (iv) are fixed, and (iv) is the $n = 2$ analogue of Theorem 17.20 of [1] (d5-4a to d5-4c).

**Proposition 7.9** (Duplicated momentum fields in the closure)**.** *Let one rotor act on two constituents with the same inertia $N = (\nu_1 \eta^{-1} + \nu_2 G \eta^{-1})/m$, whose momentum fields are a duplication of one momentum field: $\pi^{(1)} = \alpha\pi$ and $\pi^{(2)} = \beta\pi$, with $\alpha, \beta$ acting through $\mathrm{span}\{\mathbf{1}, G\}$. Then $w_1 + w_2 = (\alpha^2 + \beta^2)\, w(\pi)$ with $w = q(z(\pi))$ (equation (30) of [1]). The effective strength of*

| action | $(k_I, k_B, k_S)$ | algebra | dim | compact | content of $\mathbb{R}^{6,6}$ | spin on $S_2$ |
|---|---|---|---|---|---|---|
| spectator | $(1,0,1)$ | $\mathfrak{so}(5,3)$ | 28 | 13 | $V \oplus 8 \cdot \mathbf{1}$ | half-integer |
| binding + spectator | $(0,1,1)$ | $\mathfrak{so}(3,3)$ | 15 | 6 | $\Lambda^2 V \oplus 6 \cdot \mathbf{1}$ | integer |
| diagonal interior | $(2,0,0)$ | $\mathfrak{so}(2) \oplus \mathfrak{so}(4)$ | 7 | 7 | $2V \oplus 4 \cdot \mathbf{1}$ | integer |
| mixed | $(1,1,0)$ | $\mathfrak{so}(2)$ | 1 | 1 | $V \oplus \Lambda^2 V \oplus 2 \cdot \mathbf{1}$ | half-integer |
| binding on both | $(0,2,0)$ | $\mathfrak{so}(2,\mathbb{C})$ | 2 | 1 | $2\,\Lambda^2 V$ | integer |

Table 8: The Lorentz actions at $n = 2$ and the grade-two elements commuting with them (Theorem 10.8 of [1]), with their compact parts, the content of the vector space (Lorentz Casimir 3 on a four-vector $V$, 4 on a field strength $\Lambda^2 V$), and the effect of a $2\pi$ rotation on $S_2$. Every entry was computed in the realisation (d5-5a, d5-5b, d5-6b).

*Proposition 17.18 of [1] is therefore $R_{\text{eff}} = |\nu_1 + i\nu_2|\,|q_2(\pi)|/m$: the closure sees the holomorphic invariant of the duplicated momentum field. For the equal split with a relative quarter turn of duality, $\pi^{(2)} = G\pi^{(1)}$, equivalently $p^{(2)} = -Gp^{(1)}$, one has $w_2 = -w_1$ and $R_{\text{eff}} = 0$. The rate is then exactly constant, and nonzero when $h(0) \neq 0$, while each copy keeps $E_0(\pi^{(k)}) = \frac{1}{2}E_0(\pi) \neq 0$. On one constituent a constant nonzero rate requires $E_0(\pi_0) = 0$ (Corollary 17.14 of [1]).*

This restates Proposition 17.18 and Corollary 17.14 of [1] (fixed there). The expression of $R_{\text{eff}}$ through $q_2$ for duplicates is a one-line consequence (fixed; d5-9f). So a duplicate can do what one copy cannot: run at an exactly constant rate while its copies carry nonzero rest energy in the momentum field. Calling that rest energy a mass needs $mc^2 = \hbar\kappa$.

### 7.5 The five Lorentz actions at the second rung

What the duplication does depends on how the Lorentz algebra acts on the two copies. Theorem 10.8 of [1] lists the choices, and at $n = 2$ there are five (Table 8).

Read row by row: under the spectator action copy 2 is a passenger whose six directions join a non-compact internal algebra. With $\mathfrak{l}_G$ on copy 1, copy 2's whole $\mathfrak{so}(3,3)$ becomes internal symmetry and no spin one-half survives. The diagonal interior action gives a compact algebra containing the copy rotation $M_{12}$, but only integer spin. The mixed action keeps only $J_1 = e_2^{(1)} e_3^{(1)}$. With $\mathfrak{l}_G$ on both copies the frames are two field strengths mixed by $\mathfrak{so}(2,\mathbb{C})$. The table restates Theorem 10.8 and Propositions 10.9, 13.16 and 13.17 of [1] (fixed there), re-verified. For the mixed rung we also re-verified the Casimir identity $(C_{\text{mix}} - \frac{3}{2})(C_{\text{mix}} - \frac{11}{2}) = 0$ with ranks 16 and 48 (d5-5b). We checked the grade-three element $\Theta_F$ of Section 13.6 of [1], which identifies the frame field of copy 2 with $dA$ of copy 1. It and $\Theta_F$ followed by $G$ are invariant and span the invariants in $\Lambda^2(\text{four-plane}_1) \otimes V_2$, of dimension two as stated there. The full space of invariant grade-three elements has dimension four: the other two lie in $\Lambda^3 V_2$ of copy 2 alone, the two $\mathfrak{l}_G$-invariant trivectors of $(\Lambda^3\mathbb{R}^6)_{\mathbb{C}} = 2\,(0,0) \oplus 2\,(1,1)$ (Section 12.3 of [1]; d5-5c). This is a precision, not a correction. The mixed rung is the only action at $n = 2$ that houses both readings, at the price of the whole internal algebra.

### 7.6 Spin and statistics of the duplicate

**Proposition 7.10** (Spin of the duplicate)**.** *(i) The copy rotations $M_{12}$ (four commuting terms of square $-\frac{1}{4}$) and $C_b$ (six) have eigenvalues $i\{-2,\dots,2\}$ and $i\{-3,\dots,3\}$, and $\exp(2\pi M_{12}) = \exp(2\pi C_b) = \mathbf{1}$. A surplus rotation $\frac{1}{2}e_s e_t$ has $\exp(\pi e_s e_t) = -\mathbf{1}$.*

*(ii) A rotation through $2\pi$ is $+\mathbf{1}$ on $S_2$ under the diagonal interior action and $-\mathbf{1}$ under the spectator action. Under the mixed action it is $-\mathbf{1}$, and under $\mathfrak{l}_G$ on one or both copies it is $+\mathbf{1}$. The spin Casimir is 0 on 16 and 2 on 48 dimensions for the diagonal action ($j = 0, 1$), and $\frac{3}{4}$ on all of $S_2$ for the spectator action.*

*(iii) Under the diagonal interior action the Lorentz Casimir is* $0, 3, 4$ *on* $8, 32, 24$ *dimensions and* $C_{\mathrm{cop}} = -M_{12}^2$ *is* $0, 1, 4$ *on* $24, 32, 8$*, with* $C_{\mathrm{cop}} + C_\Delta = 4$*. The exchange* $\tau_K$ *commutes with the diagonal Lorentz action and anticommutes with* $M_{12}$*, and its* $\pm 1$ *eigenspaces carry identical Lorentz content,* $0, 3, 4$ *on* $4, 16, 12$ *dimensions each.*

This restates Proposition 10.9, Theorem 10.4 at $n = 2$ ($\frac{1}{2}n(n+2) = 4$) and the last paragraph of Section 13.6 of [1] (fixed there), with the $2\pi$ eigenvalues computed (d5-6a, d5-6b). That two spin-$\frac{1}{2}$ constituents make an integer-spin composite is inherited [12]: under the diagonal action the duplicate of a fermion is a boson, and fermions need an odd number of interior constituents. The Lorentz isomorphism of the two exchange sectors in (iii) is fixed (d5-6c). It holds because exchange reverses the copy charge, pairing charge $\pm m$, and splits the neutral sector evenly; there is no "singlet antisymmetric, triplet symmetric" rule once the surplus directions and the Koszul signs are included.

### 7.7 What the second rung adds: the vacuum and the imaginary unit

**Proposition 7.11** (The photon vacuum and the imaginary unit, re-verified)**.** *(i) At* $n = 1$ *the commutant of* $\mathfrak{l}_{\mathrm{int}}$ *and* $Q = \frac{1}{2}e_2e_3$ *in* $M_8(\mathbb{R})$ *has dimension four, and its symmetric part is spanned by* $\mathbf{1}$ *and* $\mathcal{V} = e_1 \cdots e_6$*. The rank-four projectors in it are* $\frac{1}{2}(\mathbf{1} \pm \mathcal{V})$*, and every bivector commutes with* $\mathcal{V}$*, so they are rigid (Proposition 26.4 of [1]).*

*(ii) (Theorems 26.6 and 26.7 of [1]) With* $a, b = e_2^{(1)}, e_3^{(1)}$ *and* $c, d, e = e_1^{(2)}, e_2^{(2)}, e_3^{(2)}$*:* $H = abcd$ *has* $H^2 = \mathbf{1}$*,* $H^\top = H$ *and* $\operatorname{Tr} H = 0$*, and* $\Pi_0 = \frac{1}{2}(\mathbf{1} + H)$ *commutes with* $Q = \frac{1}{2}ab$*, with the four interior gammas and with the internal volume. Moreover* $\pi = [\frac{1}{2}be, \Pi_0] = -\frac{1}{2}acde$ *is symmetric with* $\operatorname{Tr}\pi^2 = 16$ *and* $[Q, \pi] = \frac{1}{2}bcde$*. The* $Z_i = I_5 v_i$ *are symmetric of grade four with* $\{Z_i, Z_j\} = 2\delta_{ij}$*.* $\Pi(n)$ *is a rank-*$32$ *projector commuting with the gammas (checked at* $n = (2, 3, 6, 0, 0)/7$*). The tangents* $T_i = \frac{1}{2}I_5 v_i$ *have* $\operatorname{Tr} T_i T_j = 16\delta_{ij}$*,* $[Q, T_a] = -T_b$*,* $[Q, T_b] = T_a$ *and* $[Q, T_c] = [Q, T_d] = 0$*.*

*(iii) The grade-four blades in the five positive internal generators that commute with* $Q$ *are exactly* $abcd$*,* $abce$ *and* $abde$*. Every such vacuum straddles the two copies.*

*(iv) (Theorem 4.1 of [1] at* $n = 2$*) The product of the eight internal generators is* $J = (e_2e_3) \otimes \omega$*, with* $J^2 = -\mathbf{1}$*. The joint commutant of* $\mathfrak{l}_{\mathrm{int}}^{(1)}$ *and the* $28$ *internal bivectors in* $M_{64}(\mathbb{R})$ *is spanned by the blades* $\mathbf{1}, \Omega, J, \Omega J$*. The solutions of* $x^2 = -\mathbf{1}$ *in it are exactly* $\pm\Omega$ *and* $\pm J$*, and only* $\pm J$ *commute with the gammas.*

Parts (i), (ii) and (iv) restate results of [1] (fixed there), re-verified in the realisation of $\mathrm{Cl}(6, 6)$ (d5-7a to d5-7c, d5-8). Part (iii) is fixed but elementary (d5-7d). The Lorentz-invariant, uncharged, non-rigid vacuum needs four positive internal directions that commute with $Q$ in pairs, and one copy offers only the surplus pair; it exists because there are two copies. Part (iv) adds one identity: the imaginary unit of the duplicate is the imaginary unit of copy 1 times the volume element of copy 2, which is that copy's fermion parity up to sign (Proposition 7.2). It is neither the embedded one-rung charge $J_1 \otimes \mathbf{1}$ nor $J^{(1)} + J^{(2)}$.

### 7.8 The copy rotation as a $b$-field between two manifolds

On one copy a $b$-field commuting with $G$ is removable along a worldline, and an anticommuting one tears the binding (Section 5, Theorem 5.10). On two copies the binding term is $K \otimes G$ with $K = \operatorname{diag}(\kappa_1, \kappa_2)$. That is a multiple of $G_2$ only when the rates agree, and the following theorem measures what changes. The interior Lagrangian of each copy is that of Section 5, with $S = K \otimes G + b$; the frame field obeys $\dot{X} = -SX + \pi/m$ and the momentum field $\dot{\pi} = S^\dagger \pi$.

**Theorem 7.12** (The inter-copy $b$-field)**.** *Let* $\kappa_1, \kappa_2 \geq 0$*,* $\beta \in \mathbb{R}$*,* $b = \beta\, C_v$*, and* $M = \left(\begin{smallmatrix} -S & \mathbf{1}/m \\ 0 & S^\dagger \end{smallmatrix}\right)$ *on* $(X, \pi)$*.*

*(i)* *Up to scale, $b$ is the only element of $\mathfrak{so}(6,6)\cap\mathfrak{so}(12)$ commuting with $G_2$ and the diagonal rotations. It is compact, and $S^\dagger = K\otimes G - b$.*

*(ii)* $\det(\lambda - S) = \big(\lambda^4 + (\kappa_1^2+\kappa_2^2+2\beta^2)\lambda^2 + (\kappa_1\kappa_2-\beta^2)^2\big)^3$, *so* $\operatorname{spec} S = \{\pm i\omega_\pm\}$ *with* $\omega_\pm = \frac{1}{2}(\kappa_1+\kappa_2) \pm \sqrt{\frac{1}{4}(\kappa_1-\kappa_2)^2+\beta^2}$; *equivalently, $\omega_\pm$ are the eigenvalues of* $\left(\begin{smallmatrix}\kappa_1 & i\beta\\ -i\beta & \kappa_2\end{smallmatrix}\right)$.

*(iii)* $[K\otimes G, b] = \beta(\kappa_2-\kappa_1)\,\sigma_x\otimes G$, *so $b$ is removable along the worldline (Larmor [33]) iff $\kappa_1=\kappa_2$. The centraliser of $K\otimes G$ in $\mathfrak{so}(6,6)$ has dimension 30 if $\kappa_1=\kappa_2\neq 0$, 12 if $\kappa_1\neq\kappa_2$ are both nonzero, and 21 if exactly one vanishes.*

*(iv)* $(M^2+\omega_+^2)(M^2+\omega_-^2) = 0$ *identically in $\kappa_1,\kappa_2,\beta,m$.*

*(v)* *Every orbit is bounded iff either $\kappa_1=\kappa_2>0$, or $\kappa_1\neq\kappa_2$ and $\kappa_1\kappa_2\neq\beta^2$. In particular, for $\kappa_1=0<\kappa_2$ the orbits are bounded iff $\beta\neq 0$: the inter-copy field confines a frame with no rate of its own. For $\kappa_1=\kappa_2=0$ they are unbounded for every $\beta$.*

*(vi)* *For $\kappa_1=0<\kappa_2$, $\omega_+\omega_- = -\beta^2$ and $\omega_+ + \omega_- = \kappa_2$. For $\beta\ll\kappa_2$ the slow line is $|\omega_-| = \beta^2/\kappa_2 - \beta^4/\kappa_2^3+\cdots$, the seesaw. For $\beta\gg\kappa_2$ both lines approach $\pm\beta$, and the orbit size grows like $2/(m\kappa_2)$.*

*Proof.* (i) is Proposition 7.4(i), and $C_v$ is $\eta_2$-antisymmetric while $K\otimes G$ is self-adjoint. (ii) On the sector $G_2=i$, $S=(iK+\beta\varepsilon)\otimes\mathbf{1}_3$, whose characteristic polynomial is $\lambda^2 - i(\kappa_1+\kappa_2)\lambda - \kappa_1\kappa_2+\beta^2$; the sector $G_2=-i$ is its conjugate. (iii) $K\varepsilon-\varepsilon K = (\kappa_2-\kappa_1)\sigma_x$. If $\kappa_1=\kappa_2=\kappa$, then $X=e^{-b\tau}Y$ and $\pi = e^{-b\tau}\varpi$ turn the system into $b=0$ with rate $\kappa$; otherwise this substitution makes the binding $e^{b\tau}(K\otimes G)e^{-b\tau}$ time-dependent. The centraliser dimensions are computed. (iv) is a symbolic identity (d5-9c). (v) If $\kappa_1\neq\kappa_2$ and $\kappa_1\kappa_2\neq\beta^2$, then $\omega_+\neq\omega_-$, $\omega_-\neq 0$ and $\omega_+ + \omega_- = \kappa_1+\kappa_2 > 0$, so the polynomial of (iv) has four simple imaginary roots and $M$ is semisimple with imaginary spectrum. If $\kappa_1=\kappa_2=\kappa>0$, use (iii) and the boundedness of the $b=0$ system (Theorem 16.2 of [1]); if $\kappa=0$ the substituted system is free drift. If $\kappa_1\neq\kappa_2$ and $\kappa_1\kappa_2=\beta^2$, then $\omega_-=0$, and $M$ is semisimple at 0 iff $\ker S^\dagger \subset \operatorname{im} S$. Because $S^\top = -S$ and $S^\dagger = -\eta_2 S\eta_2$, this says that $\ker S$ is $\eta_2$-totally isotropic. Complexifying, $\ker S$ is spanned by the real parts of $v\otimes c$ in the $G_2=i$ sector, with $c\in\mathbb{C}^3$ in the basis $e_j - ie_{j+3}$ and $v=(\beta,i\kappa_1)$ (or $v=(1,0)$ when $\beta=\kappa_1=0$), and $\eta_2(\operatorname{Re}(v\otimes c),\operatorname{Re}(v\otimes c')) = \operatorname{Re}\big[(v\cdot v)(c\cdot c')\big]$. This vanishes for all $c,c'$ iff $v\cdot v = \beta^2-\kappa_1^2 = 0$, which together with $\kappa_1\kappa_2=\beta^2$ means $\kappa_1=\kappa_2$ or $\beta=\kappa_1=0$, and in the second case $v=(1,0)$ is not isotropic; so there is a Jordan block. (vi) Vieta and expansion. The orbit sizes are numerical (d5-9e, d5-9g). □

Parts (ii) and the seesaw in (vi) are inherited: two-level mixing with detuning $\frac{1}{2}(\kappa_1-\kappa_2)$ and coupling $\beta$, the generalised Rabi splitting, and the seesaw formula [37]. Part (i) restates Proposition 13.14 of [1] (fixed there). Parts (iii)–(v) are fixed (d5-9a to d5-9d). The exact checks are: bounded at $(\kappa_1,\kappa_2,\beta) = (1,3,\frac{2}{5}), (0,5,\frac{1}{2}), (2,2,\frac{1}{3}), (1,1,1), (2,2,0)$ and $(0,\frac{1}{100},1)$; unbounded at $(0,0,\frac{1}{2}), (1,4,2), (1,9,3), (0,3,0)$. At $(0,5,\frac{1}{2})$, $\omega_- = -0.04951$ against $\beta^2/\kappa_2 = 0.05$, and $\sup_{t\le 600}\|e^{Mt}\| = 19.86$ against $1/|\omega_-| = 20.2$. At $(0,\frac{1}{100},1)$ the lines are 1.00501 and $-0.99501$, and the supremum over a beat period is $200.0 = 2/(m\kappa_2)$. This confirms the coordinator's preliminary computation and decides its open point: the seesaw orbit is bounded, with amplitude of order $1/(m|\omega_-|)$.

**Remark 7.13** (The author's first remark, as far as this track reaches)**.** On one manifold the no-go of Section 5 stands. Between two manifolds with different rates it does not: the compact inter-copy field is not removable, it changes the spectrum, and it confines a frame with $\kappa_1=0$. In that sense the $b$-field is a "deal breaker" at $n\ge 2$ (fixed). Three qualifications remain. The confinement needs a second frame with $\kappa_2\neq\kappa_1$: for $\kappa_1=\kappa_2=0$, $b$ produces only a removable rotation (obstructed). The field can set the rates, which approach $\pm\beta$ when $\beta\gg\kappa_2$, but the size of the orbit is set by $\kappa_2$ and diverges as $\kappa_2\to 0$. And reading a rate as a rest energy still needs $mc^2=\hbar\kappa$ (an added identification).

| process | mathematics | preserves | class |
|---|---|---|---|
| splitting one field into two (beam splitter) | $\iota_{\alpha,\beta}$, $S^3 = U(2)/U(1)$; circle if $\eta$-isometric | $E_6$, $T$, $\sum E_0^{(k)}$; $q_2$ only on the circle | inherited; reading fixed |
| second quantisation | $\mathrm{Cl}(V \oplus V) = \mathrm{Cl}(V)\hat{\otimes}\mathrm{Cl}(V)$; Fock space; Jordan–Wigner strings | Clifford relations | inherited (forced) |
| two neutral copies into one charged field | $C_v$ as $i$, $w = X^{(1)} + iX^{(2)}$ | $\eta_2$, $E_6 \oplus E_6$ | inherited; exists only at $n \geq 2$: fixed |
| two-body composite | diagonal interior action; exchange $\tau_K$ | integer spin | inherited; restated |
| pair creation from null energy | linear covariant duplication keeps null copies; two non-collinear null fields have $E_{\mathrm{pair}} > 0$ | $T$ | inherited |
| cloning, $z \mapsto z \otimes z$ | quadratic, not isometric and linear | — | obstructed (inherited) |
| closure balance | $\pi^{(2)} = G\pi^{(1)}$, $R_{\mathrm{eff}} \propto \lvert q_2(\pi)\rvert = 0$ | constant rate, $E_0^{(k)} \neq 0$ | restated (Prop. 17.18) |
| inter-copy $b$-field | $S = K \otimes G + \beta C_v$ | bounded iff Theorem 7.12(v) | fixed; mixing inherited |

Table 9: Processes that take one copy of the arena to two.

### 7.9 Assessment

**What is forced, what is chosen, what is added.** *Forced* (inherited): the algebra step, $\mathrm{Cl}(V \oplus V) = \mathrm{Cl}(V)\hat{\otimes}\mathrm{Cl}(V)$, with the Klein factor determined up to conjugation; $S_2 = S \otimes S$ as the Fock space of six modes; and the exchange as the Koszul swap. *Chosen* (constructed): the Lorentz action, one of five; the energy duplication, a point of the three-sphere, or of the circle if $\eta$ is to be respected, with the angle $\phi$ free; and the vacuum point on $S^4$. *Added* at $n = 2$ (fixed): the copy algebra $\mathfrak{so}(2, \mathbb{C})$ on two binding frames (dimension 0 at $n = 1$); an $\eta$-isometric complex structure and a compact, Lorentz-invariant copy charge (none at $n = 1$); a Lorentz-compatible energy symmetry $\mathfrak{so}(6) \supset \mathfrak{u}(3)'$, against $\mathfrak{so}(3)$; a closure with constant rate and massive copies; a non-rigid, Lorentz-invariant vacuum; the mixed rung, which houses both readings; and an inter-copy $b$-field that confines (d5-10 collects the dimensions). *Lost*: spin one-half under the diagonal interior action, and the uniqueness of the photon direction (Corollary 6.6 of [1]).

**Processes.** Table 9 lists the physical processes that take one copy to two, and what each preserves.

**The precise sense of "a unitary energy results in a duplication".** Three statements, in increasing order of what the arena adds. First, the undivided energy $(\mathbb{C}^3, E_6, U(3))$ embeds isometrically in $(\mathbb{C}^3 \otimes \mathbb{C}^2, E_6 \oplus E_6, U(6))$. The Lorentz-covariant embeddings form the three-sphere of a beam splitter, and the diagonal $U(3)$ and the copy $U(2)$ are a dual pair (inherited). Second, compatibility with $\eta$ cuts the copy $U(2)$ to the copy rotation $SO(2)$ and $U(6)$ to $SO(6)$. The copy rotation is the one structure that preserves the energy, the metric, the binding root and the diagonal Lorentz group at once, and it does not exist on one copy (fixed). Third, the duplicate's rest energy splits into a stress-energy part, preserved by every duplication, and a holomorphic part, $|q_2|$, preserved only by the copy rotation and seen by the closure (fixed). Read literally as a multiplication of the energy, the phrase fails: a linear duplication divides the undivided energy, and copying it is impossible (no-cloning, inherited).

| statement | class | checks |
|---|---|---|
| Proposition 7.1: graded tensor product, $\mathrm{Cl}(6,6) = M_{64}(\mathbb{R})$, Klein factor forced | inherited (restated: eq. (2)) | d5-1a to d5-1d |
| Proposition 7.2: Fock space, Jordan–Wigner strings, pole operators $= \mathbf{1} - 2N_i$ | inherited (reading of Prop. 2.5) | d5-1e |
| Proposition 7.3: halves, $\pi/4$ copy rotation, $\tau_K = g_1 \cdots g_6$, what each half carries | inherited; identities fixed | d5-1f, d5-2a to d5-2c |
| Proposition 7.4: $\mathfrak{u}(2)$, $\mathfrak{so}(2,\mathbb{C})$, $\mathfrak{so}(2)$; dual pair | fixed (restated: Prop. 13.14); inherited | d5-3a, d5-3b |
| Theorem 7.5: duplication maps, $S^3$ and circle, $q_2$ versus $T$ | inherited; reading fixed | d5-3c to d5-3e |
| Remark 7.6: two rest energies; Gram invariants; pair invariant; no cloning | fixed; inherited | d5-3f, d5-3i |
| Proposition 7.7: isometric complex structure only at $n \geq 2$; $\mathfrak{so}(6) \supset \mathfrak{u}(3)'$ | inherited (parity); fixed | d5-3g, d5-3h |
| Proposition 7.8: $\mathrm{Cent}(G_n) = \mathfrak{so}(3n,\mathbb{C})$; $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{5}$; copy algebra; analogue of Thm 17.20 | fixed (restated: Thm 17.19, Prop. 13.14) | d5-4a to d5-4c |
| Proposition 7.9: closure sees $q_2(\pi)$; balance with massive copies | fixed (restated: Prop. 17.18, Cor. 17.14) | d5-9f |
| Table 8: five Lorentz actions; mixed rung; $\Theta_F$ (dimension 2 within $\Lambda^2 E_1 \otimes V_2$, 4 in all) | fixed (restated: Thm 10.8, Props. 13.16–13.17) | d5-5a to d5-5c |
| Proposition 7.10: spin and statistics, Casimirs, exchange sectors | fixed (restated: Prop. 10.9, Thm 10.4); inherited | d5-6a to d5-6c |
| Proposition 7.11: rigid first rung, $S^4$ vacuum, straddling, $J = J_1 \otimes \omega$ | fixed (restated: Prop. 26.4, Thms 26.6, 26.7, 4.1) | d5-7a to d5-7d, d5-8 |
| Theorem 7.12: inter-copy $b$-field: uniqueness, spectrum, removability, boundedness criterion, seesaw | fixed; mixing and seesaw inherited | d5-9a to d5-9e, d5-9g |
| Remark 7.13: the deal-breaker reading | fixed; obstructed for $\kappa_1 = \kappa_2 = 0$; $mc^2 = \hbar\kappa$ an identification | — |
| Section 7.9: comparison $n = 1$ against $n = 2$ | fixed | d5-10 |

Table 10: Claims of Section 7, their classes, and the checks printed by `d5_duplication_math.py`. "Restated": fixed in [1], restated and re-verified here.

# 8 Duplication in motion: the balanced pair

Two further remarks are tested here: that the $b$-particle of Section 5 becomes decisive once two or more manifolds are present, and that one should find the process by which a unitary (undivided) energy results in a duplication, from $\mathrm{Cl}(3,3)$ to $\mathrm{Cl}(6,6)$. Duplication is taken literally: two copies of the arena whose tangent spaces add, $V_2 = V \oplus V \cong \mathbb{R}^{6,6}$, which is grade one of $\mathrm{Cl}(6,6) \cong \mathrm{Cl}(3,3)\hat{\otimes}\mathrm{Cl}(3,3)$ (equation (2) of [1]; the graded tensor product is inherited). It is followed in motion, inside the closure of Section 17 of [1], and at a single field point. In brief: (1) with one constituent and Lorentz-class inertia a constant nonzero rate forces a null momentum field, while with two or more a codimension-two balance surface carries massive constituents at an exactly constant rate with bounded frames (Theorem 8.1, restating Proposition 17.18 of [1] with the dimension count added). This is the precise sense in which a duplicate does what one copy cannot, and it needs no $b$-field. (2) The closure never creates a pair (Corollary 8.3). (3) The admissible $b$-field between the manifolds, the copy rotation, duplicates a field literally, and no Lorentz-covariant operation makes a massive pair out of a single null field (Proposition 8.5); the rotation used in the task's example is a different, non-covariant element (Example 8.8). (4) The rest energy of a duplicated field point is $M_{\mathrm{pair}} = \frac{1}{2}(\sigma_1 + \sigma_2)$, and the copy rotation can move all of it into the two copies (Theorem 8.6 and Example 8.7): the kinematics of $\gamma\gamma \to e^+e^-$, inherited. (5) The rest energy of each copy separately is not gauge invariant, and with equal inertias the closure's own symmetry maps every balanced pair to one null constituent (Proposition 8.9): pair creation is physical only where the copies are distinguished. (6) On the Dirac side the graded duplication adds Dirac masses in quadrature (Proposition 8.11). (7) The zero-point energy stays

zero, but an inter-manifold $b$ at unequal rates can cancel one rate exactly, and there the motion is unbounded (Proposition 8.12).

**Conventions.** As in Section 4 for one constituent. For two, $V_2 = V \oplus V$ with coordinates $(X^{(1)}; X^{(2)})$, $\eta_2 = \mathbf{1}_2 \otimes \eta$, $G_2 = \mathbf{1}_2 \otimes G$; in the tower $e_a^{(1)} = e_a \otimes \mathbf{1}$, $e_a^{(2)} = \omega \otimes e_a$, $\omega = e_1 \cdots e_6$, and $V_2$ is spanned by the twelve $e_a^{(k)}$. $Z = (z_1, z_2) \in \mathbb{C}^2 \otimes \mathbb{C}^3$, $z_k = z(X^{(k)})$, so $G_2$ acts as $i$; per copy $E_6^{(k)}, E_0^{(k)}, \mathbf{S}^{(k)}$ and $q_k = z_k \cdot z_k$ are those of (1), and $q_{\mathrm{tot}} = q_1 + q_2$. The Gram matrix $Q_{kl} = z_k \cdot z_l$ (complex bilinear, symmetric) has singular values $\sigma_1 \geq \sigma_2 \geq 0$; $H_{\mathrm{sum}} = \sum_k z_k z_k^\dagger$. The *copy rotation* is generated by $E := \epsilon \otimes \mathbf{1}_6$, $\epsilon = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$: $R_\varphi = e^{\varphi E}$ maps $(z_1, z_2) \mapsto (\cos\varphi\, z_1 - \sin\varphi\, z_2,\ \sin\varphi\, z_1 + \cos\varphi\, z_2)$, identically in all six directions; $E^2 = -\mathbf{1}$, $[E, G_2] = 0$, $E$ is antisymmetric for $\eta_2$ and for the Euclidean form. $T(z)$ is the stress tensor of Proposition 2.5, $P(t) = T(z)t$ the density seen by a unit future timelike observer $t$, $P_{\mathrm{sum}}(t) = \sum_k T(z_k)t$, and

$$M_{\mathrm{pair}} := \inf_t |P_{\mathrm{sum}}(t)|,$$

the rest energy of the duplicated field point. In the closure the per-copy quantities are those of the momentum fields $\pi^{(k)} = \eta^{-1} p^{(k)}$, $w_k = q(z(\pi^{(k)}))$; inertias are of Lorentz class, $N_k = (\nu_{k,1}\eta^{-1} + \nu_{k,2} G\eta^{-1})/m_k$, with $c_k := \nu_k/m_k \in \mathbb{C}$ and $R_{\mathrm{eff}} e^{i\varphi_{\mathrm{eff}}} := \sum_k c_k w_k(0)$ (Proposition 17.18 of [1]). Two Lorentz actions occur: the per-copy algebras $\mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)}$, and the diagonal $\mathfrak{l}_{G,\Delta}$ ($k_B = 2$ in Theorem 10.8 of [1]). Rates are rates: $mc^2 = \hbar\kappa$ is an added identification (Section 4). Identifiers d6-$n$ refer to `verify/d6_pair_balance.py` (integrations DOP853, relative tolerance $10^{-12}$; d6-0 checks the conventions).

## 8.1 Mass with a constant rate needs a second constituent

**Theorem 8.1** (Constant rate with massive constituents)**.** *Let one rotor act on $n$ constituents with Lorentz-class inertias and all $c_k \neq 0$.*

*(i) Along every solution $w_k(\tau) = e^{2i\theta(\tau)} w_k(0)$, so $R_{\mathrm{eff}}$ is constant. The rate $h/I_0$ is constant and nonzero iff $R_{\mathrm{eff}} = 0$ and $h(0) \neq 0$; then $\theta = \kappa\tau$ and every frame moves on the circle $x_k(\tau) = x_k(0) + (e^{2\kappa\tau G} - \mathbf{1})(2\kappa G)^{-1} N_k p^{(k)}(0)$, of diameter $|N_k p^{(k)}(0)|/|\kappa|$.*

*(ii) For $n = 1$, $R_{\mathrm{eff}} = 0$ iff $E_0(\pi) = 0$: no datum with a constant nonzero rate has a massive momentum field.*

*(iii) For $n \geq 2$ the balance set $\mathcal{B} = \{\sum_k c_k w_k = 0\} \subset \mathbb{R}^{6n}$ is, away from 0, a smooth submanifold of codimension two; its part on which every $E_0(\pi^{(k)})$ vanishes has dimension $4n < 6n - 2$, so the data with a constant nonzero rate and some $E_0(\pi^{(k)}) > 0$ are open and dense in $\mathcal{B} \times \{h \neq 0\}$. For $n = 2$ both constituents are then massive, with $|c_1| E_0(\pi^{(1)}) = |c_2| E_0(\pi^{(2)})$.*

*Proof.* (i) With a common rotor the momenta evolve as $p^{(k)} = e^{-\theta G} p^{(k)}(0)$ (Proposition 17.8 and the closing remark of Theorem 17.7 of [1], $\mathfrak{s}_n = \mathbb{R} \ltimes_{G_n} V_n$), so $w_k = e^{2i\theta} w_k(0)$ (Theorem 4.1) and, by Theorem 17.11(ii) of [1] per constituent, $U(\theta) = \sum_k \frac{1}{2} p^{(k)\top} N_k p^{(k)} = \frac{1}{2} \mathrm{Re}(e^{2i\theta} \sum_k c_k w_k(0))$, whence $I_0 \ddot{\theta} = R_{\mathrm{eff}} \sin(2\theta + \varphi_{\mathrm{eff}})$. If $R_{\mathrm{eff}} = 0$, $\ddot{\theta} = 0$; if $R_{\mathrm{eff}} > 0$ and $h$ is constant, then $\sin(2\theta + \varphi_{\mathrm{eff}}) \equiv 0$, so $\theta$ is constant and $h = 0$. Lorentz-class $N_k$ anticommutes with $G$, so $\dot{x}_k = e^{\theta G} N_k e^{-\theta G} p^{(k)}(0) = e^{2\theta G} N_k p^{(k)}(0)$ (Theorem 17.12 of [1]), which integrates to the circle. (ii) $R_{\mathrm{eff}} = |c|\,|w(p)| = 2|c| E_0(\pi)$. (iii) $F(p) = \sum_k c_k w(p^{(k)})$ has $dF_p(\delta) = 2\sum_k c_k\, z(\pi^{(k)}) \cdot z(\delta\pi^{(k)})$. At $p \neq 0$ some $z(\pi^{(k)}) \neq 0$, and $u \mapsto 2c_k z(\pi^{(k)}) \cdot u$ maps $\mathbb{C}^3$ onto $\mathbb{C}$, so 0 is a regular value of $F$ on $\mathbb{R}^{6n} \setminus 0$. The null set $\{w = 0\} \subset \mathbb{R}^6$ is the cone $|\mathbf{B}| = |\mathbf{E}|$, $\mathbf{B} \cdot \mathbf{E} = 0$, of dimension four, and the all-null part of $\mathcal{B}$ is its $n$-fold product. Points with $w(p^{(1)}) = 1$, $w(p^{(2)}) = -c_1/c_2$ show that the massive part is not empty. □

Items (i) and (ii) restate Proposition 17.18 and Corollary 17.14 of [1] (fixed there); (iii) is fixed, the dimension count being added. Checks: $\frac{1}{2} p^\top N p = \mathrm{Re}(\nu w(p))/(2m)$ exactly, and

the rank of $dF$ at random balanced points for $n = 1, 2, 3$ (d6-1a,b); integrations of the full two-constituent closure, rotor, momenta and frames (d6-2a–d). With equal inertias and $p^{(2)} = -Gp^{(1)}$ ($E_0(\pi^{(k)}) = 4.95$) the rate stays constant to $1.6 \times 10^{-14}$ over $\tau \in [0, 60]$ and the frames lie on the circles of (i) to $4 \times 10^{-12}$; a generic balanced pair, and unequal inertias ($m = 1, 2.5$; $\nu = 1 + 0.3i,\ 0.4 - 0.7i$), give the same to $10^{-13}$. Off the surface ($R_{\text{eff}} = 0.0752$, rotating) the rotor follows $I_0\ddot{\theta} = R_{\text{eff}} \sin(2\theta + \varphi_{\text{eff}})$ to $2 \times 10^{-10}$ and frame 1 drifts by 0.1795 per half turn, exactly additively (Theorem 17.16 of [1] for several constituents).

**Remark 8.2** (The Lorentz requirement is what empties (ii))**.** $N = \operatorname{diag}(1, -1, 0, -1, 1, 0)/m$ anticommutes with $G$ but is not $\mathfrak{l}_G$-invariant. For $p_0 = e^1 + e^2$, $z(\pi_0) = (1, 1, 0)$ and $E_0(\pi_0) = 1$, while $U(\theta) = \frac{1}{2} p_0^\top e^{2\theta G} N p_0 \equiv 0$: the rate is exactly constant and the frame moves on a circle (d6-1c). One constituent can thus combine a constant rate with a massive momentum field only through a cancellation between directions that breaks Lorentz invariance; the balance surface is the Lorentz-invariant form of the same cancellation, between constituents. Fixed, as a counterexample to (ii) without the hypothesis.

**Corollary 8.3** (The closure creates no pair)**.** *For every Hamiltonian on $\mathfrak{s}_n^*$, including any coupling of the constituents through $H$, $p^{(k)}(\tau) = e^{-\theta(\tau)G} p^{(k)}(0)$ with one angle $\theta$. Hence $E_6$, $E_0$ and $\mathbf{S}$ of each $\pi^{(k)}$, the Hermitian Gram matrix $\bar{z}_k \cdot z_l$ and the moduli $|Q_{kl}|$ of the momentum fields are constants of motion, and $Q \mapsto e^{2i\theta} Q$. No closure Hamiltonian turns a null constituent into a massive one, fills an empty copy, or moves rest energy between constituents.*

*Proof.* The Lie–Poisson equations of $\mathfrak{s}_n$ are (7) with $G_n = \bigoplus_k G$; the first integrates to $p = e^{-\theta G_n} p_0$, $\dot{\theta} = H_h$. □

Obstructed, for the momentum field and every $H$; it extends Theorem 4.1 to $n$ constituents. With a random Hamiltonian that couples the two copies (quartic in $h$, trigonometric in $p$) these quantities stay constant to $2 \times 10^{-13}$ and $Q$ follows $e^{2i\theta} Q$ to the same accuracy (d6-3). Inside the closure a pair is therefore a label of the initial data, carried and not generated, and the balance surface is where such labels are compatible with a constant rate.

## 8.2 The balanced pairs

**Proposition 8.4** (Balanced pairs)**.** *Let $z_k = z(\pi^{(k)})$.*

*(i) Equal inertias. With $a = \frac{1}{2}(z_1 - iz_2)$ and $b = \frac{1}{2}(z_1 + iz_2)$, $q_1 + q_2 = 4\, a \cdot b$ and $q_1 = a \cdot a + 2a \cdot b + b \cdot b$. The balance surface is the quadric $a \cdot b = 0$, and on it $q_2 = -q_1$ and $E_0^{(1)} = E_0^{(2)} = \frac{1}{2}|a \cdot a + b \cdot b|$.*

*(ii) $EG_2 = \epsilon \otimes G$ is an $\eta_2$-antisymmetric involution, the noncompact generator of $\mathfrak{g}(0, 2, 0) \cong \mathfrak{so}(2, \mathbb{C})$. Its eigenspaces $V_\pm = \{X^{(2)} = \pm GX^{(1)}\}$ are six-dimensional and totally $\eta_2$-null, and $E = \mp G_2$ on $V_\pm$. A momentum field lies in $V_+$ iff $p^{(2)} = -Gp^{(1)}$ iff $b = 0$, and in $V_-$ iff $p^{(2)} = Gp^{(1)}$ iff $a = 0$; so $V_\pm$ lie in the balance surface, and on them the copy rotation $R_\varphi$ is the duality rotation $e^{\mp\varphi G_2}$ of both copies.*

*(iii) On $V_+$, $z_2 = iz_1$: $w_2 = -w_1$, and the two momentum fields have equal $E_6$, $\mathbf{S}$ and $E_0$; when $q_1 > 0$ the first is of magnetic and the second of electric type. $T(z_2) = T(z_1)$, so the summed stress tensor is $2T(z_1)$ and $M_{\text{pair}} = 2E_0^{(1)}$, although $q_{\text{tot}} = 0$.*

*(iv) Unequal inertias. If $q_1 \neq 0$ the balanced pairs are $z_2 = \rho\Lambda z_1$ with $\Lambda \in SO(3, \mathbb{C})$ and $\rho^2 = -c_1/c_2$, and $|c_1| E_0^{(1)} = |c_2| E_0^{(2)}$.*

*Proof.* (i) Expand. (ii) $(\epsilon \otimes G)^2 = \epsilon^2 \otimes G^2 = \mathbf{1}$ and $(\epsilon \otimes G)^\dagger = \epsilon^\top \otimes G^\dagger = -\epsilon \otimes G$. The eigenvalue $+1$ means $-GX^{(2)} = X^{(1)}$, i.e. $X^{(2)} = GX^{(1)}$, and then $\eta_2(X, Y) = \eta(X^{(1)}, Y^{(1)}) + \eta(GX^{(1)}, GY^{(1)}) = 0$ since $G$ is an anti-isometry; on $V_+$, $E(X^{(1)}, X^{(2)}) = (-GX^{(1)}, X^{(1)}) =$

$-G_2(X^{(1)}, X^{(2)})$. Finally $\eta^{-1}p^{(2)} = G\eta^{-1}p^{(1)}$ iff $p^{(2)} = -Gp^{(1)}$, as $\eta G = -G\eta$, and $z_2 = iz_1$ iff $b = 0$. (iii) $T$ is duality invariant (Proposition 2.5), and Proposition 2.5 gives $|2T(z_1)t| = 2E_0^{(1)}$ for every $t$. (iv) $c_1q_1 + c_2\rho^2 q(\Lambda z_1) = c_1(q_1 - q_1) = 0$; conversely $q(z_2/\rho) = q_1 \neq 0$, and $SO(3,\mathbb{C})$ is transitive on each complex sphere $\{q = q_1\}$ (Theorem 2.3). □

The first sentence of (iii) restates the last sentence of Proposition 17.18 of [1] (fixed there); (i), (ii) and (iv) are fixed (that $V_\pm$ are null, and are exactly where the copy rotation is a duality rotation, could have failed); "magnetic and electric pair" and "duality-balanced" are readings. Checked exactly (d6-4a–c) and numerically (d6-4d). So the balanced pair of Proposition 17.18 of [1] is a magnetic and an electric constituent a quarter duality turn apart, with equal rest energies, whose duality phases the rotor turns together. Its total $q$ vanishes, but $q_{\mathrm{tot}}$ is not the rest energy of the pair: its summed density has rest energy $2E_0^{(1)}$ for every observer. Thus $\frac{1}{2}|q_{\mathrm{tot}}|$ is what the rotor sees ($R_{\mathrm{eff}} = |\nu|\,|q_{\mathrm{tot}}|/m$ for equal inertias) and $M_{\mathrm{pair}} \geq \frac{1}{2}|q_{\mathrm{tot}}|$ is what an observer of the two copies sees (Theorem 8.6). The block-diagonal inertia is the per-copy Lorentz requirement: the invariant symmetric forms on $V_2^*$ number four under $\mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)}$, six under $\mathfrak{l}_{G,\Delta}$ alone (cross terms, $U = \frac{1}{2}\operatorname{Re}\operatorname{tr}(CQ(\pi))$, $C$ complex symmetric, balance $\operatorname{tr}(CQ(\pi)) = 0$, again of codimension two) and two under $\mathrm{Cent}(G_2) \cap \mathfrak{so}(6,6)$, all of Lorentz class (d6-4e; the $n = 2$ analogue of Theorem 17.24 of [1]).

### 8.3 What a copy rotation can and cannot duplicate

**Proposition 8.5** (Lorentz-covariant duplication). *(i) The real linear maps of $V_2$ commuting with $\mathfrak{l}_{G,\Delta}$ are the $A \otimes \mathbf{1}_3$, $A \in M_2(\mathbb{C})$, acting on $Z$ with $G_2 = i$. Inside $\mathfrak{so}(6,6)$ they form* $\mathrm{span}\{E, EG_2\} \cong \mathfrak{so}(2,\mathbb{C})$, *which is $\mathfrak{g}(0,2,0)$ of Theorem 10.8 of [1]; its compact part is $\{R_\varphi\}$.*

*(ii) $A \otimes \mathbf{1}_3$ maps $Q \mapsto AQA^\top$ and $Z \mapsto ZA^\top$ ($Z$ the $3 \times 2$ matrix of columns $z_k$). Hence $R_\varphi$ preserves $H_{\mathrm{sum}}$, so every summed density $P_{\mathrm{sum}}(t)$, and $q_{\mathrm{tot}}$, $\det Q$, $\sigma_1$, $\sigma_2$. The unitary $A$, the group $U(2)$ of energy-preserving duality rotations of two free Maxwell fields (end of Section 13.5 of [1]; [16]), preserve $H_{\mathrm{sum}}$, $\sigma_1$ and $\sigma_2$ but not $q_{\mathrm{tot}}$, and $\mathfrak{u}(2) \cap \mathfrak{so}(6,6) = \mathbb{R}E$.*

*(iii) Every map $F : \mathbb{C}^6 \to \mathbb{C}^6$, linear or not, commuting with the diagonal $SO(3,\mathbb{C})$ sends $(z,0)$ with $q(z) = 0 \neq z$ to $(\alpha z, \beta z)$: two null constituents. In particular $R_\varphi(z,0) = (\cos\varphi\, z, \sin\varphi\, z)$.*

*Proof.* (i) $V$ is an irreducible $\mathfrak{l}_G$-module with commutant $\mathrm{span}\{\mathbf{1}, G\} \cong \mathbb{C}$ (Proposition 17.4 of [1]), so the commutant of $\mathfrak{l}_{G,\Delta}$ on $\mathbb{R}^2 \otimes V$ is $M_2(\mathbb{R}) \otimes \mathrm{span}\{\mathbf{1}, G\} = M_2(\mathbb{C})$; antisymmetry for $\eta_2$ leaves $A \in \mathbb{C}\epsilon$, which is Proposition 13.14 of [1] (row "diagonal $\mathfrak{l}_G$", dimension 2). (ii) $Q = Z^\top Z$ and $H_{\mathrm{sum}} = ZZ^\dagger$, and $T$ is linear in $H$ (Proposition 2.4). (iii) For $\Lambda \in SO(3,\mathbb{C})$ fixing $z$, $F(z,0) = F(\Lambda z, 0) = \Lambda F(z,0)$. In the action $u \mapsto w \times u$ of $\mathfrak{so}(3,\mathbb{C})$ ((8)) the stabiliser algebra of a null $z$ is $\{w : w \times z = 0\} = \mathbb{C}z$, whose common fixed vectors satisfy $z \times v = 0$, i.e. $v \in \mathbb{C}z$. □

Item (i) restates Proposition 13.14 and Theorem 10.8 of [1] (fixed there), (ii) is inherited linear algebra and restates the remark on $U(n)$ at the end of Section 13.5 of [1], and (iii) is an obstruction; all checked exactly (d6-5a–c). This settles the task's reading in the negative. The copy rotation acting identically in all six directions is the admissible $b$-field between the two manifolds, and it duplicates a field literally: the undivided energy and the rest energy of $(z,0)$ are shared in the same ratio $\cos^2\varphi : \sin^2\varphi$, a null field becomes two null fields and a massive one two massive replicas. It cannot produce a balanced massive pair from a single null field, and no other Lorentz-covariant operation can.

**Theorem 8.6** (The rest energy of a duplicated field point). *Let $z_1, z_2 \in \mathbb{C}^3$, with Gram matrix $Q$ of singular values $\sigma_1 \geq \sigma_2$.*

*(i)* $T(z_1)\cdot T(z_2) = |z_1\cdot z_2|^2$, *and as mixed tensors* $\big(T(z_1)+T(z_2)\big)^2 = \frac{1}{4}\|Q\|_F^2\,\mathbf{1} + T(z_1\times z_2)$, *with* $(z_1\times z_2)\cdot(z_1\times z_2) = \det Q$.

*(ii)* $|P_{\mathrm{sum}}(t)|^2 = \frac{1}{4}\|Q\|_F^2 + u_t(z_1\times z_2)$ *for every observer* $t$, $u_t(w)$ *being the energy density of* $w$ *seen by* $t$. *Hence* $M_{\mathrm{pair}} = \frac{1}{2}(\sigma_1+\sigma_2)$, *attained exactly in the rest frames of* $z_1\times z_2$ *when it is not null.*

*(iii)* *In every frame,* $E_0^{(1)} + E_0^{(2)} \le M_{\mathrm{pair}} \le |P_{\mathrm{sum}}(t)| \le E_6^{(1)} + E_6^{(2)}$, *and* $M_{\mathrm{pair}} \ge \frac{1}{2}|q_{\mathrm{tot}}|$.

*(iv)* $\max_\varphi(E_0^{(1)} + E_0^{(2)})(R_\varphi Z) = M_{\mathrm{pair}}$.

*(v)* *For two null fields* $M_{\mathrm{pair}} = |z_1\cdot z_2| = (E_6^{(1)}E_6^{(2)})^{1/2}(1-\cos\vartheta_{12})$, $\vartheta_{12}$ *the angle between their Poynting vectors; for* $(z,0)$ *with* $z$ *null,* $M_{\mathrm{pair}} = 0$.

*Proof.* (i) Polynomial identities, verified symbolically in twelve real variables (d6-6a); in spinor form [44], $T = \sum_k \phi_k\bar\phi_k$ and $\phi^k_{AB}\phi^{l\,B}{}_C = \frac{1}{2}(\phi^k\cdot\phi^l)\epsilon_{AC} + \chi^{kl}_{AC}$ with $\chi^{kl}$ antisymmetric in $(k,l)$, so the cross terms of $T^2$ pair the symmetric $Q$ with the antisymmetric $\chi$ and drop out. The last identity is the complex Lagrange identity. (ii) $T$ is $g$-self-adjoint, so $|P(t)|^2 = t\cdot T^2 t$; insert (i). By Theorem 2.6, $u_t(w) \ge E_0(w) = \frac{1}{2}|\det Q|$, with equality iff $t$ is a rest frame of $w$, and $\frac{1}{4}\|Q\|_F^2 + \frac{1}{2}|\det Q| = \frac{1}{4}(\sigma_1+\sigma_2)^2$. For null $w \ne 0$ the infimum is approached by Doppler boosts; for $w = 0$, $|P(t)| = \frac{1}{2}\sigma_1$ for all $t$. (iii) Each $P_k(t)$ is future causal of length $E_0^{(k)}$ (Proposition 2.5), future causal vectors obey $|P_1+P_2| \ge |P_1| + |P_2|$ and $|P| \le P^0$, and $\sigma_1+\sigma_2 \ge |\operatorname{tr} Q|$. (iv) Write $q_{1,2}(\varphi) = t \pm d(\varphi)$ and $Q_{12}(\varphi) = r(\varphi)$ with $t = \frac{1}{2}q_{\mathrm{tot}}$. The pair $(d,r)$ turns by $2\varphi$, $d^2+r^2 = t^2 - \det Q$ is fixed, and $(|q_1|+|q_2|)^2 = 2|t|^2 + 2|d|^2 + 2|t^2-d^2| \le 2|t|^2+2|d|^2+2|r|^2+2|\det Q| = (\sigma_1+\sigma_2)^2$, by $|t^2-d^2| = |\det Q + r^2| \le |\det Q| + |r|^2$. Equality needs $r(\varphi)^2$ to be a non-negative multiple of $\det Q$; as $\varphi$ varies, $r(\varphi)$ runs over an ellipse centred at 0 or a segment through 0, so such $\varphi$ exists. (v) $Q = \left(\begin{smallmatrix} 0 & r \\ r & 0\end{smallmatrix}\right)$, $r = z_1\cdot z_2$; and $T(z_k) = k_k\otimes k_k$ with $k_k$ future null, $k_k^0 = (E_6^{(k)})^{1/2}$, so $|r| = k_1\cdot k_2$ by (i), and $|\mathbf{S}^{(k)}| = E_6^{(k)}$. □

Inherited: ordinary electrodynamics of two field strengths, with the reverse triangle inequality as the kinematics of invariant mass; (i), (ii) and (iv) are not stated in [1]. Checked symbolically (i), at the rest frames of $z_1\times z_2$ to $10^{-10}$ against 300 random observers (ii), by maximising over $\varphi$ (relative deviation $5\times10^{-16}$) (iv), and on random and null pairs (iii), (v) (d6-6a–d).

**Example 8.7** (Two counter-propagating null fields)**.** $z_1 = (1,i,0)$ and $z_2 = (1,-i,0)$ are null with $\mathbf{S}^{(1)} = -e_3 = -\mathbf{S}^{(2)}$, so $\mathbf{S}_{\mathrm{sum}} = 0$, $E_6^{\mathrm{tot}} = 2$, $Q = \left(\begin{smallmatrix} 0 & 2 \\ 2 & 0\end{smallmatrix}\right)$ and $M_{\mathrm{pair}} = 2$. The rotation $R_{\pi/4}$ gives $z_1' = (0,\sqrt{2}\,i,0)$ and $z_2' = (\sqrt{2},0,0)$: a pure electric and a pure magnetic field, both at rest, with $q = -2,+2$ and $E_0 = 1$ each. Every inequality of Theorem 8.6(iii) is an equality: the balanced pair is produced at rest in the centre-of-momentum frame and takes the whole invariant mass (d6-7). Inherited: the kinematics of $\gamma\gamma \to e^+e^-$ [4]; the same superposition is the standing wave of Proposition 2.9(3).

**Example 8.8** (The rotation of one direction)**.** Let $R_\varphi^{(1)}$ rotate the planes $(e_1^{(1)}, e_1^{(2)})$ and $(e_4^{(1)}, e_4^{(2)})$ through $\varphi$. It is compact, commutes with $G_2$ and lies in $\mathrm{Cent}(G_2)\cap\mathfrak{so}(6,6) \cong \mathfrak{so}(6,\mathbb{C})$, the $n=2$ analogue of Theorem 17.19 of [1]. On $((1,i,0),0)$ it gives $z_1' = (\cos\varphi, i, 0)$, $z_2' = (\sin\varphi,0,0)$: $q_1' = -\sin^2\varphi$, $q_2' = \sin^2\varphi$, a balanced pair with $E_0 = \frac{1}{2}\sin^2\varphi$ each. $E_6^{\mathrm{tot}} = 1$ and $q_{\mathrm{tot}} = 0$ are conserved, $\mathbf{S}_{\mathrm{sum}}$ goes from $-e_3$ to $-\cos\varphi\, e_3$ and $M_{\mathrm{pair}}$ from 0 to $\sin\varphi$, so $E_6^{\mathrm{tot}} = 1 \ge M_{\mathrm{pair}} = \sin\varphi \ge \sum_k E_0^{(k)} = \sin^2\varphi$, with equality throughout at $\varphi = \pi/2$ (a magnetic field along $e_1$ and an electric one along $e_2$, at rest). $R^{(1)}$ does not commute with $\mathfrak{l}_{G,\Delta}$, and lies neither in $\mathfrak{g}(0,2,0)$ nor in $\mathfrak{g}(0,1,1) = \mathfrak{so}(3,3)$ (Lorentz algebra on copy 1, copy 2 a spectator): it is a candidate gauge transformation under no Lorentz action of Theorem 10.8 of [1] with $k_I = 0$ (d6-8). Read kinematically it is an infinitely heavy third body, absorbing momentum

and no energy, and $E_6 \geq E_0^{(1)} + E_0^{(2)}$ is the field-point form of the threshold $E_\gamma \geq 2mc^2$ for pair production in the field of a nucleus [3]; the kinematics is inherited, and the agent is constructed: it exists, and nothing in the arena supplies it.

The rotation in the task's example is $R_\varphi^{(1)}$, not the copy rotation $R_\varphi$: the conservation of $E_6^{\mathrm{tot}}$ and $q_{\mathrm{tot}}$ and the non-conservation of $\mathbf{S}_{\mathrm{sum}}$ hold for it as stated, but it is not "identical in all six directions", and the copy rotation that is conserves $\mathbf{S}_{\mathrm{sum}}$ and the whole summed stress tensor.

### 8.4 Is the rest energy of one copy observable?

**Proposition 8.9** (Gauge status of the per-copy rest energy)**.** *(i) $R_\varphi$ is admissible for the diagonal action: compact, commuting with $G_2$ and $\mathfrak{l}_{G,\Delta}$, and the compact part of $\mathfrak{g}(0,2,0)$. On $S_2$ it is generated by $M = \frac{1}{2}\sum_a \eta_{aa} e_a^{(1)} e_a^{(2)}$, with $\operatorname{ad} M = -E$ on $V_2$ and eigenvalues $i\{-3,\dots,3\}$; $U_\varphi = \prod_a (\cos\frac{\varphi}{2} + \sin\frac{\varphi}{2}\,\eta_{aa} e_a^{(1)} e_a^{(2)})$ has $U_{2\pi} = \mathbf{1}$, so the copy rotation acts through $SO(2)$; and $U_{\pi/2} J^{(1)} U_{\pi/2}^{-1} = J^{(2)}$, $J^{(k)} = e_2^{(k)} e_3^{(k)}$.*

*(ii) $R_\varphi$ preserves $H_{\mathrm{sum}}$, $q_{\mathrm{tot}}$, $\det Q$, $\sigma_{1,2}$, $M_{\mathrm{pair}}$ and the total charge $J^{(1)} + J^{(2)}$; it does not preserve $E_0^{(k)}$, $q_k$ or the per-copy charges (Example 8.7, (i)). On $V_\pm$ it is a duality rotation of both copies, and there the $E_0^{(k)}$ are invariant.*

*(iii) Equal inertias. The two-constituent closure is invariant under $O(\eta_2) \cap \mathrm{Cent}(G_2) \cong O(6,\mathbb{C})$, whose compact part, the real orthogonal group of $\mathbb{C}^6$, acts transitively on $\{q_{\mathrm{tot}} = 0,\ |Z| = r\}$: every balanced pair is carried to one null constituent and an empty copy at the same $E_6^{\mathrm{tot}}$; for instance $(e_1, ie_1) \mapsto ((1,i,0),0)$, with $M_{\mathrm{pair}}$ going from 1 to 0.*

*(iv) If $c_1 \neq c_2$, no nonzero element of $\mathfrak{g}(0,2,0)$ preserves the inertia.*

*Proof.* (i) Each term pairs two generators of the same square, hence is compact (Theorem 2.3 of [1]), squares to $-\frac{1}{4}$, and the six terms commute; exponentiate term by term. (ii) Proposition 8.5(ii) and Proposition 8.4(ii). (iii) Every $\Lambda$ commuting with $G_2$ preserves the Lie–Poisson structure, and for $\Lambda \in O(\eta_2)$, $\Lambda\eta_2^{-1}\Lambda^\top = \eta_2^{-1}$ and $\Lambda G_2 \eta_2^{-1} \Lambda^\top = G_2 \eta_2^{-1}$, so the equal Lorentz-class inertia is invariant. A nonzero $Z$ with $q_{\mathrm{tot}}(Z) = 0$ is $u + iv$ with $u, v \in \mathbb{R}^6$, $|u| = |v|$, $u \perp v$, and $SO(6)$ is transitive on such pairs. (iv) $sE + tEG_2$ acts on the copy index as $(s+it)\epsilon$, and $\epsilon^\top \operatorname{diag}(c_1,c_2) + \operatorname{diag}(c_1,c_2)\,\epsilon = (c_2 - c_1)\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, which must vanish for the form $\sum_k c_k\, z_k \cdot z_k$ to be invariant. □

Item (i) restates Theorem 10.8 and Proposition 10.9 of [1] (fixed there), with the lift computed; (ii)–(iv) are fixed; checked in integer arithmetic (d6-9, d6-10). This decides whether "mass creation in pairs" is physical. *If the copy rotation is gauged*, the rest energy of each copy is not an observable: Example 8.7 is a change of gauge, not a process, and two counter-propagating null fields and an electric–magnetic pair at rest are one physical state, whose gauge-invariant content $M_{\mathrm{pair}} = 2$ was there before. *In the equal-inertia closure alone*, its own compact symmetry identifies every balanced pair with one null constituent, and the rotor sees only $\frac{1}{2}|q_{\mathrm{tot}}| = 0$; the mass of the pair is invisible to the rate and becomes meaningful only through a per-copy or diagonal Lorentz action, which the closure does not select (the Lorentz action at several constituents is open, Theorem 17.20 of [1]). *If the copies are distinguished* — by unequal inertias, which by (iv) break $\mathfrak{g}(0,2,0)$, by unequal binding rates, by a projector, or by a Lorentz action on one copy only, whose algebra $\mathfrak{g}(0,1,1)$ contains no copy mixing — the per-copy rest energies are observables, but then the closure creates no pair (Corollary 8.3). In the interior with $S = \operatorname{diag}(\kappa_1,\kappa_2) \otimes G + \beta E$ the momentum field obeys $\dot\pi = S^\dagger \pi$, i.e. $z(\pi)(\tau) = e^{ih\tau} z(\pi)(0)$ with $h = \left(\begin{smallmatrix} \kappa_1 & -i\beta \\ i\beta & \kappa_2 \end{smallmatrix}\right)$ on the copy index. For $\kappa_1 = \kappa_2$ this is $e^{i\kappa\tau} R_{-\beta\tau}$, and the per-copy rest energies change only by the gauge rotation (Larmor [33]; Section 5). For $\kappa_1 \neq \kappa_2$ they change physically:

for the pair of Example 8.7, $E_0(\pi^{(1)})(\tau) = |z_1 \cdot z_2|\,|U_{11}U_{12}|$, $U = e^{ih\tau}$, which for $\kappa = (1,3)$, $\beta = 0.4$ peaks at 0.6897, the Rabi value $2\sqrt{s(1-s)}$ with $s = 4\beta^2/((\kappa_2-\kappa_1)^2+4\beta^2)$, at fixed $E_6^{\mathrm{tot}}$; from $(z,0)$ with $z$ null both copies stay null for all $\tau$ (d6-11). This is two-level mixing [47], inherited, and it moves rest energy between the manifolds only where a second, non-collinear field supplies it.

**Remark 8.10** (A copy rotor)**.** The uniqueness argument of Theorem 17.7 of [1] uses that equivariant maps between distinct constituents vanish, which holds for $\mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)}$ but not for $\mathfrak{l}_{G,\Delta}$: $E$ is $\mathfrak{l}_{G,\Delta}$-equivariant, commutes with $G_2$ and rotates $p$ without dilating it, so a second rotor $\hat{c}$ with $[\hat{c}, X] = EX$, $[\hat{c}, \hat{g}] = 0$ gives an equivariant closure $\mathbb{R}^2 \ltimes V_2$, with Lie–Poisson equations $\dot{p} = (H_h G_2^\top + H_{h_c} E^\top)p$, $\dot{h} = -p^\top G_2 \nabla H$, $\dot{h}_c = -p^\top E \nabla H$. With equal Lorentz-class inertias $\dot{h}_c = 0$, the copy angle turns uniformly, and the per-copy $E_0(\pi^{(k)})$ oscillate while the summed momentum tensor, $M_{\mathrm{pair}}(\pi)$ and $|q_{\mathrm{tot}}(\pi)|$ stay constant; $((1,i,0),0)$ stays null–null (d6-12). The inter-manifold $b$-field made dynamical moves rest energy between the manifolds at fixed pair invariants and creates none. Constructed.

## 8.5 The Dirac side

**Proposition 8.11** (Pairs of Dirac states in the tower)**.** *On $S_2 = \mathbb{R}^8 \otimes \mathbb{R}^8$ with the tower generators:*

*(i) $J^{(1)} = J \otimes \mathbf{1}$ and $J^{(2)} = \mathbf{1} \otimes J$ (the Klein factors cancel on bivectors); they commute with each other and with the eight interior gammas $e_a^{(k)}$, $a \in \{1,4,5,6\}$. With an external $i$, $Q_{\mathrm{tot}} = -i(J^{(1)} + J^{(2)})$ has spectrum $\{-2,0,2\}$ with multiplicities $16, 32, 16$.*

*(ii) Let $\psi, \phi \in S \otimes \mathbb{C}$ be particles at rest, $e_1\psi = \psi$, $J\psi = i\psi$ ($H(0) = me_1$, equation (8) of [1]), and $C = e_2$ (Theorem 5.1 of [1]). Then $\Psi = \psi \otimes C\phi$ has charges $(1,-1)$, $Q_{\mathrm{tot}}\Psi = 0$, and $-i\big(JH(0) \otimes \mathbf{1} + \mathbf{1} \otimes JH(0)\big)\Psi = 2m\Psi$.*

*(iii) $C^{(1)}C^{(2)} = e_2^{(1)}e_2^{(2)}$ anticommutes with $J^{(1)}$ and $J^{(2)}$, but $(C^{(1)}C^{(2)})^2 = -\mathbf{1}$: no state is mapped to $\pm$ itself. The Klein-corrected $C \otimes C = -\omega^{(1)}C^{(1)}C^{(2)}$, $\omega^{(1)} = \omega \otimes \mathbf{1}$, squares to $\mathbf{1}$, and $\Psi_\pm = \psi \otimes C\psi \pm C\psi \otimes \psi$ satisfy $(C \otimes C)\Psi_\pm = \pm\Psi_\pm$ and $Q_{\mathrm{tot}}\Psi_\pm = 0$.*

*(iv) The tower's mass terms anticommute: $(e_1^{(1)} + e_1^{(2)})^2 = 2\,\mathbf{1}$, so $m(e_1^{(1)} + e_1^{(2)})$ has spectrum $\{\pm\sqrt{2}\,m\}$, whereas the two-particle operator $m(e_1 \otimes \mathbf{1} + \mathbf{1} \otimes e_1) = m(e_1^{(1)} + \omega^{(1)}e_1^{(2)})$ has spectrum $\{-2m, 0, 2m\}$, multiplicities $16, 32, 16$.*

*(v) Under the diagonal interior action ($k_I = 2$) a rotation through $2\pi$ is $+\mathbf{1}$ on $S_2$.*

*Proof.* (i) $(\omega \otimes e_2)(\omega \otimes e_3) = \mathbf{1} \otimes J$ as $\omega^2 = \mathbf{1}$; the commutations are the blade rule (equation (3) of [1]); $(J^{(1)} + J^{(2)})^2 = -2(\mathbf{1} - J^{(1)}J^{(2)})$, where $\frac{1}{2}(\mathbf{1} - J^{(1)}J^{(2)})$ is a projector of trace 32. (ii) $C$ anticommutes with $e_1$ and $J$, and $J$ commutes with $H(0)$. (iii) Generators of distinct constituents anticommute and $e_2^2 = \mathbf{1}$; $\mathbf{1} \otimes e_2 = \omega^{(1)}e_2^{(2)}$. (iv) The same anticommutation; $e_1 \otimes \mathbf{1}$ and $\mathbf{1} \otimes e_1$ are commuting traceless involutions with traceless product. (v) Each copy's rotation through $2\pi$ is $-\mathbf{1}$, and the two commute. □

Inherited: Klein–Jordan–Wigner signs [26], the charge-conjugation parity of a particle–antiparticle pair [57], and integer spin for two spin-$\frac{1}{2}$ constituents, which is Proposition 10.9 of [1] at $k_I = 2$ (restated, fixed there); checked exactly (d6-13a–e). Two points are specific to the tower. The total charge is invariant under the copy rotation while $U_{\pi/2}$ exchanges $J^{(1)}$ and $J^{(2)}$ (Proposition 8.9(i)), so which copy carries the particle is gauge dependent exactly as the per-copy rest energy is. And by (iv) the graded duplication $\mathrm{Cl}(3,3)\hat{\otimes}\mathrm{Cl}(3,3) = \mathrm{Cl}(6,6)$ adds Dirac masses in quadrature, like two anticommuting Clifford directions — one particle of mass $\sqrt{2}\,m$ — while the pair threshold $2m$ appears only for Klein-corrected, ungraded operators: duplication in the tower's sense is not pair creation.

## 8.6 Zero-point energy of the duplicated frame

**Proposition 8.12** (Zero-point energy). *(i) For* $\mathcal{L} = \sum_k \frac{m_k}{2}\eta(\dot{X}^{(k)} + \kappa_k G X^{(k)}, \dot{X}^{(k)} + \kappa_k G X^{(k)})$ *each rate* $\kappa_k$ *carries Krein signature* $(3,3)$*; with positive-norm quantisation the zero-point energy is* $\sum_k \frac{\hbar\kappa_k}{2}(3-3) = 0$ *for all* $\kappa_1, \kappa_2, m_1, m_2$.

*(ii) With an inter-manifold* $b = \beta E$ *(*$m_1 = m_2$*,* $\kappa_k \geq 0$ *not both zero),* $S = \mathrm{diag}(\kappa_1, \kappa_2) \otimes G + \beta E$ *has spectrum* $\pm i\mu_\pm$*,* $\mu_\pm = \frac{1}{2}(\kappa_1+\kappa_2) \pm \left(\frac{1}{4}(\kappa_1-\kappa_2)^2+\beta^2\right)^{1/2}$*, each three times. Every solution is bounded unless* $\beta^2 = \kappa_1\kappa_2$ *and* $\kappa_1 \neq \kappa_2$*; when bounded, each line has Krein signature* $(3,3)$ *and the zero-point energy vanishes. At* $\kappa_1 = 0$*,* $|\mu_-| \approx \beta^2/\kappa_2$.

*(iii) At* $\beta^2 = \kappa_1\kappa_2$*,* $\kappa_1 \neq \kappa_2$*,* $\mu_- = 0$ *and the phase-space generator is not semisimple: some solutions grow linearly in* $\tau$.

*Proof.* (i) Proposition 3.10 per copy, or Theorem 5.14. (ii) On the $G_2 = i$ sector $S$ acts on the copy index as $i\,\mathrm{diag}(\kappa_1, \kappa_2) + \beta\epsilon = i(K - i\beta\epsilon)$ with $K - i\beta\epsilon$ Hermitian. $S^\dagger = K \otimes G - \beta E$ commutes with $G_2$, so for $\mu \neq 0$ its eigenspaces lie in the $G_2 = \pm i$ sectors, which are totally isotropic for $h$ (a vector with $GY = iY$ has $\mathbf{E} = -i\mathbf{B}$); Theorem 5.5(ii) then gives boundedness except at $\mu_- = 0$, and Theorem 5.14 the signatures. (iii) At $\beta^2 = \kappa_1\kappa_2$ the kernel of $S^\dagger$ consists of the $(v_1 c, v_2 c)$, $c \in \mathbb{C}^3$, with $v_2 = -i\kappa_1 v_1/\beta$, on which $\eta_2 = \mathrm{Re}\big((v_1^2 + v_2^2)\, c \cdot c\big)$ and $v_1^2 + v_2^2 = v_1^2(1 - \kappa_1/\kappa_2) \neq 0$; so the kernel is not $\eta_2$-null and Theorem 5.5(i) fails. □

Item (i) restates Theorem 18.6 of [1] (fixed there) for each copy: duplication preserves the neutral signature, and the cancellation survives unequal rates and masses. The lines in (ii) are inherited two-level mixing, with the seesaw value 0.0495097568 at $(\kappa_1, \kappa_2, \beta) = (0, 5, \frac{1}{2})$ against $\beta^2/\kappa_2 = 0.05$; the Krein statement is Theorem 5.14, fixed; (iii) is fixed and new, the inter-manifold analogue of the exceptional ratio of Theorem 5.10. Checked numerically for the signatures (d6-14a,b) and with rational data for semisimplicity: not semisimple at $(1, 4, 2)$, semisimple at $(1, 4, 1)$, $(2, 2, 2)$ and $(1, 3, \frac{2}{5})$ (d6-14c). Read with the added identification $mc^2 = \hbar\kappa$: an inter-manifold $b$ can give an unbound manifold the rate $\beta^2/\kappa_2$ borrowed from the binding of the other, or cancel a rate exactly at the price of unbounded motion; it never produces a rate without an existing binding, since at $\kappa_1 = \kappa_2 = 0$ the lines $\pm\beta$ are removed by $X \mapsto e^{-\beta E\tau} X$.

## 8.7 Assessment

*On the b-particle with two manifolds.* The remark is partly borne out. With two manifolds there is an admissible $b$-field between them, the copy rotation, the compact part of $\mathfrak{g}(0, 2, 0)$; it commutes with the binding, keeps every orbit bounded except at $\beta^2 = \kappa_1\kappa_2 \neq \kappa_1^2$, and moves rest energy from one manifold to the other, which on one manifold only the anti-diagonal part of $b$ can do, and only by tearing or by screening (Proposition 2.14 and Theorem 5.10); with unequal rates it mixes them, lending a rate to an unbound manifold or cancelling one. It does not generate an invariant mass: $M_{\mathrm{pair}}$, the summed stress tensor and $q_{\mathrm{tot}}$ are invariant under it; with equal rates it is removable; the rest energy of each copy is gauge dependent; and the one thing two constituents do in the closure that one cannot, carrying rest energy at a constant rate, needs no $b$-field at all: it is the balance surface.

*On how a unitary energy results in a duplication.* The mathematical process is the graded tensor product (inherited), under which the undivided energy becomes $E_6^{\mathrm{tot}} = \frac{1}{2}|Z|^2$; the Lorentz-covariant, energy–momentum-preserving duplications form $U(2) \otimes \mathbf{1}_3$, of which only the copy rotation $SO(2)$ is a metric, hence admissible, transformation (Proposition 8.5). The physical process it describes shares the energy of one copy between two as literal replicas, preserving the type; a balanced magnetic–electric pair requires either a second, non-collinear null field, with the inherited kinematics of $\gamma\gamma \to e^+e^-$ ($\sum_k E_0^{(k)} \leq M_{\mathrm{pair}}$, saturated), or a non-covariant agent that takes the momentum, with the kinematics of pair production on a nucleus

| claim | statement | class | check |
|---|---|---|---|
| Thm. 8.1(i) | rate constant iff $R_{\text{eff}} = 0$; frames on circles | fixed (restated, Prop. 17.18) | d6-2a–d |
| Thm. 8.1(ii) | $n = 1$: constant rate forces $E_0(\pi) = 0$ | fixed (restated, Cor. 17.14) | d6-1a |
| Thm. 8.1(iii) | $n \geq 2$: codimension-two balance set, massive data dense | fixed | d6-1b |
| Rem. 8.2 | without the Lorentz requirement $n = 1$ suffices | fixed (counterexample) | d6-1c |
| Cor. 8.3 | no closure Hamiltonian creates or moves a pair | obstructed | d6-3 |
| Prop. 8.4 | $a \cdot b = 0$; $V_\pm$ null, $E = \mp G_2$; magnetic–electric pair, $M_{\text{pair}} = 2E_0$, $q_{\text{tot}} = 0$ | fixed (part restated); readings interpretive | d6-4a–e |
| Prop. 8.5(i,ii) | commutant $M_2(\mathbb{C})$, $\mathfrak{g}(0, 2, 0)$, invariants of $R_\varphi$ and $U(2)$ | restated / inherited | d6-5a,b |
| Prop. 8.5(iii) | no covariant map makes a pair from one null field | obstructed | d6-5c |
| Thm. 8.6 | $M_{\text{pair}} = \frac{1}{2}(\sigma_1 + \sigma_2)$, the chain, $\max_\varphi$ | inherited | d6-6a–d |
| Ex. 8.7 | two null fields $\to$ electric and magnetic at rest | inherited | d6-7 |
| Ex. 8.8 | the task's rotation: non-covariant third body | constructed | d6-8 |
| Prop. 8.9 | $R_\varphi$ admissible; per-copy $E_0$ not invariant; $O(6, \mathbb{C})$ maps pairs to null fields | fixed (part restated) | d6-9, 10 |
| interior | distinguished copies: rest energy moved by two-level mixing | inherited | d6-11 |
| Rem. 8.10 | a dynamical copy rotor moves $E_0^{(k)}$ at fixed pair invariants | constructed | d6-12 |
| Prop. 8.11 | Klein signs, $C$-parity, masses in quadrature | inherited | d6-13a–e |
| Prop. 8.12 | zero-point energy 0; exceptional ratio $\beta^2 = \kappa_1\kappa_2$ | fixed ((i) restated) | d6-14a–c |
| answers | duplication as copy rotation, pair creation and balance | interpretive | — |

Table 11: The claims of Section 8, their classes and the checks of `d6_pair_balance.py` (d6-0 verifies the conventions).

($\sum_k E_0^{(k)} \leq E_6^{\text{tot}}$). In motion the closure conserves the content of each copy and carries the pair, on the balance surface, at a constant rate. The decisive caveat is Proposition 8.9: every statement that a mass has been created in one copy needs a structure that distinguishes the copies, and the symmetric closure and the gauged copy rotation provide none. Whether such a structure is physical is open.

# 9 The $b$-field between manifolds

The author's first remark is that the $b$-particle is a deal breaker when two manifolds or more are in question. Theorem 7.12 tested it for two binding frames and one field, the copy rotation. This section asks the question for any number of manifolds and any field that can act between them, and answers it in five steps. First, which fields exist depends on how the Lorentz algebra acts: a common action admits the complex orthogonal copy algebra $\mathfrak{so}(n, \mathbb{C})$, a spectator action decouples the Lorentz-carrying manifold from every binding-compatible field (fixed, partly restated from [1]; Section 9.1). Second, the closure itself admits such a field: with a common Lorentz action the rotor may act by any element of $\mathfrak{u}(n)$, which is a copy rotation plus a matrix of rates (constructed; Section 9.2). Third, every binding-compatible field turns the frame flow into a Schrödinger flow on $\mathbb{C}^{3n}$, unitary for the undivided energy exactly when the field is compact. The rates are gauge invariants, the field redistributes the frequencies at fixed sum,

and it is removable exactly when it preserves the rate eigenspaces (fixed; Sections 9.3 and 9.4). Fourth, three manifolds carry an invariant that two do not: the sign of $B_{12}B_{23}B_{31}$, odd under interior time reversal, the analogue of Jarlskog's invariant (fixed; Section 9.5). Fifth, the copy isospin is integral: frames carry isospin one, never one-half. The Yang–Mills doublet acting on two copies of the undivided energy is a connection in one of its three directions and a rate term in the other two, and doublets exist only under the spectator action, where unequal internal rates break $\mathfrak{so}(4)_\Delta$ of [1] to a diagonal $\mathfrak{so}(3)$ (fixed; Section 9.6). The verdict is in Section 9.7. Identifiers d7-$n$ refer to the `PASS` lines of `verify/d7_bfield_between.py` (Table 12).

**Notation.** As in Section 7, with $n$ copies: $V_n = \mathbb{R}^n \otimes \mathbb{R}^6$ in the copy-major basis, $\eta_n = \mathbf{1}_n \otimes \eta$, $G_n = \mathbf{1}_n \otimes G$, and $z = (z_1, \dots, z_n) \in \mathbb{C}^{3n}$ with $z_k = \mathbf{B}_k + i\mathbf{E}_k$, on which $G_n$ acts as $i$. $E_{kl}$ are the matrix units on the copy index. The *copy rotation* $C^{kl} = (E_{lk} - E_{kl}) \otimes \mathbf{1}_6$ maps copy $k$ to copy $l$, and the *copy boost* is $(E_{lk} - E_{kl}) \otimes G$. A *copy field* is $b = B \otimes \mathbf{1}_6$ with $B \in \mathfrak{so}(n)$ real antisymmetric, and $J := B_{12}B_{23}B_{31}$ at $n = 3$. The binding term with rates $\kappa_k \geq 0$ is $K \otimes G$ with $K = \mathrm{diag}(\kappa_1, \dots, \kappa_n)$, and $S = K \otimes G + b$, $\dot{X} = -SX + \pi/m$, $\dot{\pi} = S^\dagger \pi$, with $M$ as in Theorem 7.12. A field is *binding-compatible* if it commutes with $G_n$. Rates are rates, and reading one as a rest energy needs $mc^2 = \hbar\kappa$ (Proposition 4.10).

## 9.1 Which fields can act between manifolds

**Proposition 9.1** (Inter-manifold fields and the Lorentz action)**.** *(i)* Common binding action *(the diagonal $\mathfrak{l}_G$). The elements of $\mathfrak{so}(3n, 3n)$ that commute with it and with $G_n$ are $\{B \otimes \mathbf{1}_6 + B' \otimes G : \ B, B' \in \mathfrak{so}(n)\} \cong \mathfrak{so}(n, \mathbb{C})$, of dimension $0, 2, 6, 12$ for $n = 1, \dots, 4$. The compact part consists of the copy rotations, of dimension $0, 1, 3, 6$; the copy boosts $B' \otimes G$ are noncompact.*

*(ii)* Common interior action *(the diagonal $\mathfrak{l}_{\mathrm{int}}$, the action $\Delta$ of Section 10 of [1]). The elements of $\mathfrak{so}(3n, 3n)$ commuting with it and with $G_n$ are the copy rotations alone, of dimension $0, 1, 3$ for $n = 1, 2, 3$; no copy boost commutes with $\mathfrak{l}_{\mathrm{int}}$. At $n = 2$ the copy rotation is $C^{12} = M_{12} + C_{\mathrm{surp}}$, where $M_{12} = \varepsilon \otimes P_{\mathrm{int}}$ is the copy rotation of the interior four-planes, the copy algebra of Theorem 10.2 of [1], and $C_{\mathrm{surp}} = \varepsilon \otimes P_{\mathrm{surp}}$ that of the surplus pairs. Both commute with the diagonal $\mathfrak{l}_{\mathrm{int}}$, and neither commutes with $G_2$.*

*(iii)* Spectator action *($\mathfrak{l}_G^{(1)}$ or $\mathfrak{l}_{\mathrm{int}}^{(1)}$ on copy 1 alone). The elements of $\mathfrak{so}(3n, 3n)$ commuting with it and with $G_n$ form $\mathfrak{so}(3n - 3, \mathbb{C})$ of copies $2, \dots, n$, of dimension 6 and 30 at $n = 2, 3$. Every one of them vanishes on copy 1 and has no component into it.*

Part (i) restates the third row of Proposition 13.14 of [1] (fixed there) and Proposition 7.8(iii) at $n = 2$. Part (iii) restates Theorem 17.20 of [1] at $n = 3$ (fixed there), and Proposition 7.8(iv) at $n = 2$. Its proof is Schur's lemma: $V$ is an irreducible, nontrivial module of the Lorentz algebra of copy 1, and the other copies are trivial modules of it. Part (ii) and the decomposition of $C^{12}$ are fixed (d7-1a to d7-1d). Two consequences matter below. A binding-compatible, Lorentz-invariant field can couple the manifold that carries the Lorentz algebra to the others only when that algebra acts on all of them in the same way. Under the spectator action copy 1 is out of reach, and the inter-manifold fields act among the internal manifolds only. And the paper's copy algebra $\mathfrak{k}_{\mathrm{cop}}$ does not respect the binding: the binding-compatible copy rotation is $M_{kl}$ plus the surplus copy rotation.

## 9.2 The closure admits an inter-manifold field

Theorem 17.7 of [1] shows that, on one constituent, the only equivariant closure is $\mathbb{R} \ltimes_G V$: the rotor acts by $G$. After it the paper extends the result to several constituents "factor by factor, since equivariant maps between distinct constituents vanish", with a common rate. That

is correct for a Lorentz algebra acting on each constituent separately. For a common action it is not the only possibility.

**Proposition 9.2** (Rotor generators on $n$ binding frames)**.** *The endomorphisms of $V_n$ commuting with the diagonal $\mathfrak{l}_G$ form $\mathfrak{gl}(n,\mathbb{C}) = \mathfrak{gl}(n) \otimes \mathrm{span}\{\mathbf{1}, G\}$, of dimension $2n^2$. Those that rotate the frame covector without dilating it, the condition of Theorem 17.7 of [1], form*

$$\mathfrak{u}(n) = \mathfrak{so}(n) \otimes \mathbf{1}_6 \;\oplus\; \mathrm{Sym}(n) \otimes G,$$

*of dimension $n^2$. The first summand is $\eta_n$-antisymmetric (copy rotations) and the second is $\eta_n$-self-adjoint (rate matrices). The paper's rotor $G_n$ spans the centre. For $\mathfrak{s}_n = \mathbb{R} \ltimes_A V_n$ with $A = K \otimes G + B \otimes \mathbf{1}_6 \in \mathfrak{u}(n)$ and the rotor clamped at $\theta = \Omega\tau$, the substitution $X = e^{-\theta A}x$ of Section 17.3 of [1] gives $e^{-\theta A}\dot{x} = \dot{X} + \Omega A X$, so the shift is $S = \Omega A$.*

At $n = 2$ the algebra is the energy symmetry $\mathfrak{u}(2)$ of Proposition 7.4(i). The dimensions and the decomposition are fixed (d7-2a); the reading as rotor generators is new. The semidirect product is a Lie algebra for every $A$, so the closure exists. That a non-central $A$ is admissible is constructed: nothing in the algebra selects it, and Theorem 17.7 of [1] selects $G_n$ only once the action is taken per constituent, or a common rate is imposed. With a non-central $A$ the copy field and unequal rates enter the closure together, as the two halves of one generator. Their ratios are fixed by $A$ and their scale by the rotor. This is the one place found where the arena itself supplies the unequal rates that Theorem 7.12 needs. The paper's one-rotor closure, with $A = G_n$, gives every constituent the same rate (Proposition 17.18 of [1]).

### 9.3 The frame flow is a Schrödinger flow

**Theorem 9.3** (Binding-compatible fields between $n$ manifolds)**.** *Let $b \in \mathrm{Cent}(G_n) \cap \mathfrak{so}(3n, 3n)$. By Proposition 7.8(i), $b$ acts on $z$ as $R + iQ$ with $R, Q \in \mathfrak{so}(3n)$ real: $R$ is its paired-rotation part and $Q$ its paired-boost part. Put $\tilde{K} = K \otimes \mathbf{1}_3$ and $H := \tilde{K} - iR + Q$.*

*(i) $S$ acts on $z$ as $iH$, so the free frame flow is $z(\tau) = e^{-iH\tau}z(0)$.*

*(ii) The flow preserves $E_6 = \frac{1}{2}|z|^2$ for every $K$ and every initial value iff $Q = 0$, that is, iff $b$ is compact; then $H$ is Hermitian. The compact fields, the only ones admissible as Yang–Mills fields (Corollary 2.4 of [1]), are exactly those whose frame flow is unitary for the undivided energy.*

*(iii) The rates are gauge invariants. Under any $\eta_n$-isometric change of frame $X = U(\tau)Y$, the $\eta_n$-self-adjoint part of $S$ is conjugated, so its spectrum $\{\pm i\kappa_k\}$ is unchanged. No field, compact or not, changes a rate.*

*(iv) For compact $b$: $\mathrm{tr}\, H = 3\sum_k \kappa_k$ and $\mathrm{tr}\, H^2 = 3\sum_k \kappa_k^2 + \|R\|^2$. The field redistributes the frequencies of the frame at fixed sum and widens their spread (level repulsion).*

*(v) $b$ is removable along the worldline iff $[\tilde{K}, R + iQ] = 0$, that is, iff it preserves every rate eigenspace. At $n = 3$ the compact binding-compatible fields, of dimension 36, have a removable part of dimension 36, 18 and 9 for rates $(\kappa,\kappa,\kappa)$, $(\kappa,\kappa,\kappa')$ and three distinct rates. Within the copy $\mathfrak{so}(3)$ the dimensions are 3, 1 and 0.*

*(vi) (Boundedness) Let $b$ be compact. Every orbit of $(X, \pi)$ is bounded iff, for every $\lambda$ with $\lambda, -\lambda \in \mathrm{spec}\, H$ (including $\lambda = 0$), the eigenspaces $\mathcal{E}_\lambda(H)$ and $\mathcal{E}_{-\lambda}(H)$ are orthogonal for the bilinear form $v \cdot w = \sum_a v_a w_a$ on $\mathbb{C}^{3n}$.*

*Proof.* (i) On the sector $G_n = i$, $K \otimes G$ acts as $i\tilde{K}$, a paired rotation as $R$ and a paired boost as $iQ$ (Propositions 5.3 and 7.8); $\dot{X} = -SX$ becomes $\dot{z} = -iHz$. (ii) $E_6 = \frac{1}{2}|X|^2$, so it is conserved for all $X$ iff $S$ is Euclidean antisymmetric. $K \otimes G$ and the paired rotations are, and

the paired boosts are Euclidean symmetric. (iii) is Theorem 5.2(ii) on $V_n$. The self-adjoint part transforms as a tensor, $U^{-1}(K\otimes G)U$, while $U^{-1}bU+U^{-1}\dot{U}$ stays in $\mathfrak{so}(3n,3n)$. (iv) $\operatorname{tr}(\tilde{K}R)=0$ because $\tilde{K}$ is symmetric and $R$ antisymmetric, and $\operatorname{tr}(-iR)^2=\|R\|^2$. (v) is Proposition 5.7: $e^{-b\tau}$ intertwines the systems iff $b$ commutes with $K\otimes G$. (vi) is Theorem 5.5 with $S^{\top}=-S$ (both summands are Euclidean antisymmetric). Then $\mathcal{E}_\mu(S^\dagger)=\eta_n\mathcal{E}_{-\mu}(S)$, and the condition reads $\eta_n(\mathcal{E}_\nu(S),\mathcal{E}_{-\nu}(S))=0$ for every $\nu$. The sectors $G_n=\pm i$ are $\eta_n$-orthogonal, and on the sector $G_n=i$, $\eta_n=2\,v\cdot w$ in the coordinates $z$. The eigenspaces of $S$ there are those of $iH$, and on the other sector their conjugates. □

Parts (i), (ii), (iv) and (v) are fixed but elementary: the arena's version of the fact that a compact gauge field acts unitarily (d7-3a, d7-3b, d7-3d, d7-3e). Part (iii) is Theorem 5.2(ii) restated for $n$ copies, checked exactly on a Cayley curve (d7-3c). Part (vi) is a corollary of Theorem 5.5. On five cases at $n=3$, exact semisimplicity of $M$, the criterion and the growth of $\|e^{Mt}\|$ agree (d7-3f). Bounded: rates $(1,2,4)$ with $(B_{12},B_{23},B_{31})=(\frac12,\frac13,\frac15)$; the seesaw $(0,3,5)$ with $(1,\frac12,\frac13)$, a frame without rate confined by two couplings; equal rates $(1,1,1)$ with $B_{12}=1$. Unbounded: $(1,2,3)$ with $(1,1,1)$, where $\operatorname{spec}H=\{0,2,4\}$ and the zero mode has $v\cdot v\neq 0$, a resonance created by the field between manifolds with different rates; and zero rates with $(1,1,1)$, where the field produces only drift. Part (vi) generalises Theorem 7.12(v): at $n=2$, $\lambda_+ +\lambda_-=\kappa_1+\kappa_2$, so the only coincidence is $\lambda_-=0$, at $\beta^2=\kappa_1\kappa_2$.

### 9.4 Copy boosts: metric connections, not gauge fields

**Proposition 9.4** (The copy boost between two manifolds)**.** *Let $n=2$ and $b=\beta'\varepsilon\otimes G$, so that $H=K+\beta'\varepsilon$ on the copy index (Theorem 9.3(i)), real and not symmetric. Its eigenvalues are $\frac12(\kappa_1+\kappa_2)\pm\sqrt{\frac14(\kappa_1-\kappa_2)^2-\beta'^2}$.*

*(i) Below the exceptional point, $|\beta'|<\frac12|\kappa_1-\kappa_2|$, the spectrum is real, $M$ is semisimple with imaginary spectrum, and every orbit is bounded. The flow then preserves a positive form $z^\dagger\mathcal{Q}z$ with $H^\dagger\mathcal{Q}=\mathcal{Q}H$, which is not $E_6$.*

*(ii) At $|\beta'|=\frac12|\kappa_1-\kappa_2|$, $M$ is not semisimple; above it, $M$ has eigenvalues with real part $\pm\sqrt{\beta'^2-\frac14(\kappa_1-\kappa_2)^2}$.*

*(iii) At equal rates the threshold is zero: every $\beta'\neq 0$ grows exponentially, although $b$ commutes with $K\otimes G$ and is removable, by a boosted frame $e^{-b\tau}$ that is itself unbounded.*

This is fixed, checked exactly at rates $(1,4)$ with $\beta'=\frac12,1,\frac32,2$, and at $(2,2)$ with $\beta'=\frac13$ (d7-4a, d7-4b). It is the copy analogue of the pairing boosts of Proposition 16.7 of [1] and of Section 5.4. The spectral pattern is inherited: the non-Hermitian dimer with an exceptional point [27, 2], and the conserved metric of a pseudo-Hermitian Hamiltonian [39]. A copy boost is a metric connection, legitimate in the interior Lagrangian, but not a gauge field (Corollary 2.4 of [1]). Where it keeps the motion bounded it does so at the price of the undivided energy, which it replaces by another positive form.

### 9.5 Three manifolds: a time-odd invariant

**Theorem 9.5** (The copy field between three manifolds)**.** *Let $n=3$, distinct rates, and $b=B\otimes\mathbf{1}_6$, so $H=(K-iB)\otimes\mathbf{1}_3$.*

*(i) $\det(x-K+iB)=x^3-(\sum\kappa)x^2+(\sum_{k<l}\kappa_k\kappa_l-\sum B^2)x-(\kappa_1\kappa_2\kappa_3-\kappa_3B_{12}^2-\kappa_1B_{23}^2-\kappa_2B_{31}^2)$: the spectrum depends on the couplings only through their squares.*

*(ii) The plaquette $H_{12}H_{23}H_{31}=iJ$ is purely imaginary, and $\det[H,K]=2iJ(\kappa_1-\kappa_2)(\kappa_1-\kappa_3)(\kappa_2-\kappa_3)$.*

*(iii) The endomorphisms commuting with $G_3$, the diagonal $\mathfrak{l}_G$ and $K \otimes G$ are $\sum_k (a_k + b_k G)$ on copy $k$, and the $\eta_3$-isometries among them are exactly the eight copy reflections $\mathrm{diag}(\pm 1, \pm 1, \pm 1) \otimes \mathbf{1}_6$. Under them $B_{kl} \mapsto s_k s_l B_{kl}$, so the invariants of the field are $|B_{12}|, |B_{23}|, |B_{31}|$ and the sign of $J$.*

*(iv) With $P = \mathbf{1}_3 \otimes \mathrm{diag}(\mathbf{1}_3, -\mathbf{1}_3)$ ($\mathbf{E} \mapsto -\mathbf{E}$) and $T = -P$ ($\mathbf{B} \mapsto -\mathbf{B}$), parity maps $(K, B)$ to $(-K, B)$, and interior time reversal, $X(\tau) \mapsto TX(-\tau)$, maps $(K, B)$ to $(K, -B)$. At $n = 2$ a copy reflection undoes $B \mapsto -B$; at $n = 3$ with $J \neq 0$ none does, since $J \mapsto -J$. For general $n$, $(K, B)$ and $(K, -B)$ are related by a copy reflection iff the graph of the nonzero couplings is bipartite.*

*(v) With energy started in copy 1, the copy energies $P_k(\tau) = |z_k(\tau)|^2/|z(0)|^2$ obey $P_k(\tau; -B) = P_k(-\tau; B)$. $P_1$ is the same for $B$ and $-B$, and $P_2, P_3$ are not. At rates $(1, 2, 4)$ with $B_{12} = B_{23} = B_{31} = 1$ and $\tau = \frac{1}{2}$, $P_2 = 0.293$ for $B$ against $0.116$ for $-B$: the sign of $J$ is the sense in which the energy circulates around the three manifolds.*

*Proof.* (i), (ii) Symbolic expansion (d7-5a, d7-5b). For a $3 \times 3$ Hermitian matrix, $\det H$ contains $2\,\mathrm{Re}(H_{12}H_{23}H_{31})$, which vanishes here. (iii) The commutant is exact (d7-5c). $(a + bG)^\top \eta (a + bG) = (a^2 - b^2)\eta + 2ab\,\eta G$ because $G^\top \eta = \eta G$ and $G^2 = -\mathbf{1}$, so $ab = 0$ and $a^2 - b^2 = 1$. (iv) $PGP^{-1} = -G$, $P$ commutes with $B \otimes \mathbf{1}_6$, and $\dot{X} = -SX$ goes to $\dot{X}' = -S'X'$ with $S' = -TST^{-1}$. A copy reflection $s$ maps $B$ to $-B$ iff $s_k s_l = -1$ on every edge, which is a two-colouring (d7-5d, d7-5e). (v) $|(e^{-i\bar{H}\tau})_{k1}| = |(e^{iH\tau})_{k1}|$, and $P_1$ is even in $\tau$ for Hermitian $H$ (d7-5f). □

The statements are fixed. Their pattern is inherited three times over. With two families there is no rephasing-invariant phase and with three there is one, the Kobayashi–Maskawa phase [31], whose invariant measure is Jarlskog's commutator determinant [25], the form of (ii). A purely imaginary hopping around a triangle is a flux $\pm\pi/2$ through the plaquette, the Peierls phase [43] of a lattice gauge field. And time-reversal-odd hopping makes a quantum walk chiral [64]. What the arena adds is the identification. The copy field is an $\eta$-compatible gauge field on the three-site "lattice" of manifolds, with $J$ its Wilson loop. Its phase is fixed at $\pm\pi/2$ because the field is real: per-copy duality rotations, which would move it, are not $\eta$-isometries (Proposition 4.10(ii)). On one manifold the binding is odd under parity and even under interior time reversal. At two manifolds the copy field is even under both, up to relabelling. From three on, a field whose couplings close a triangle is odd under interior time reversal, and no relabelling removes it.

### 9.6 Isospin one, not one-half

Yang and Mills' $b$-field acts on the isotopic doublet of nucleons [63]. The question is whether an inter-manifold field can.

**Proposition 9.6** (Copy isospin)**.** *(i) On the tower module, the spin lift of the copy rotation of a set $D$ of directions is $-\frac{1}{2}\sum_{a \in D} \eta_{aa} e_a^{(k)} e_a^{(l)}$, a sum of $|D|$ commuting terms of square $-\frac{1}{4}$, so $\exp(2\pi\,\cdot) = (-1)^{|D|}$. At $n = 2$: for the six binding directions the eigenvalues are $i\{-3, \ldots, 3\}$ and $\exp(2\pi C) = +\mathbf{1}$. For the four interior directions ($M_{12}$) they are $i\{-2, \ldots, 2\}$, $+\mathbf{1}$. For the three temporal directions they are $i\{\pm\frac{1}{2}, \pm\frac{3}{2}\}$, $-\mathbf{1}$, and that rotation does not commute with $G_2$. A binding-compatible copy rotation pairs each temporal direction with its partner, so $|D|$ is even.*

*(ii) At $n = 3$ the three binding copy rotations close on $\mathfrak{so}(3)$ on the 512-dimensional module, with Casimir $j(j+1) = 0, 2, 6, 12$ on $40, 216, 200, 56$ dimensions ($j = 0, \ldots, 3$), and $\exp(2\pi C^{12}) = +\mathbf{1}$. The frames form the vector, $j = 1$.*

*(iii) The doublet* $\mathfrak{u}(2)$ *acting on* $(z_1, z_2)$ *preserves* $E_6 \oplus E_6$ *and commutes with* $G_2$ *and the diagonal* $\mathfrak{l}_G$*. In real form* $i\mathbf{1} \mapsto \mathbf{1} \otimes G$*,* $i\sigma_x \mapsto \sigma_x \otimes G$*,* $i\sigma_y \mapsto -\varepsilon \otimes \mathbf{1}_6$ *and* $i\sigma_z \mapsto \sigma_z \otimes G$*. Only* $i\sigma_y$*, the copy rotation, is* $\eta_2$*-antisymmetric. The other three are* $\eta_2$*-self-adjoint, terms of the same type as the binding. For* $S = \kappa G_2 + b_{\mathrm{YM}}$*,* $b_{\mathrm{YM}} = \sum_a \beta_a\,(i\sigma_a)$*, the lines are* $\kappa \pm |\boldsymbol{\beta}|$*, and the rates, the spectrum of the self-adjoint part, are* $\kappa \pm (\beta_x^2 + \beta_z^2)^{1/2}$*.*

This is fixed (d7-6a to d7-6c). Part (i) restates Proposition 10.9 of [1] ("copy rotations act through orthogonal groups") and Proposition 7.10(i), and adds that the binding is what makes the lift single-valued. Under a common Lorentz action the inter-manifold field is therefore an $SO(n)$ gauge field. At $n = 3$ it acts on triplets, as in the model of Georgi and Glashow [18], and never on doublets. Part (iii) restates Proposition 7.4(i) in the Yang–Mills reading and adds the spectra; it is the precise form of "Yang–Mills between two copies". The doublet group exists as a symmetry of the undivided energy (the copy $\mathfrak{u}(2)$), but $\eta_2$ cuts it to its circle. The two directions it removes are not connections at all: they belong to the symmetric part, where the binding $c^2$ lives in the dictionary of Section 1. A doublet field that splits the rates between two manifolds does so only through those two directions. In the arena that is a modification of the symmetric part, not a $b$-field.

**Proposition 9.7** (The weak isospin of the paper between unequal constituents)**.** *At* $n = 3$*, with the numbering of Section 17 of [1] and* $T_i^{\pm}$ *as in Proposition 5.16, let* $K = \mathrm{diag}(\kappa_1, \kappa_2, \kappa_3)$ *with* $\kappa_2 \neq \kappa_3$*.*

*(i) The part of* $\mathfrak{so}(4)_\Delta$ *commuting with* $K \otimes G$ *is* $\mathrm{span}\{\mathrm{rot}(13,14), \mathrm{rot}(13,15), \mathrm{rot}(14,15)\} = \mathrm{span}\{T_1^+ + T_1^-,\ T_2^+ - T_2^-,\ T_3^+ - T_3^-\}$*, the graph of an isomorphism* $\mathfrak{su}(2)_+ \to \mathfrak{su}(2)_-$*: a diagonal* $\mathfrak{so}(3)$*. The three directions* $\mathrm{rot}(7,13), \mathrm{rot}(7,14), \mathrm{rot}(7,15)$*, which join constituent two to constituent three, are not removable.*

*(ii) For* $\kappa = (5,3,5)$ *and* $b = 2T_3^+$*,*

$$\det(\lambda - S) = (\lambda^2+9)^2(\lambda^2+16)(\lambda^2+25)^3(\lambda^2+36)(\lambda^4+36\lambda^2+196),$$

*and* $M$ *is semisimple. The quartic carries the lines* $4 \pm \sqrt{2}$*, the two-level mixing of* $z_7$ *and* $z_{13}$ *with rates* $3, 5$ *and coupling* $1$ *(Theorem 7.12(ii)). At equal rates* $5$ *the same field gives* $(\lambda^2+25)^5(\lambda^2+16)^2(\lambda^2+36)^2$*, the removable Zeeman pattern of Proposition 5.17(i).*

This is fixed (d7-7a, d7-7b). It completes Proposition 5.17, which used equal rates. The paper's binding-compatible non-abelian algebra is itself an inter-manifold field: four of its six generators join constituent two to constituent three. It carries doublets ($(\frac{1}{2}, \frac{1}{2})$ on $W$) because the spectator action leaves constituents two and three free of the Lorentz algebra (Proposition 9.1(iii)). With equal rates it is removable along a worldline, as Section 5.6 found. With unequal rates it breaks to a diagonal $\mathfrak{so}(3)$, and its three remaining directions become physical. The pattern $\mathfrak{su}(2) \oplus \mathfrak{su}(2) \to \mathfrak{su}(2)_{\mathrm{diag}}$ is that of chiral symmetry breaking and of the custodial symmetry of the Higgs sector [52]. That is an interpretive parallel only: here the breaking is by a constant rate matrix, not by a vacuum.

## 9.7 Assessment

**Remark 9.8** (The deal breaker, settled)**.** (a) *True, in this sense* (fixed). Between manifolds with unequal rates, a compact binding-compatible field cannot be removed. It shifts and splits the lines, it can confine a frame that has no rate of its own, it can create a resonance, and from three manifolds on it is non-abelian and carries a time-odd invariant. The one-manifold no-go of Section 5 does not extend to $n \geq 2$ (Theorems 7.12, 9.3 and 9.5).

(b) *False, in this sense* (fixed). The field never creates or changes a rate. Rates are the spectrum of the self-adjoint part, a tensor under every change of frame. The field only redistributes

| statement | class | checks |
|---|---|---|
| Proposition 9.1(i), (iii): $\mathfrak{so}(n,\mathbb{C})$ under a common action; copy 1 decoupled under the spectator action | fixed (restated: Prop. 13.14, Thm 17.20) | d7-1a, d7-1d |
| Proposition 9.1(ii): $\mathfrak{so}(n)$ under the interior action; $C^{12} = M_{12} + C_{\mathrm{surp}}$ | fixed | d7-1b, d7-1c |
| Proposition 9.2: rotor generators $\mathfrak{u}(n) = \mathfrak{so}(n) \oplus \mathrm{Sym}(n) \otimes G$ | fixed; non-central rotor constructed | d7-2a |
| Theorem 9.3(i), (ii), (iv), (v): Schrödinger form, unitarity iff compact, sum rules, removability | fixed (elementary) | d7-3a, d7-3b, d7-3d, d7-3e |
| Theorem 9.3(iii): rates are gauge invariants | fixed (restated: Theorem 5.2) | d7-3c |
| Theorem 9.3(vi): bilinear boundedness criterion; five cases at $n = 3$ | fixed (corollary of Theorem 5.5) | d7-3f |
| Proposition 9.4: copy boost, exceptional point, conserved positive form | fixed; pattern inherited | d7-4a, d7-4b |
| Theorem 9.5: spectrum, plaquette, Jarlskog determinant, reflections, $P$ and $T$, bipartite criterion, circulation | fixed; parallels inherited | d7-5a to d7-5f |
| Proposition 9.6: integral copy isospin; the doublet is one connection and two rate terms | fixed (restated: Prop. 10.9, Proposition 7.4) | d7-6a to d7-6c |
| Proposition 9.7: $\mathfrak{so}(4)_\Delta \to$ diagonal $\mathfrak{so}(3)$ at unequal rates | fixed; custodial parallel interpretive | d7-7a, d7-7b |
| Remark 9.8: the deal breaker; the commutator; the arrow reversed | fixed; (c) constructed; (e) interpretive | d7-8 |

Table 12: Claims of Section 9, their classes, and the checks printed by `d7_bfield_between.py`.

the frequencies, at fixed sum. With equal rates every binding-compatible field, abelian or the paper's $\mathfrak{so}(4)_\Delta$, is removable along a worldline. With zero rates it produces only drift (Theorem 9.3(iii)–(vi)).

(c) *Where unequal rates come from.* Not from the paper's closure: its single rotor acts through $G_n$ at one rate, so within [1] as it stands the inter-manifold field is removable (fixed for that closure). Under a common Lorentz action the closure admits a rotor acting by a non-central element of $\mathfrak{u}(n)$. The copy field and the rate differences then arrive together, as two halves of one generator (constructed: admitted, not selected; Proposition 9.2).

(d) *Isospin one-half* (fixed). Under a common Lorentz action the copy isospin is integral, and the doublet field on two copies is a connection in one of its three directions only. Doublets need the spectator action, where the Lorentz-carrying manifold is decoupled (Propositions 9.1, 9.6 and 9.7).

(e) *The arrow reversed* (interpretive). The non-removable part of the copy field is $[K \otimes G, B \otimes \mathbf{1}_6] = [K, B] \otimes G$, with $[K, B]_{kl} = (\kappa_k - \kappa_l) B_{kl}$ and $\|[K \otimes G, b]\|^2 = 6 \sum_{k \neq l} (\kappa_k - \kappa_l)^2 B_{kl}^2$ (d7-8, fixed). Formally this is the gauge-boson mass term $g^2 \|[T, \langle \Phi \rangle]\|^2$ of the Englert–Brout–Higgs–Kibble mechanism [14, 23, 28], with the rate matrix in the role of an adjoint vacuum value. The arena thus reverses the arrow of the hypothesis. It is not the $b$-field that differentiates the rates; the differentiation of the rates, which breaks the copy symmetry, is what gives the $b$-field an effect. The parallel is not dynamical: here $b$ is a constant connection without kinetic term, and the rates are inputs.

## 10 Synthesis of the second question: duplication, and the $b$-field between manifolds

We return to the author's two remarks of Section 1: the $b$-particle is a deal breaker when two manifolds or more are in question, and some process makes a unitary energy result in a

duplication, from $\mathrm{Cl}(3,3)$ to $\mathrm{Cl}(6,6)$. The verdicts are collected in Table 13. As in Section 6, a rate becomes a rest energy only through the added identification $mc^2 = \hbar\kappa$.

## 10.1 How a unitary energy results in a duplication

**The mathematical process is forced on the algebra and chosen on the energy.** Doubling the vector space forces the graded tensor product $\mathrm{Cl}(V \oplus V) = \mathrm{Cl}(V)\hat{\otimes}\mathrm{Cl}(V)$. Its Klein factor is determined up to conjugation, it is the Jordan–Wigner sign, and the spinor module of the duplicate is the Fock space of six fermionic modes (Section 7.1; inherited). The energy is not forced. The undivided energy $(\mathbb{C}^3, E_6, U(3))$ embeds isometrically in $(\mathbb{C}^3 \otimes \mathbb{C}^2, E_6 \oplus E_6, U(6))$. The Lorentz-covariant embeddings form the three-sphere of a lossless beam splitter, and those that also respect $\eta$ form one circle, the copy rotation (Theorem 7.5). Read literally as a multiplication of the energy, the phrase fails: a linear duplication divides the undivided energy between the copies, and a quadratic copy $z \mapsto z \otimes z$ is excluded by the no-cloning theorem (inherited).

**The physical processes** that take one copy to two are listed in Table 9. Two of them concern mass. A duplication that preserves the Lorentz action cannot turn a single null field into a massive pair (Proposition 8.5). Making a pair needs either a second, non-collinear null field, which gives the kinematics of $\gamma\gamma \to e^+e^-$ (Theorem 8.6 and Example 8.7), or a non-covariant third body that takes the momentum, as a nucleus does in pair production (Example 8.8). All of this is inherited kinematics in the arena's variables. The closure never creates a pair (Corollary 8.3).

**What the duplicate can do that one copy cannot** (fixed, partly restated from [1]). It carries an $\eta$-isometric complex structure and a compact, Lorentz-invariant copy charge (Proposition 7.7), the copy algebra $\mathfrak{so}(2,\mathbb{C})$ on two binding frames (Proposition 7.8), a non-rigid Lorentz-invariant vacuum on $S^4$ and the imaginary unit $J = J_1 \otimes \omega$ (Proposition 7.11). In the closure, massive constituents can turn at an exactly constant rate on a codimension-two balance surface (Theorem 8.1). This last point is the precise dynamical sense in which the duplicate does what one copy cannot, and it needs no $b$-field. The duplicate also loses two things: spin one-half under the diagonal interior action (two fermions make a boson, Proposition 7.10), and the uniqueness of the photon direction.

**The caveat.** The rest energy of each copy separately is not gauge invariant. With equal inertias, the closure's own symmetry maps every balanced pair to one null constituent (Proposition 8.9). A statement that a mass has been created in one copy therefore needs a structure that distinguishes the copies. The symmetric closure and the gauged copy rotation provide none. Unequal rates do provide one, and that is where the second remark enters.

## 10.2 The $b$-field between manifolds

**It exists, and where it lives depends on the Lorentz action** (fixed; Proposition 9.1). If the Lorentz algebra acts on all $n$ binding frames in the same way, the binding-compatible fields between them form $\mathfrak{so}(n,\mathbb{C})$, with compact part the copy rotations $\mathfrak{so}(n)$: abelian for two manifolds, non-abelian ($\mathfrak{so}(3)$) from three on. If the Lorentz algebra acts on one manifold only, that manifold is decoupled from every binding-compatible field, and the fields act among the internal manifolds, as the paper's $\mathfrak{so}(4)_\Delta$ does.

**The remark is true for transmission** (fixed). Between manifolds with unequal rates a compact inter-manifold field cannot be removed along a worldline. It acts as a Hermitian coupling in the Schrödinger form of the frame flow. It repels the lines, confines a frame without rate of its own by a seesaw, and at special ratios creates a resonance with secular growth

(Theorems 7.12 and 9.3). From three manifolds on it carries an invariant that two manifolds lack: the sign of $B_{12}B_{23}B_{31}$, odd under interior time reversal and fixing the sense in which energy circulates. This is the arena's analogue of Jarlskog's invariant (Theorem 9.5). None of this can happen on one manifold, where the no-go of Section 5 stands.

**The remark is false for creation** (fixed). The rates are the spectrum of the self-adjoint part of the shift, which transforms as a tensor, so no field, between manifolds or on one, creates or changes a rate. The field redistributes the frequencies at fixed sum. With equal rates every binding-compatible field is removable, and that includes the non-abelian $\mathfrak{so}(4)_\Delta$. With zero rates the field produces only drift (Theorem 9.3). The deal breaker therefore presupposes a differentiation of the rates. The paper's one-rotor closure does not supply it, since it gives every constituent the same rate. A closure in which the rotor acts by a non-central element of $\mathfrak{u}(n)$ does supply it, and there the copy field and the rate differences arrive together, as the two halves of one generator (Proposition 9.2; constructed, not selected).

**Isospin one-half does not cross between manifolds that share the Lorentz algebra** (fixed). The copy isospin is integral on the whole tower module, because a binding-compatible copy rotation turns an even number of planes. The Yang–Mills doublet acting on two copies of the undivided energy is a connection in one of its three directions only. Its other two directions are self-adjoint, of the same type as the binding (Proposition 9.6). Doublets need the spectator action. There, unequal internal rates break $\mathfrak{so}(4)_\Delta = \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ to a diagonal $\mathfrak{so}(3)$ and make three directions physical (Proposition 9.7).

### 10.3 Back to $E = mc^2$

The second half sharpens the verdict of Section 6 on the clause "$m$ is generated from the $b$-particle". On one manifold the $b$-field is removable or tears the binding. Between manifolds it acts, but on frequencies, never on rates. Read in the dictionary of Section 1, the rates belong to the symmetric part, the $c^2$ of the hypothesis. The directions of a Yang–Mills doublet that would split them are not connections but symmetric terms. And the part of the copy field that survives every change of frame is the commutator $[K, B]_{kl} = (\kappa_k - \kappa_l)B_{kl}$, proportional to the rate differences (Remark 9.8). In the arena, then, the order of the hypothesis is reversed, as it was for the Pati–Salam bosons of [1], which acquire mass rather than generate it. The symmetric part carries the rates. Their differentiation, which is what a mass would be under $mc^2 = \hbar\kappa$, breaks the copy symmetry. The $b$-field between the manifolds is the gauge field of that broken symmetry, and it has an effect only because the symmetry is broken. Formally this is the structure of the Englert–Brout–Higgs–Kibble mechanism with the rate matrix in the role of an adjoint vacuum value (interpretive: here $b$ has no kinetic term and the rates are inputs).

What remains open is the origin of the rate differences. Within the arena there are two candidates: a rotor generator in $\mathfrak{u}(n)$ that is not central, which is admitted but not selected, and a quantum scale for the $b$-field by dimensional transmutation (Conjecture 5.19), which is conjectural.

## 11 How the statements were verified

Every finite-dimensional statement of Sections 2 to 5 and 7 to 9 was checked by computation, in two independent passes. The scripts are in `investigations/rest_energy/verify/` and run with `python3` from any directory, using `numpy`, `sympy`, `scipy` and `mpmath`.

1. *Derivation scripts* `d1_unitary_energy.py` to `d7_bfield_between.py`, one per section, were written together with the sections. They rebuild the realisation of [1], Appendix A, from its definition and print one `PASS` line per claim of their section, with the claim's label.

| statement | what the arena gives | verdict |
|---|---|---|
| the algebra is duplicated | $\mathrm{Cl}(6,6) = \mathrm{Cl}(3,3)\hat{\otimes}\mathrm{Cl}(3,3)$, Klein factor, Fock space (Section 7.1) | forced; inherited |
| the energy is duplicated | a beam splitter $S^3$, an $\eta$-circle; copying excluded (Theorem 7.5) | chosen; cloning obstructed |
| the duplicate adds structure | complex structure, copy charge, $\mathfrak{so}(2,\mathbb{C})$, $S^4$ vacuum, balance surface | fixed (partly restated) |
| a pair from one null field | impossible covariantly; two null fields or a third body (Proposition 8.5 and Theorem 8.6) | obstructed; inherited |
| the mass of one copy | gauge dependent unless the copies are distinguished (Proposition 8.9) | fixed; distinction open |
| a $b$-field between manifolds | $\mathfrak{so}(n,\mathbb{C})$, compact $\mathfrak{so}(n)$; spectator manifold decoupled (Proposition 9.1) | fixed |
| it is a deal breaker | non-removable at unequal rates; seesaw, confinement, resonance; time-odd at $n = 3$ | fixed |
| it generates a rate | rates are gauge invariants (Theorem 9.3(iii)) | obstructed |
| the rates differ | not in the paper's closure; a non-central rotor in $\mathfrak{u}(n)$ (Proposition 9.2) | fixed / constructed; origin open |
| isospin one-half between manifolds | integral copy isospin; doublet = one connection + two rate terms | obstructed (common action) |
| the arrow of the hypothesis | $[K, B] \propto$ rate differences makes $b$ physical | reversed; interpretive |

Table 13: Verdicts on the second question.

2. *Adversarial scripts* `x1_check.py` to `x7_check.py` were written afterwards by separate authors whose task was to refute the claims. They import nothing from the derivation scripts, rebuild every convention from the paper's sources, and wherever possible use a different method: symbolic where the first pass was numerical, random tests where it used special cases, a different integrator for the dynamics. Their findings were applied before the note was assembled: for the first part one refuted claim and several reclassifications, for the second part the corrections listed in the commit history of `investigations/rest_energy/`.

3. `run_all.py` runs all fourteen scripts and writes `RESULTS.md`; it exits with status zero only if every script does.

Algebraic statements were checked exactly, in integer or rational arithmetic, since every blade of the realisation is a signed permutation matrix. Statements about spectra and Jordan structure were checked with rational data; floating-point inputs were not used for semisimplicity tests, where they produce false negatives. Dynamical statements were checked by numerical integration with relative tolerances of $10^{-11}$ to $10^{-12}$, and elliptic and Mathieu quantities with `mpmath` and `scipy.special`. Statements of [1] that are cited without re-verification are marked as such in the text. The corrections made to [1] itself in the course of this work are listed, with their own checks, in `investigations/errata_2026-10-17/`.

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